

Esercitazione N° 1: **La miostatina**

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Creiamo il nostro profilo su Ubuntu

questo è un commento!

lanciamo il terminale per aggiungere un nuovo profilo

ctrl+alt+t

```
$ sudo adduser nome_battesimo -uid 663
```

```
$ pw di labx
```

per eliminare un profilo utente (in caso di errore)

```
$ sudo deluser --remove-home nome_utente
```

```
$ pw di labx
```

per recuperare la passwd dimenticata

sostituire <user name> con il proprio nome utente

```
$ sudo passwd user name
```

```
$ pw di labx
```

```
$ nuova pw
```

Creiamo il nostro profilo su Ubuntu

acquisiamo i privilegi dell'amministratore di sistema modificando il file di configurazione del Sistema Operativo (SO) `/etc/sudoers`

\$ **sudo visudo**

questo comando lancia l'editor di testo **nano** per creare ed aprire il file `/etc/sudoers.tmp` in modalità scrittura (per poterlo modificare); tale file è una copia di backup di `/etc/sudoers`

Creiamo il nostro profilo su Ubuntu

```
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults        env_reset
Defaults        mail_badpass
Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"

# This fixes CVE-2005-4890 and possibly breaks some versions of kdesu
# (#1011624, https://bugs.kde.org/show\_bug.cgi?id=452532)
Defaults        use_pty

# This preserves proxy settings from user environments of root
# equivalent users (group sudo)
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"

# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:%sudo env_keep += "EDITOR"

# Completely harmless preservation of a user preference.
#Defaults:%sudo env_keep += "GREP_COLOR"

# While you shouldn't normally run git as root, you need to with etckeeper
#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_*"

# Per-user preferences; root won't have sensible values for them.
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"

# "sudo scp" or "sudo rsync" should be able to use your SSH agent.
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification
```

[Lette 59 righe]

^G Guida
^X Esci**^O** Salva
^R Inserisci**^W** Cerca
^ Sostituisci**^K** Taglia
^U Incolla**^T** Esegui
^J Giustifica**^C** Posizione
^/ Vai a riga**M-U** Annulla
M-E Ripeti**M-A** Contrassegna
M-6 Copia**M-]** Parentesi
^Q Cerca ind.**M-Q** Precedente
M-W Successiva**^B** Indietro
^F Avanti

Creiamo il nostro profilo su Ubuntu

```
GNU nano 7.2 /etc/sudoers.tmp
# This file is edited by editors that preserve environment. If you
# want to control what environment is passed to sudoers you will have to
# edit the file in /etc/sudoers.d/
# Defaults: %sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"

# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
# Defaults: %sudo env_keep += "EDITOR"

# Completely harmless preservation of a user preference.
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# Ditto for GPG agent
# Defaults: %sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL
piro    ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin   ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

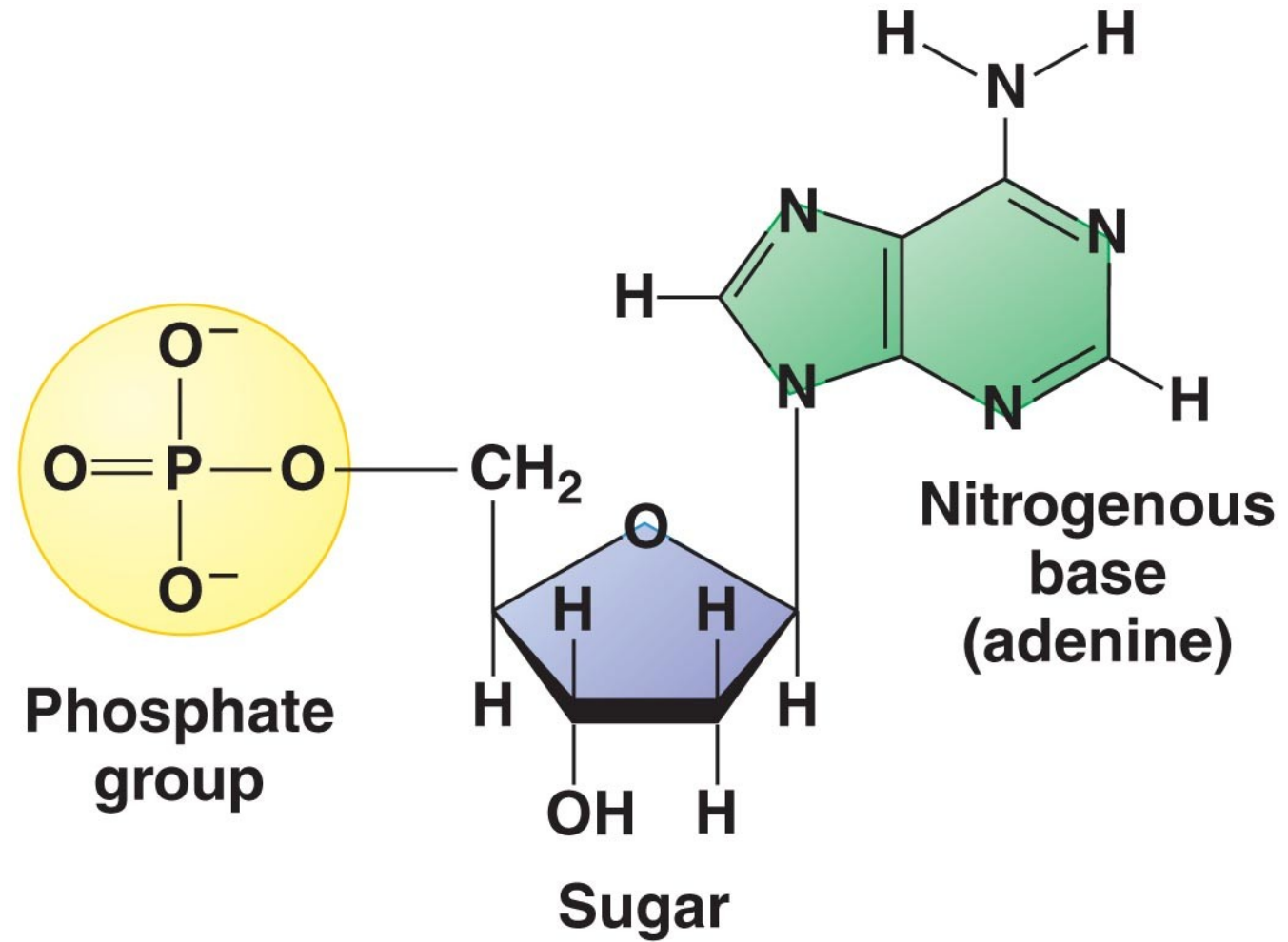
# See sudoers(5) for more information on "@include" directives:

@include /etc/sudoers.d
```

Elenco degli amministratori di sistema
portarsi con il cursore sull'ultima riga della lista
alt-6 # copia
ctrl-u # incolla l'intera riga
inserire il proprio nome utente
ctrl-o # per salvare
cancellare l'estensione .tmp
ctrl-x # per chiudere **nano**
alt-u # in caso di errore
(per annullare l'ultimo comando inserito)

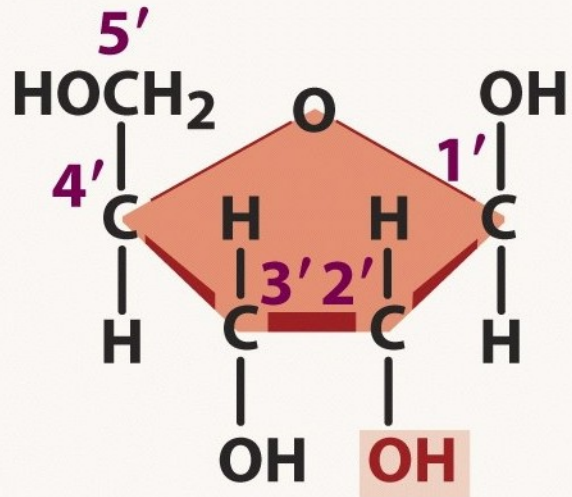
Elenco degli amministratori di sistema

Struttura di un nucleotide

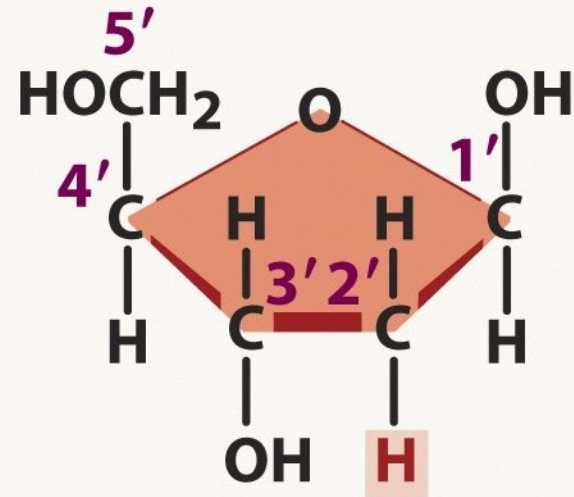


Zuccheri

Sugars



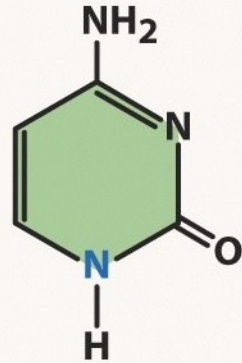
Ribose



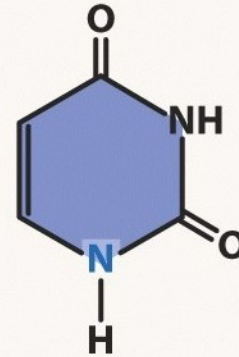
Deoxyribose

Basi pirimidiniche e puriniche

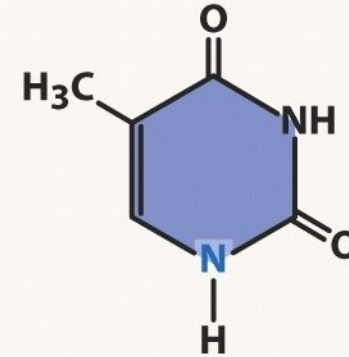
Nitrogen-containing bases



Cytosine (C)

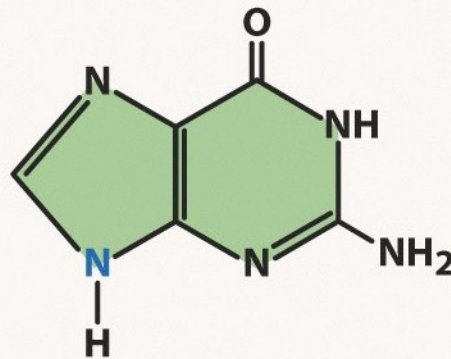


Uracil (U)

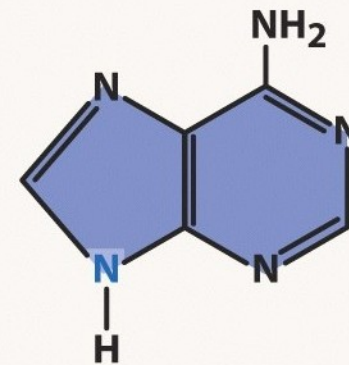


Thymine (T)

Pyrimidines



Guanine (G)



Adenine (A)

Purines

RNA

&

DNA

are chains of nucleotides

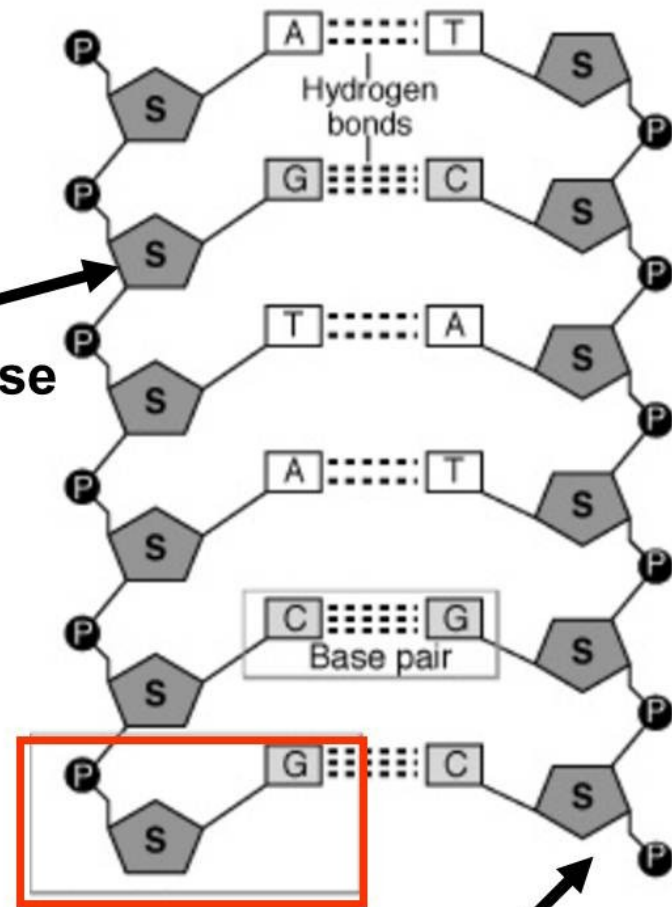
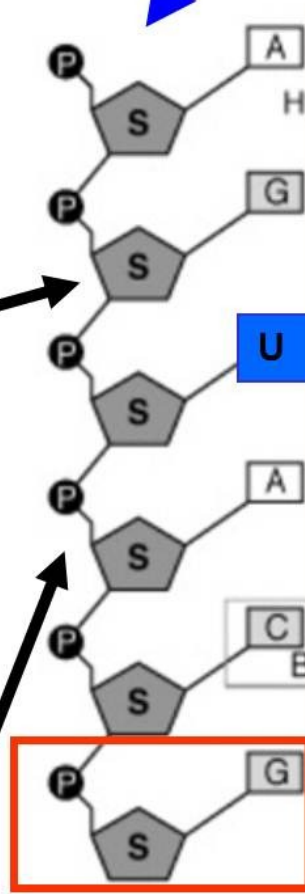
Ribose
(sugar)

Deoxyribose
(sugar)

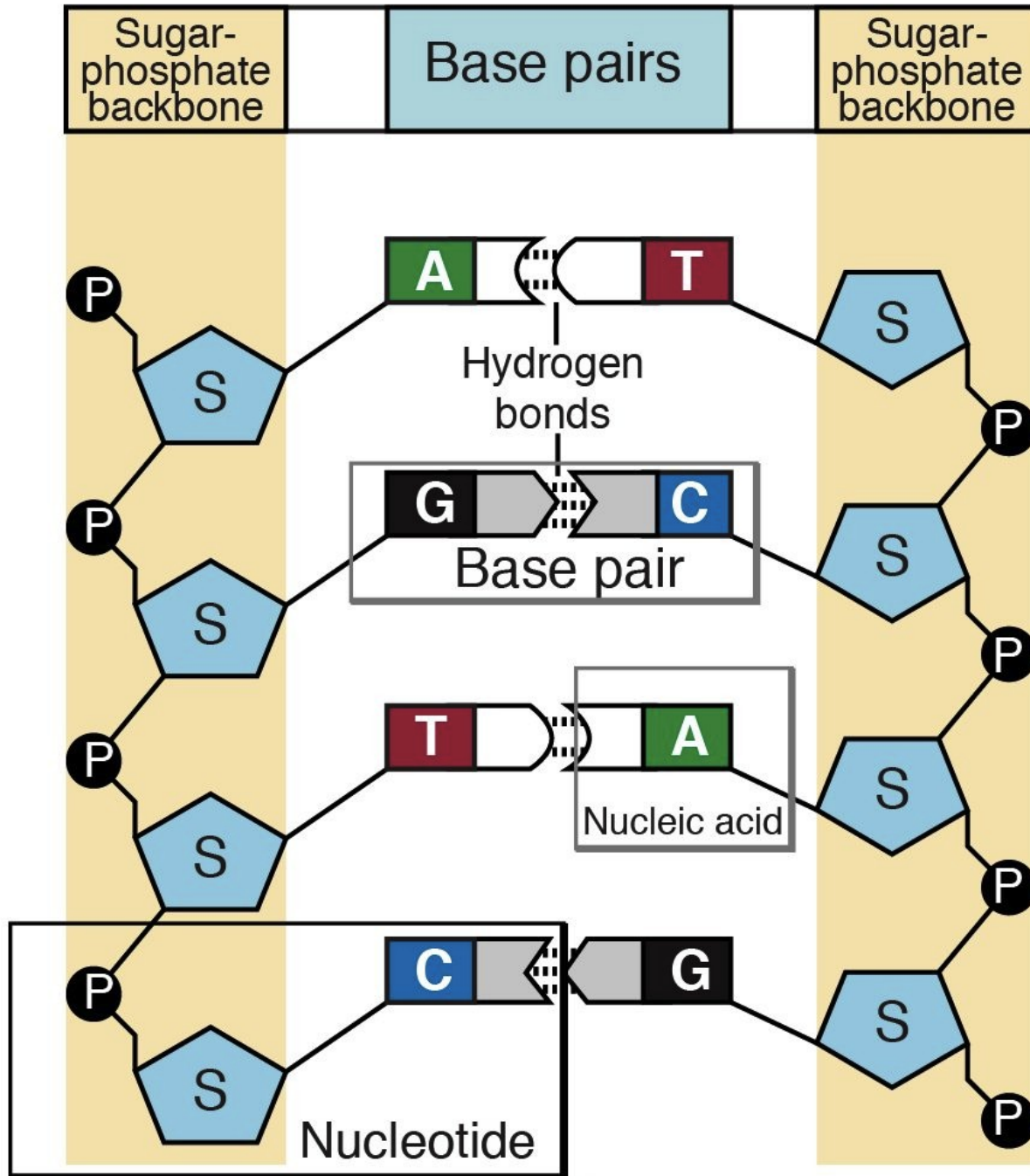
Sugar-phosphate
backbone

Sugar-phosphate
backbone

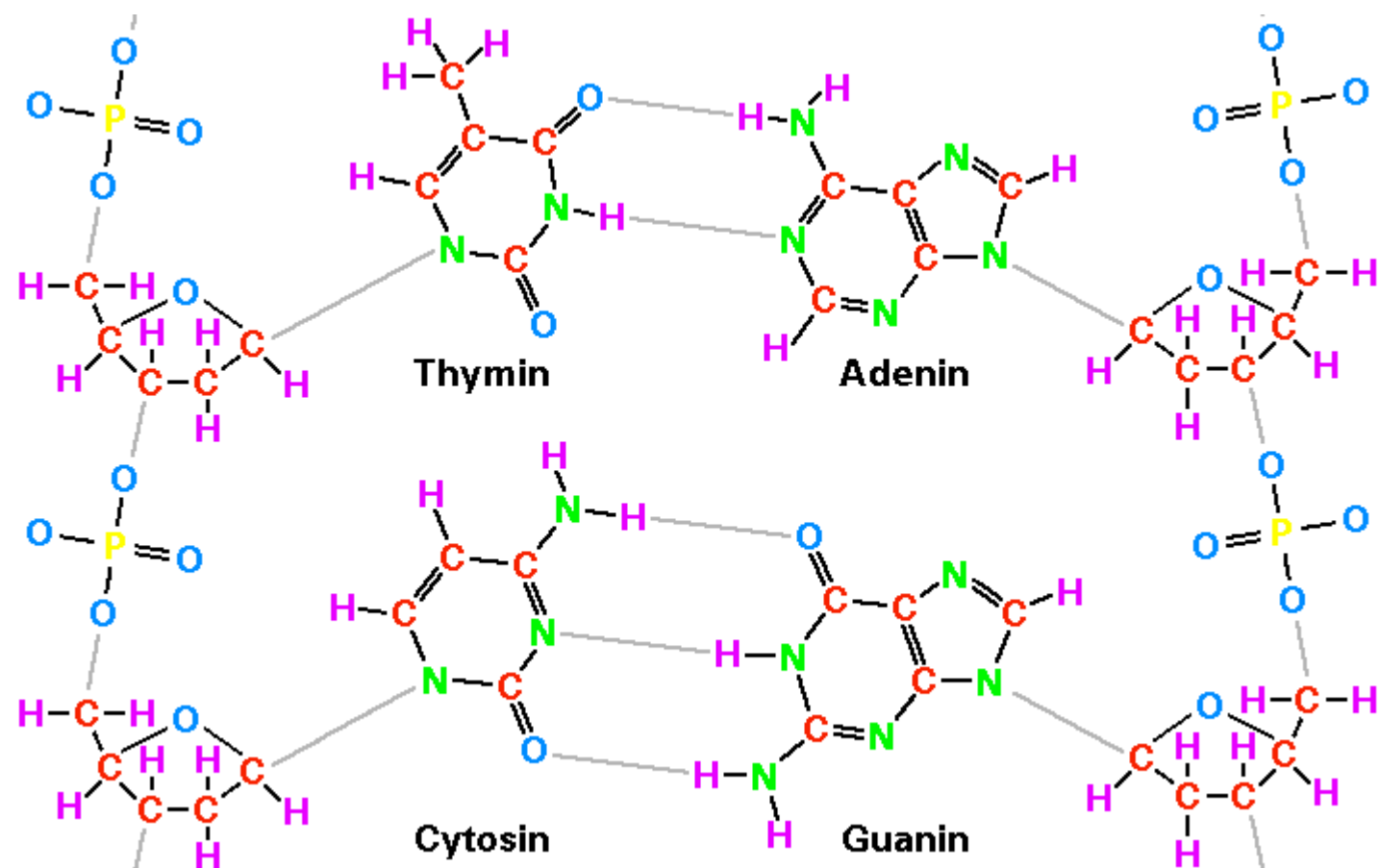
Nucleotide
Unit



Deoxyribonucleic Acid (DNA)

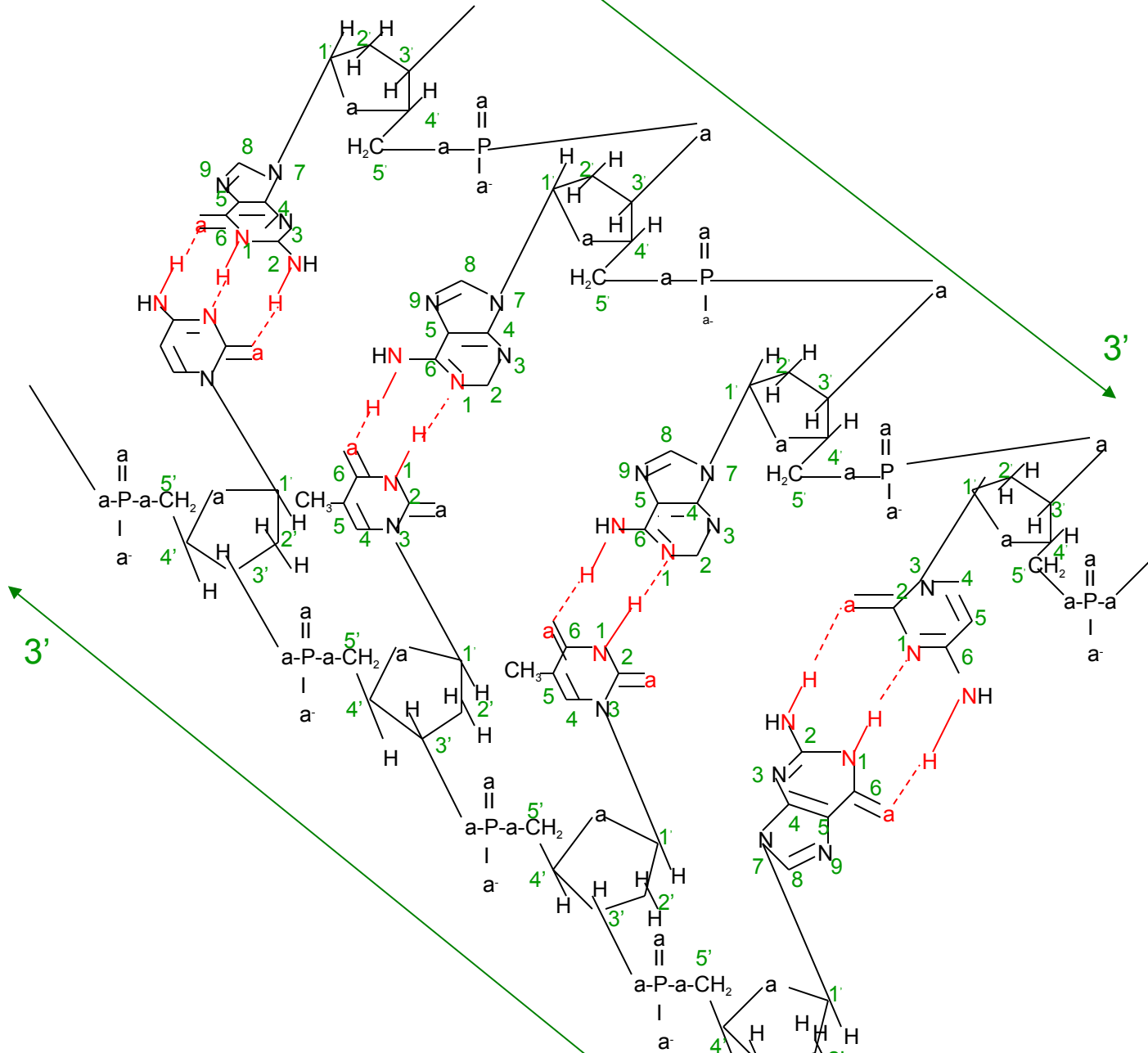


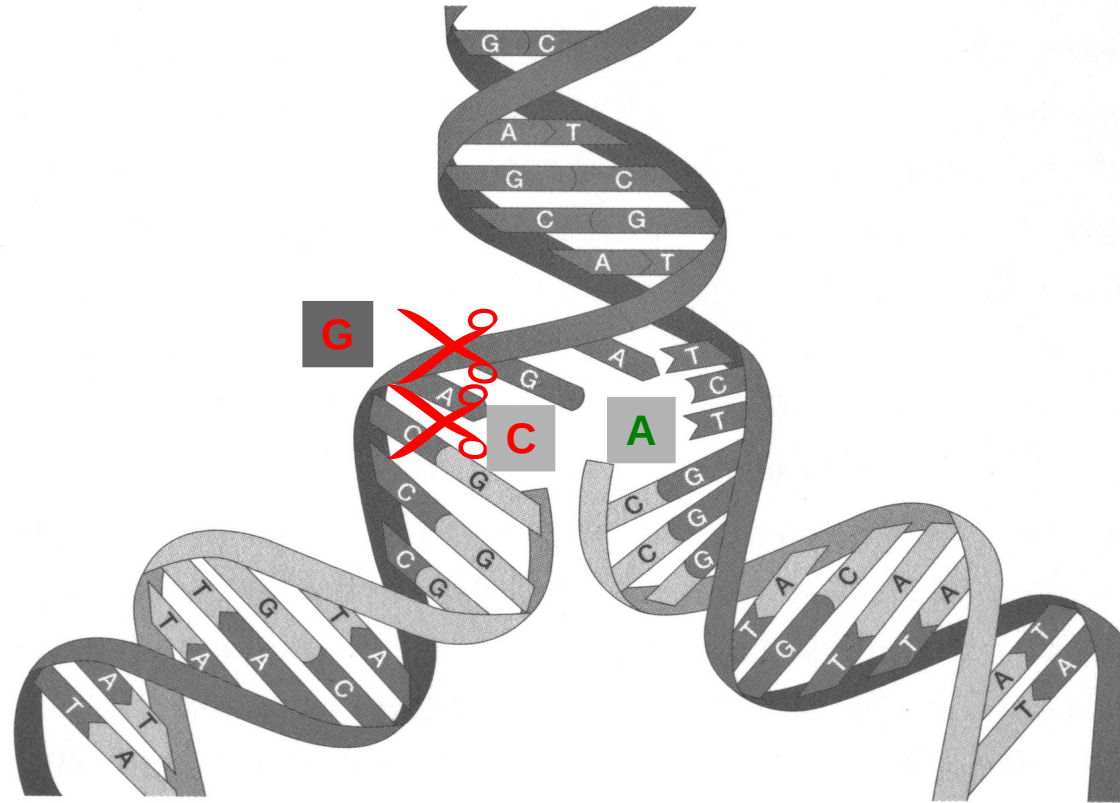
- A** Adenine
- T** Thymine
- C** Cytosine
- G** Guanine



antiparallelismo

Tratto di DNA formato da 4 paia di basi (CTTG nella catena inferiore, GaaC nella catena antiparallela ad essa). Le frecce verdi indicano la direzione dei legami fra i nucleotidi, che è opposta nelle due catene.





L'incrocio industriale o F1 toro da carne X vacca da latte

- **Razza Belga** = razza incrociante più utilizzata in Piemonte, allevata in purezza solo in Belgio



- Vitelli ibridi spuntano prezzi ≤ 400 Euro
[Piemontese puro ~ 500 E, frisione puro (*gaiot*) 50 E]
- Quante bovine da latte posso inseminare con il Belga senza compromettere la **rimonta interna**?



La razza Belga e la miostatina

Phenotype of myostatin null vertebrates.

- a) upper forelimb muscles of wild-type mouse;
- B) upper forelimb of mouse that is myostatin null;
- C) bovine with myostatin mutation;
- D) and E) human with myostatin mutation as a neonate (D) and at 7 mo of age.



La razza Belga e la miostatina



The image is a screenshot of the Google Scholar website. At the top, the browser's address bar shows the URL `https://scholar.google.com`. Below the address bar, there are navigation links: a hamburger menu icon, "Il mio profilo" (My profile), and "La mia biblioteca" (My library). On the right side of the navigation bar, there is a link that says "ACCEDI" (Log in). The main heading in the center is "Google Scholar", with "Google" in its multi-colored font and "Scholar" in grey. Below the heading is a search bar containing the text "Double muscling in cattle due to mutations in the myostatin gene". To the right of the search bar is a blue button with a magnifying glass icon. Below the search bar, there are two radio buttons for language selection: "Qualsiasi lingua" (Any language), which is selected, and "Pagine in Italiano" (Pages in Italian). At the bottom of the page, there is a green text link that says "Sali sulle spalle dei giganti" (Salt on the shoulders of giants).

← → ↻ `https://scholar.google.com` 133% ☆

☰ Il mio profilo ★ La mia biblioteca ACCEDI

Google Scholar

Double muscling in cattle due to mutations in the myostatin gene 🔍

☒ Qualsiasi lingua ☐ Pagine in Italiano

[Sali sulle spalle dei giganti](#)



RESEARCH ARTICLE | GENETICS | 



Double muscling in cattle due to mutations in the myostatin gene

Alexandra C. McPherron and Se-Jin Lee [Authors Info & Affiliations](#)

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Q GenBank



the GenBank dat... 1

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Genetics



Double muscling in cattle due to mutations in the myostatin gene

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La razza Belga e la miostatina

Data deposition: The sequences reported in this paper have been deposited in the **GenBank** database [baboon (accession no. AF019619), bovine (accession no. AF019620), chicken (accession no. AF019621), ovine (accession no. AF019622), porcine (accession no. AF019623), rat (accession no. AF019624), turkey (accession no. AF019625), zebrafish (accession no. AF019626), and human (accession no. AF019627)].

A commentary on this article begins on page 12249.

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La razza Belga e la miostatina

lanciamo il terminale per usare R

ctrl+alt+t

\$ R

> install.packages("ape")

> library(ape)

> x = paste("AF0196", 19:27, sep = "")

ctrl+MAIUSC+v per incollare nel terminale o in R

```
> myostatin.numbers = c('NM_001001525.3', 'XM_010862903.1', 'XM_005907427.1', 'GQ184147.1', 'AF019620.1',
'XM_061432883.1', 'BC134563.1', 'AY160688.1', 'XM_027555492.1', 'XM_019969486.1', 'AF019761.1',
'NM_001290967.1', 'XM_055574715.1', 'KP057613.1', 'AY629304.1', 'XM_055397770.1', 'XM_061135030.1',
'XM_043894881.1', 'XM_043488279.1', 'AY629307.1', 'XM_020900528.1', 'XM_052635569.1', 'EF629535.1',
'NM_001009428.3', 'XM_040227852.1', 'JN662463.1', 'AM992883.1', 'AY629305.1', 'AF019622.1', 'KM371731.1',
'JX840482.1', 'JN639006.1', 'HM462261.1', 'HM462259.1', 'NM_001285737.1', 'FM207636.1', 'AY629309.1',
'AY629306.1', 'AY436347.1', 'AM992884.1', 'GU377303.1', 'XM_007104442.2', 'XM_022568276.1', 'XM_061203308.1',
'XM_057551429.1', 'XM_036859085.1', 'AY629308.1', 'XM_029225345.1', 'XM_027114903.1', 'XM_024733638.1',
'XM_004323907.2', 'XM_030847723.2', 'XM_060151388.1', 'XM_060015612.1', 'XM_060106598.1', 'XM_004276934.2',
'XM_032637329.1', 'XM_059051852.1', 'XM_059928344.1', 'XM_057745280.1', 'XM_047770464.1', 'HM241657.1',
'AF019623.1', 'JN630464.1', 'EF612791.1', 'NM_214435.2', 'XM_006209983.3', 'XM_006189223.3', 'XM_028125258.1',
'XM_027590490.1', 'XM_025887402.1', 'XM_004426733.2', 'XM_010960036.2', 'XM_021703117.1', 'XM_004392403.2',
'XM_046659551.1', 'XM_046650639.1', 'XM_014837576.2', 'XM_032417983.1', 'XM_010991955.2', 'XM_058548651.1',
'XM_008516515.1', 'XM_036075506.1', 'NM_001081817.1', 'XM_006726961.2', 'XM_019626478.1', 'XM_054562700.1',
'XM_036253439.1', 'XM_019740713.1', 'MZ169554.1', 'XM_035007537.1', 'XM_026490450.4', 'XM_045868351.2',
'XM_045806119.1', 'XM_008687262.2', 'XM_029932956.1', 'XM_016215509.1', 'XM_036881965.2', 'XM_047086026.1',
'XM_045480514.1')
```

La razza Belga e la miostatina

```
> x = paste("AF0196", 19:27, sep = "")  
> x = x[-c(1:8)]  
> x  
[1] "AF019627"  
> x = x[9]  
> x  
[1] "AF019627"  
>
```

perché l'output di questi due comandi è il medesimo?

La razza Belga e la miostatina

```
> y = c(x, myostatin.numbers)
> myostatin.seq = read.GenBank(x, as.character = TRUE)
> install.packages("seqinr")
> library(seqinr)
```

nel terminale

```
$ mkdir DNA_SEQUENCES
$ ls
$ pwd
```

in R

alt gr+ì → ~

```
> setwd("~/DNA_SEQUENCES")
> getwd()
> write.fasta(sequences = myostatin.seq, names = names(myostatin.seq), nbchar
=80, file.out = "myostatin.fas")
```

nel terminale

```
$ cd DNA_SEQUENCES
$ gedit myostatin.fas
```

La razza Belga e la miostatina

dopo
>AF019620

Aggiungere

>bovine_belgian
atgcaaaaactgcaaatctctgtttatatatttacctatttatgctgattgttgctgggccagtggatctgaatgagaacag
cgagcagaaggaaaatgtggaaaaagaggggctgtgtaatgcatgtttgtggagggaaaacactacatcctcaagactag
aagccataaaaaatccaaatcctcagtaaacttcgcctggaaacagctcctaacatcagcaaagatgctatcagacaactt
ttgcccagggtcctccactcctggaactgattgatcagttcgatgtccagagagatgccagcagtgacggctccttgga
agacgatgactaccacgccaggacggaaacgggtcattaccatgcccacggagtctgatcttctaacgcaagtggaaggaa
aacccaaagtgttgcttctttaaatcttagctctaagatacaatacaataaactagtaaaggcccaactgtggatataatctg
aggcctgtcaagactcctgcgacagtgtttgtgcaaatcctgagactcatcaaaccatgaaagacgggtacaaggtatac
tggaatccgatctctgaaacttgacatgaaccaggcactggatatttggcagagcattgatgtgaagacagtgttgcaga
actggctcaaacaacctgaatccaacttaggcattgaaatcaaagctttagatgagaatggccatgatcttgctgtaacc
ttcccagaaccaggagaagatggactgactccttttttagaagtcaaggtaacagacacacaaaaagatctaggagaga
ttttgggcttgattgtgacagaatctcgatgctgtcgttaccctctaactgtggattttgaagcttttggtgggattgg
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ccatcttgctgcaccaagcaaaccagaggttcagccggccccctgctgtactcctacaaagatgtctccaattaatatgc
tatattttaatggcgaaggacaaataatatcggggaagattccagccatggtagtagatcgctgtgggtgttcatga
>bovine_piedmontese
atgcaaaaactgcaaatctctgtttatatatttacctatttatgctgattgttgctgggccagtggatctgaatgagaacag
cgagcagaaggaaaatgtggaaaaagaggggctgtgtaatgcatgtttgtggagggaaaacactacatcctcaagactag
aagccataaaaaatccaaatcctcagtaaacttcgcctggaaacagctcctaacatcagcaaagatgctatcagacaactt
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agacgatgactaccacgccaggacggaaacgggtcattaccatgcccacggagtctgatcttctaacgcaagtggaaggaa
aacccaaagtgttgcttctttaaatcttagctctaagatacaatacaataaactagtaaaggcccaactgtggatataatctg
aggcctgtcaagactcctgcgacagtgtttgtgcaaatcctgagactcatcaaaccatgaaagacgggtacaaggtatac
tggaatccgatctctgaaacttgacatgaaccaggcactggatatttggcagagcattgatgtgaagacagtgttgcaga
actggctcaaacaacctgaatccaacttaggcattgaaatcaaagctttagatgagaatggccatgatcttgctgtaacc
ttcccagaaccaggagaagatggactgactccttttttagaagtcaaggtaacagacacacaaaaagatctaggagaga
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gatgggattggattattgcacctaaaagatataggccaattactgctctggagaatatgaatttgtatttttgcaaaag
taccctcatacccatcttgctgcaccaagcaaaccagaggttcagccggccccctgctgtactcctacaaagatgtctcc
aattaatatgctatatattttaatggcgaaggacaaataatatcggggaagattccagccatggtagtagatcgctgtgggt
gttcatga

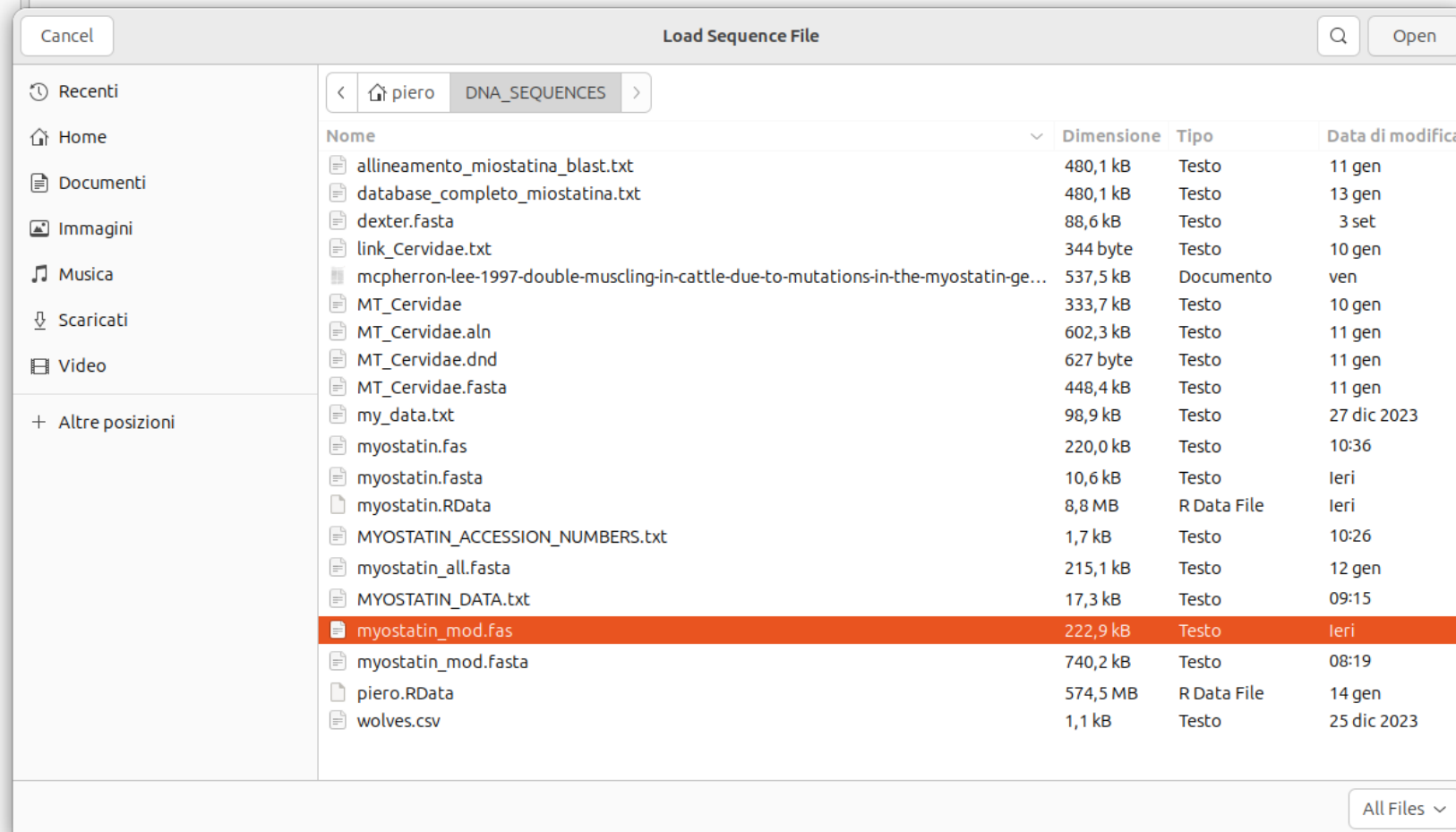
La razza Belga e la miostatina

ctrl+MAIUSC+s

save as myostatin_mod.fas

\$ **clustalx**

Multiple Alignment Mode Font: 10



La razza Belga e la miostatina

https://www.ncbi.nlm.nih.gov/search/all/?term=AF019620

An official website of the United States government [Here's how you know](#) ▾

NIH National Library of Medicine
National Center for Biotechnology Information

Search NCBI

AF019620 × **Search**

Results found in 5 databases

NUCLEOTIDE SEQUENCE

Bos taurus myostatin (MSTN) mRNA, complete cds

Bos taurus

1,128 bp mRNA sequence

AF019620.1

 [FASTA](#) [Gene](#)

[BLAST](#) [Download](#)

FASTA ▾

Send to: ▾

Change region shown

Customize view

Bos taurus myostatin (MSTN) mRNA, complete cds

GenBank: AF019620.1

[GenBank](#) [Graphics](#)

```
>AF019620.1 Bos taurus myostatin (MSTN) mRNA, complete cds
ATGCAAAACTGCAAATCTCTGTTTATATTTACCTATTTATGCTGATTGTTGCTGGCCCAGTGGATCTGA
ATGAGAACAGCGAGCAGAAGGAAAATGTGGAAAAAGAGGGGCTGTGTAATGCATGTTTGTGGAGGGAAAA
CACTACATCCTCAAGACTAGAAGCCATAAAAATCCAAATCCTCAGTAACTTCGCCTGGAAACAGCTCCT
AACATCAGCAAAGATGCTATCAGACAACTTTTGCCCAAGGCTCCTCCACTCCTGGAAGTATTGATCAGT
TCGATGTCCAGAGAGATGCCAGCAGTGACGGCTCCTTGGAAAGACGATGACTACCACGCCAGGACGGAAAC
GGTCATTACCATGCCCACGGAGTCTGATCTTCTAACGCAAGTGGAAAGGAAAACCCAAATGTTGCTTCTTT
AAATTTAGCTCTAAGATACAATAACAATAAACTAGTAAAGGCCCAACTGTGGATATATCTGAGGCCTGTCA
AGACTCCTGCGACAGTGTTTGTGCAAATCCTGAGACTCATCAAACCCATGAAAGACGGTACAAGGTATAC
TGGAATCCGATCTCTGAAACTTGACATGAACCCAGGCACTGGTATTTGGCAGAGCATTGATGTGAAGACA
GTGTTGCAGAACTGGCTCAAACAACCTGAATCCAACCTTAGGCATTGAAATCAAAGCTTTAGATGAGAATG
GCCATGATCTTGCTGTAACCTTCCCAGAACCAGGAGAAGATGGACTGACTCCTTTTTTAGAAGTCAAGGT
AACAGACACACCAAAAAGATCTAGGAGAGATTTTGGGCTTGATTGTGATGAACACTCCACAGAATCTCGA
TGCTGTCGTTACCCTCTAACTGTGGATTTTGAAGCTTTTGGATGGGATTGGATTATTGCACCTAAAAGAT
ATAAGGCCAATTACTGCTCTGGAGAATGTGAATTTGTATTTTGCAAAAGTATCCTCATACCCATCTTGT
GCACCAAGCAAACCCAGAGGTTTCAGCCGGCCCCTGCTGTACTCCTACAAAGATGTCTCCAATTAATATG
CTATATTTTAATGGCGAAGGACAAATAATATACGGGAAGATTCCAGCCATGGTAGTAGATCGCTGTGGGT
GTTTCATGA
```

Analyze this sequence

Run BLAST

Pick Primers

Articles about the MSTN gene

Combined Transcriptome and Metabolome Analysis [Int J Mol Sci. 2

Myostatin site-directed mutation and simultaneous PPAF [J Cell Physiol. 2

Association between polymorphisms Myf5, MSTN and [Anim Biotechnol. 2

See

Reference sequence information

RefSeq mRNA

See reference mRNA sequence for t MSTN gene (NM_001001525.3)

Miostatina: allineamento multi-sequenza

```
> x = paste("AF0196", 19:27, sep = "")  
> setwd('~ /DNA_SEQUENCES/')  
> load('piero.RData')  
> library(ape)  
> library(seqinr)  
> myostatin_x.seq = read.GenBank(x, as.character = TRUE)
```

usando la funzione `<read.GenBank>` del pacchetto APE (Analysis of Phylogenetics and Evolution), viene creata una lista delle sequenze nucleotidiche corrispondenti ai numeri di accesso a **GenBank** contenuti nel vettore **x**

GenBank Overview

What is GenBank?

GenBank[®] is the NIH genetic sequence database, an annotated collection of all publicly available DNA sequences ([Nucleic Acids Research, 2013 Jan;41\(D1\):D36-42](#)). GenBank is part of the [International Nucleotide Sequence Database Collaboration](#), which comprises the DNA DataBank of Japan (DDBJ), the European Nucleotide Archive (ENA), and GenBank at NCBI. These three organizations exchange data on a daily basis.

A GenBank release occurs every two months and is available from the [ftp site](#). The [release notes](#) for the current version of GenBank provide detailed information about the release and notifications of upcoming changes to GenBank. Release notes for [previous GenBank releases](#) are also available. GenBank growth [statistics](#) for both the traditional GenBank divisions and the WGS division are available from each release.

An [annotated sample GenBank record](#) for a *Saccharomyces cerevisiae* gene demonstrates many of the features of the GenBank flat file format.

Access to GenBank

There are several ways to search and retrieve data from GenBank.

- Search GenBank for sequence identifiers and annotations with [Entrez Nucleotide](#).

GenBank Resourc

[GenBank Home](#)

[Submission Types](#)

[Submission Tools](#)

[Search GenBank](#)

[Update GenBank Rec](#)

Home > Packages > ape

ape package^{5.8-1}

Analyses of Phylogenetics and Evolution

```
install.packages("ape")
```

Copy

Functions Readme Datasets Dependencies Imports Versions

subtreeplot

Zoom on a Portion of a
Phylogeny by Successive
Clicks

subtrees

All subtrees of a
Phylogenetic Tree

summary.phylo

Print Summary of a
Phylogeny

📄 Download source package

📖 Read PDF manual

Functions for reading, writing, plotting, and
manipulating phylogenetic trees, analyses of
comparative data in a phylogenetic
framework, ancestral character analyses

where

which.edge

write.dna

Miostatina: allineamento multi-sequenza

```
> str(myostatin_x.seq)
List of 9
 $ AF019619: chr [1:1128] "a" "t" "g" "c" ...
 $ AF019620: chr [1:1128] "a" "t" "g" "c" ...
 $ AF019621: chr [1:1128] "a" "t" "g" "c" ...
 $ AF019622: chr [1:1128] "a" "t" "g" "c" ...
 $ AF019623: chr [1:1128] "a" "t" "g" "c" ...
 $ AF019624: chr [1:1131] "a" "t" "g" "a" ...
 $ AF019625: chr [1:1128] "a" "t" "g" "c" ...
 $ AF019626: chr [1:1125] "a" "t" "g" "c" ...
 $ AF019627: chr [1:1128] "a" "t" "g" "c" ...
- attr(*, "species")= chr [1:9] "Papio_hamadryas" "Bos_taurus"
"Gallus_gallus" "Ovis_aries" ...
> length(myostatin_x.seq)
[1] 9
```

Miostatina: allineamento multi-sequenza

```
> myostatin_x.seq[1]
$AF019619
  [1] "a" "t" "g" "c" "a" "a" "a" "a" "a" "a" "c" "t" "g" "c" "a" "a" "c" "t" "c"
 [19] "t" "g" "t" "g" "t" "t" "t" "a" "t" "a" "t" "t" "t" "a" "c" "c" "t" "g"
 [37] "t" "t" "t" "a" "t" "g" "c" "t" "g" "a" "t" "t" "g" "t" "t" "g" "c" "t"
   ...
[1099] "a" "t" "g" "g" "t" "a" "g" "t" "a" "g" "a" "c" "c" "g" "c" "t" "g" "c"
[1117] "g" "g" "g" "t" "g" "c" "t" "c" "a" "t" "g" "a"
```

Miostatina: allineamento multi-sequenza

```
> myostatin_x.seq = read.GenBank(x, as.character = FALSE)
> str(myostatin_x.seq)
List of 9
 $ AF019619: raw [1:1128] 88 18 48 28 ...
 $ AF019620: raw [1:1128] 88 18 48 28 ...
 $ AF019621: raw [1:1128] 88 18 48 28 ...
 $ AF019622: raw [1:1128] 88 18 48 28 ...
 $ AF019623: raw [1:1128] 88 18 48 28 ...
 $ AF019624: raw [1:1131] 88 18 48 88 ...
 $ AF019625: raw [1:1128] 88 18 48 28 ...
 $ AF019626: raw [1:1125] 88 18 48 28 ...
 $ AF019627: raw [1:1128] 88 18 48 28 ...
- attr(*, "class")= chr "DNABin"
- attr(*, "description")= chr [1:9] "AF019619.1 Papio hamadryas myostatin
(MSTN) mRNA, complete cds" "AF019620.1 Bos taurus myostatin (MSTN) mRNA,
complete cds" "AF019621.1 Gallus gallus myostatin (MSTN) mRNA, complete cds"
"AF019622.1 Ovis aries myostatin (MSTN) mRNA, complete cds" ...
- attr(*, "species")= chr [1:9] "Papio_hamadryas" "Bos_taurus"
"Gallus_gallus" "Ovis_aries" ...
>
```

Miostatina: allineamento multi-sequenza

```
> attributes(myostatin_x.seq)
$names
[1] "AF019619" "AF019620" "AF019621" "AF019622" "AF019623" "AF019624" "AF019625"
[8] "AF019626" "AF019627"

$class
[1] "DNABin"

$description
[1] "AF019619.1 Papio hamadryas myostatin (MSTN) mRNA, complete cds"
[2] "AF019620.1 Bos taurus myostatin (MSTN) mRNA, complete cds"
[3] "AF019621.1 Gallus gallus myostatin (MSTN) mRNA, complete cds"
[4] "AF019622.1 Ovis aries myostatin (MSTN) mRNA, complete cds"
[5] "AF019623.1 Sus scrofa myostatin (MSTN) mRNA, complete cds"
[6] "AF019624.1 Rattus norvegicus myostatin (MSTN) mRNA, complete cds"
[7] "AF019625.1 Meleagris gallopavo myostatin (MSTN) mRNA, complete cds"
[8] "AF019626.1 Danio rerio myostatin (MSTN) mRNA, complete cds"
[9] "AF019627.1 Homo sapiens myostatin (MSTN) mRNA, complete cds"

$species
[1] "Papio_hamadryas"      "Bos_taurus"           "Gallus_gallus"
[4] "Ovis_aries"           "Sus_scrofa"           "Rattus_norvegicus"
[7] "Meleagris_gallopavo"  "Danio_rerio"          "Homo_sapiens"

>
```

Miostatina: allineamento multi-sequenza

```
> myostatin_x.seq = read.GenBank(x, as.character = TRUE)
> attributes(myostatin_x.seq)
$names
[1] "AF019619" "AF019620" "AF019621" "AF019622" "AF019623" "AF019624"
"AF019625"
[8] "AF019626" "AF019627"

$species
[1] "Papio_hamadryas"      "Bos_taurus"           "Gallus_gallus"
[4] "Ovis_aries"           "Sus_scrofa"           "Rattus_norvegicus"
[7] "Meleagris_gallopavo"  "Danio_rerio"          "Homo_sapiens"

>
```

Miostatina: allineamento multi-sequenza

```
> library(seqinr)
> write.fasta(sequences = myostatin_x.seq, names = names(myostatin_x.seq),
nbchar = 80, file.out = "myostatin_x.fas")
>
```

uso la funzione `<write.fasta>` del pacchetto SEQINR (Biological Sequences Retrieval and Analysis) per creare il file di testo in formato «fasta» `<myostatin_x.fas>`, contenente le sequenze elencate nella list `<myostatin_x.seq>`

Home > Packages > seqinr

seqinr package 42-36

Biological Sequences Retrieval and Analysis

```
install.packages("seqinr")
```

Copy

Functions Datasets Dependencies Imports Versions

a

Converts amino-acid
three-letter code into the
one-letter one

aaa

Converts amino-acid one-
letter code into the three-
letter one

AAstat

To Get Some Protein
Statistics

⬇ Download source package

📖 Read PDF manual

Exploratory data analysis and data
visualization for biological sequence (DNA
and protein) data. Seqinr includes utilities
for sequence data management under the

acnucopen

al2bp

alllistranks

Miostatina: allineamento multi-sequenza

nel terminale

```
$ locate myostatin_x.fas
```

```
/home/piero/myostatin_x.fas
```

```
/home/piero/DNA_SEQUENCES/myostatin_x.fas
```

questo è il «percorso» (PATH) del file <myostatin_x.fas>

```
$ pwd
```

```
/home/piero
```

sono già nella home, quindi non è necessario specificare l'intero PATH

```
$ gedit DNA_SEQUENCES/myostatin_x.fas
```

oppure

```
piero@piero-XPS-9320:~$ cd DNA_SEQUENCES/
```

```
piero@piero-XPS-9320:~/DNA_SEQUENCES$ pwd
```

```
/home/piero/DNA_SEQUENCES
```

```
piero@piero-XPS-9320:~/DNA_SEQUENCES$ gedit myostatin_x.fas
```

```
1 >AF019619
2 atgcaaaactgcaactctgtgtttatatttacctgtttatgctgattgttgctggccagtggaatcctaaatgagaacag
3 tgagcaaaaagaaaatgtggaaaaagaggggctgtgtaatgcatgtacttgagacaaaactaaatcttcaagaatag
4 aagccattaaaaatacaaatcctcagtaaaacttcgtctggaaacagctcctaatacagcaaaagatgctataagacaactt
5 ttaccctaaagctcctccactcgggaactgattgatcagatgatgttcagagggatgacagcagcgatggctctttgga
6 agatgacgattatcacgctacaacggaacaatcattaccatgcctacagagcttgatttttaaatgcaagtggatggaa
7 aaccctaaatgttgcttctttaaatgtgctctaaaatacaatacaataaagtggtaaaggcccaactatggatatattg
8 agaccgctcgagactcctacaacagtggttgtgcaaatcctgagactcatcaaacctatgaaagacggtacaaggtatac
9 tggaaatccgatctctgaaacttgacatgaaccaggcactgggtatttggcagagcattgatgtgaagacagtggtgcaaa
10 attggctcaaacaacctgaatccaacttaggcattgaaataaaagccttagatgagaatggctcatgatcttgctgtaacc
11 ttcccaggaccaggagaagatgggctgaatcccttttagagggtcaaggtaacagacacacacaaaagatccagaaggga
12 ttttggcttgactgtgatgagcactcaacagaatcgcgatgctgtgctttaccctctaactgtggattttgaagctcttg
13 gatgggattggattatcgctcctaaaagatataaggccaattactgctctggagagtggtgaatttgatttttacaaaaa
14 tatctctatactcatctggtacaccaagcaaacccagagggttcagcaggcccttgctgtactcccacaagatgtctcc
15 aattaatatgctatattttaatggcaagaacaataatatatgggaaaattccagccatggtagtagaccgctgcgggt
16 gctcatga
17 >AF019620
18 atgcaaaaactgcaaatctctgtttatatttacctatttatgctgattgttgctggccagtggaatcgaatgagaacag
19 cgagcagaaggaaaatgtggaaaaagaggggctgtgtaatgcatgtttgtggagggaacactacatcctcaagactag
20 aagccataaaaaatccaaatcctcagtaaaacttcgcctggaaacagctcctaatacagcaaaagatgctatcagacaactt
21 ttgcccagggtcctccactcctggaactgattgatcaggttcgatgtccagagagatgccagcagtgacggctccttggga
22 agacgatgactaccacgcccaggacggaacgggtcattaccatgcccacggagcttgatcttctaacgcaagtggaaaggaa
23 aaccctaaatgttgcttctttaaatgtgctctaagatatacaataaaacttagtaaaggcccaactgtggatatatctg
24 aggcctgtcaagactcctgcgacagtggttgtgcaaatcctgagactcatcaaacctatgaaagacggtacaaggtatac
25 tggaaatccgatctctgaaacttgacatgaaccaggcactgggtatttggcagagcattgatgtgaagacagtggtgcaga
26 actggctcaaacacactgaatccaacttaggcattgaaatcaaaagccttagatgagaatggccatgatcttgctgtaacc
27 ttcccagaaccaggagaagatggactgactcctttttagaagtcaaggtaacagacacacacaaaagatctaggagaga
28 ttttgggcttgattgtgatgaacactccacagaatctcgatgctgtcggttaccctctaactgtggattttgaagcttttg
29 gatgggattggattattgcacctaagaatataaggccaattactgctctggagaatgtgaatttgatttttgcaaaag
30 tatctctataccatcttgtgcaccaagcaaacccagagggttcagccggcccttgctgtactcctacaagatgtctcc
31 aattaatatgctatattttaatggcgaaggacaaataataacgggaagattccagccatggtagtagatcgctgtgggt
32 gttcatga
33 >AF019621
34 atgcaaaagctagcagtcctatgtttatatttacctgttcatgcagatcgcggttgatccgggtggctctggatggcagtag
35 tcagcccacagagaacgctgaaaaagacggactgtgcaatgctgtacgtggagacagaatacaaaatcctccagaatag
36 aagccataaaaaatcaaatcctcagcaaaactgcgcctggaaacagcactaacatttagcagggacgttattaagcagctt
37 ttaccctaaagctcctccactgcaggaactgattgatcagatgatgttcagagggagcagatagcgatggctccttggga
38 agacgatgactatcatgccacaaccgagacgattatcacaatgcctacggagcttgattttctgtacaaaatggagggaa
39 aaccctaaatgttgcttctttaagtttagctctaaaatacaataaacaagtagtaaaggcacaaattatggatatacttg
40 aggcaggtccaaaacactacaacgggtgttgtgcagatcctgagactcattaagcccatgaaagacggtacaagatatac
41 tggaaatcgatctttgaaacttgacatgaaccaggcactgggtatctggcagagtattgatgtgaagacagtgctgcaaa
42 attggctcaaacagcctgaatccaatttaggcacgaaataaaagcctttgatgagactggacgagatcttgctgtcaca
43 ttcccaggaccgggtgaagatggattgaacccatttttagagggtcagagttacagacacaccgaaacgggtcccgcagaga
44 ttttggccttgactgtgatgagcactcaacggaatcccgatgttgtcgtacccgctgacagtggtttcgaagcttttg
45 gatgggactggattatagcacctaagaatataaaagccaattactgctccggagaatgcgaatttggtttctacagaaa
46 taccgcacactcacctggtacaccaagcaaatcccagagggttcagcaggcccttgctgcacaccaccaagatgtcccc
47 tataaacatgctgtatttcaatggaaaagaacaataatatatggaaaagataccagccatggttgtagatcggttgcggtt
48 gctcatga
49 >AF019622
```

Sostituire ogni numero di accesso con il
nome scientifico della specie
(ATTENZIONE:
NON DEVONO ESSERCI SPAZI!)

La razza Belga e la miostatina

dopo
>Bos_taurus

Aggiungere

```
>bovine_belgian
atgcaaaaactgcaaatctctgtttatatatttacctatttatgctgattgttgctgggccagtggatctgaatgagaacag
cgagcagaaggaaaatgtggaaaaagaggggctgtgtaatgcatgtttgtggagggaaaacactacatcctcaagactag
aagccataaaaaatccaaatcctcagtaaacttcgcctggaaacagctcctaacatcagcaaagatgctatcagacaactt
ttgcccagggtcctccactcctggaactgattgatcagttcgatgtccagagagatgccagcagtgacggctccttgga
agacgatgactaccacgccaggacggaaacgggtcattaccatgcccacggagtctgatcttctaacgcaagtggaaggaa
aacccaaagtgttgcttcttttaaatttagctctaagatacaatacaataaaactagtaaaggcccaactgtggatataatctg
aggcctgtcaagactcctgcgacagtgtttgtgcaaatcctgagactcatcaaaccatgaaagacgggtacaaggtatac
tggaatccgatctctgaaacttgacatgaaccaggcactggatatttggcagagcattgatgtgaagacagtgttgcaga
actggctcaaacaacctgaatccaacttaggcattgaaatcaaagctttagatgagaatggccatgatcttgctgtaacc
ttcccagaaccaggagaagatggactgactccttttttagaagtcaaggtaacagacacaccaaaaagatctaggagaga
ttttgggcttgattgtgacagaatctcgatgctgtcgttaccctctaactgtggattttgaagcttttggtgggattgg
attattgcacctaaaagatataggccaattactgctctggagaatgtgaatttgtattttgcaaaagtatcctcatac
ccatcttgctgcaccaagcaaaccagaggttcagccggccccctgctgtactcctacaaagatgtctccaattaatatgc
tatattttaatggcgaaggacaaataatatacgggaagattccagccatggtagtagatcgctgtgggtgttcatga
>bovine_piedmontese
atgcaaaaactgcaaatctctgtttatatatttacctatttatgctgattgttgctgggccagtggatctgaatgagaacag
cgagcagaaggaaaatgtggaaaaagaggggctgtgtaatgcatgtttgtggagggaaaacactacatcctcaagactag
aagccataaaaaatccaaatcctcagtaaacttcgcctggaaacagctcctaacatcagcaaagatgctatcagacaactt
ttgcccagggtcctccactcctggaactgattgatcagttcgatgtccagagagatgccagcagtgacggctccttgga
agacgatgactaccacgccaggacggaaacgggtcattaccatgcccacggagtctgatcttctaacgcaagtggaaggaa
aacccaaagtgttgcttcttttaaatttagctctaagatacaatacaataaaactagtaaaggcccaactgtggatataatctg
aggcctgtcaagactcctgcgacagtgtttgtgcaaatcctgagactcatcaaaccatgaaagacgggtacaaggtatac
tggaatccgatctctgaaacttgacatgaaccaggcactggatatttggcagagcattgatgtgaagacagtgttgcaga
actggctcaaacaacctgaatccaacttaggcattgaaatcaaagctttagatgagaatggccatgatcttgctgtaacc
ttcccagaaccaggagaagatggactgactccttttttagaagtcaaggtaacagacacaccaaaaagatctaggagaga
ttttgggcttgattgtgatgaacactccacagaatctcgatgctgtcgttaccctctaactgtggattttgaagcttttg
gatgggattggattatttgcacctaaaagatataggccaattactgctctggagaatatgaatttgtattttgcaaaag
taccctcatacccatcttgctgcaccaagcaaaccagaggttcagccggccccctgctgtactcctacaaagatgtctcc
aattaatatgctatatattttaatggcgaaggacaaataatatacgggaagattccagccatggtagtagatcgctgtgggt
gttcatga
```

La razza Belga e la miostatina

ctrl+MAIUSC+s

save as `myostatin_x_mod.fas`

Editor di testo

myostatin_x.fas
~/DNA_SEQUENCES

Salva

1 >Papio_hamadryas
2 atgcaaaaactgcaactctgtgtt
3 tgagcaaaaagaaaatgtggaaaa
4 aagccattaaaatacaaatcctcag
5 ttacccaaagctcctccactccggg
6 agatgacgattatcacgctacaac
7 aaccctaatgttgcttctttaaat
8 agaccgctcgagactcctacaacag
9 tggaatccgatctctgaaacttgac
10 attggctcaaacaacctgaatccaa
11 ttcccaggaccaggagaagatggg
12 ttttggcttgactgtgatgagcac
13 gatgggattggattatcgctccta
14 tatctcatactcatctggtacac
15 aattaatatgctatattttaatgg
16 gttcatga
17 >Bos_taurus
18 atgcaaaaactgcaaactctgttt
19 cgagcagaaggaaaatgtggaaaa
20 aagccataaaaatccaaatcctcag
21 ttgccaaggctcctccactcctg
22 agacgatgactaccacgccaggac
23 aaccctaatgttgcttctttaaat
24 aggcctgtcaagactcctgcgacag
25 tggaatccgatctctgaaacttgac
26 actggctcaaacaacctgaatccaa
27 ttcccagaaccaggagaagatggg
28 ttttgggcttgattgtgatgaacac
29 gatgggattggattattgcaccta
30 tatctcataccatcttgtgcacac
31 aattaatatgctatattttaatgg
32 gttcatga
33 >bovine_belgian
34 atgcaaaaactgcaaactctgttt
35 cgagcagaaggaaaatgtggaaaa
36 aagccataaaaatccaaatcctcag
37 ttgccaaggctcctccactcctg
38 agacgatgactaccacgccaggac
39 aaccctaatgttgcttctttaaat
40 aggcctgtcaagactcctgcgacag
41 tggaatccgatctctgaaacttgac
42 actggctcaaacaacctgaatccaa
43 ttcccagaaccaggagaagatggg
44 ttttgggcttgattgtgacagaat
45 attattgcacctaagaatataagg
46 ccattctgtgcaccaagcaaaccc
47 tatatttaattggcgaaggacaa
48 >bovine_piedmontese
49 atgcaaaaactgcaaactctgtttatatttacctatttatgctgattgttgctggccagtggaatcgaatgagaacag

Apri

Nome myostatin_x_mod.fas

Salva

Home

Documenti

Immagini

Musica

Scaricati

Video

DNA_SEQUENCES

+ Altre posizioni

< piero DNA_SEQUENCES >

Nome	Dimensione	Tipo	Data di modifica
Bovidae_myostatin.fasta	8,6 kB	Testo	15 nov 2024
database_completo_miostatina.txt	480,1 kB	Testo	13 gen 2024
dexter.fasta	88,6 kB	Testo	3 set 2024
fcodons.R	210 byte	Testo	11 nov 2024
link_Cervidae.txt	344 byte	Testo	10 gen 2024
MT_Cervidae	333,7 kB	Testo	10 gen 2024
MT_Cervidae.fasta	448,4 kB	Testo	11 gen 2024
my_data.txt	98,9 kB	Testo	27 dic 2023
myostatin.dnd	3,7 kB	Testo	19 nov 2024
myostatin.fas	220,0 kB	Testo	gio
myostatin.fasta	714,6 kB	Testo	19 nov 2024
myostatin.nexus	1,2 MB	Testo	1 ott 2024
MYOSTATIN_ACCESSION_NUMBERS.txt	1,7 kB	Testo	29 set 2024
MYOSTATIN_DATA.txt	17,3 kB	Testo	29 set 2024
myostatin_mod.aln	0 byte	Testo	gio
myostatin_mod.dnd	3,8 kB	Testo	gio
myostatin_mod.fas	222,3 kB	Testo	gio
myostatin_mod.fasta	727,8 kB	Testo	gio
myostatin_species_names.dnd	4,3 kB	Testo	ven
myostatin_species_names.fas	222,9 kB	Testo	ven
myostatin_species_names.fasta	740,2 kB	Testo	ven
myostatin_x.fas	12,7 kB	Testo	15:16
myostatin_x_mod.dnd	359 byte	Testo	09:50
myostatin_x_mod.fasta	13,0 kB	Testo	09:50
Phocoena_sinus_myostatin.fas	3,0 kB	Testo	21 nov 2024
piero.fas	1,2 kB	Testo	6 dic 2024
Rhinolophus_sinicus_myostatin.fas	2,9 kB	Testo	28 nov 2024
Sus_scrofa_myostatin.fas	1,2 kB	Testo	21 nov 2024
sylvia.fas	31,5 kB	Testo	14 nov 2024
wolves.csv	1,1 kB	Testo	25 dic 2023

Codifica caratteri: Locale attuale (UTF-8)

Fine riga: Unix/Linux

Tutti i file di testo

La razza Belga e la miostatina

nel terminale

```
$ clustalx
```

ClustalX 2.1

File Edit Alignment Trees Colors Quality Help

Mode: Multiple Alignment Mode Font: 10

Cancel

Load Sequence File

Q Open

Recenti

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Musica

Scaricati

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DNA_SEQUENCES

+ Altre posizioni

< piero DNA_SEQUENCES >

Nome	Dimensione	Tipo	Data di modifica
Bos_taurus_myostatin.fas	2,9 kB	Testo	15 nov 2024
Bovidae_myostatin.dnd	75 byte	Testo	15 nov 2024
Bovidae_myostatin.fas	8,6 kB	Testo	15 nov 2024
Bovidae_myostatin.fasta	8,6 kB	Testo	15 nov 2024
database_completo_miostatina.txt	480,1 kB	Testo	13 gen 2024
dexter.fasta	88,6 kB	Testo	3 set 2024
fcodons.R	210 byte	Testo	11 nov 2024
link_Cervidae.txt	344 byte	Testo	10 gen 2024
mcpherron-lee-1997-double-muscling-in-cattle-due-to-mu...	537,5 kB	Documento	27 set 2024
MT_Cervidae	333,7 kB	Testo	10 gen 2024
MT_Cervidae.fasta	448,4 kB	Testo	11 gen 2024
my_data.txt	98,9 kB	Testo	27 dic 2023
myostatin.dnd	3,7 kB	Testo	19 nov 2024
myostatin.fas	220,0 kB	Testo	gio
myostatin.fasta	714,6 kB	Testo	19 nov 2024
myostatin.nexus	1,2 MB	Testo	1 ott 2024
myostatin.RData	8,8 MB	R Data File	28 set 2024
MYOSTATIN_ACCESSION_NUMBERS.txt	1,7 kB	Testo	29 set 2024
MYOSTATIN_DATA.txt	17,3 kB	Testo	29 set 2024
myostatin_mod.aln	0 byte	Testo	gio
myostatin_mod.dnd	3,8 kB	Testo	gio
myostatin_mod.fas	222,3 kB	Testo	gio
myostatin_mod.fasta	727,8 kB	Testo	gio
myostatin_species_names.dnd	4,3 kB	Testo	ven
myostatin_species_names.fas	222,9 kB	Testo	ven
myostatin_species_names.fasta	740,2 kB	Testo	ven
myostatin_x.fas	12,7 kB	Testo	15:16
myostatin_x_mod.dnd	359 byte	Testo	09:50
myostatin_x_mod.fas	12,7 kB	Testo	15:22
myostatin_x_mod.fasta	13,0 kB	Testo	09:50

All Files

La razza Belga e la miostatina

[illegible]

	GAOPEN	GAPEXTENSION	DNA TRANSITION WEIGHT
PAIRWISE ALIGNMENT PARAMETERS	10.0	0.2	
MULTIPLE ALIGNMENT PARAMETERS	10.0	0.1	0.5

Mode: Multip

Do Complete Alignment Ctrl+L
Do Guide Tree Only Ctrl+G

Do Alignment from Guide Tree

Realign Selected Sequences

Realign Selected Residue Range

Align Profile 2 to Profile 1

Align Profiles from Guide Trees

Align Sequences to Profile 1

Align Sequences to Profile 1 from Tree

Alignment Parameters

Iteration

Output Format Options

Set All Parameters to default

☐ Reset New Gaps before Alignment

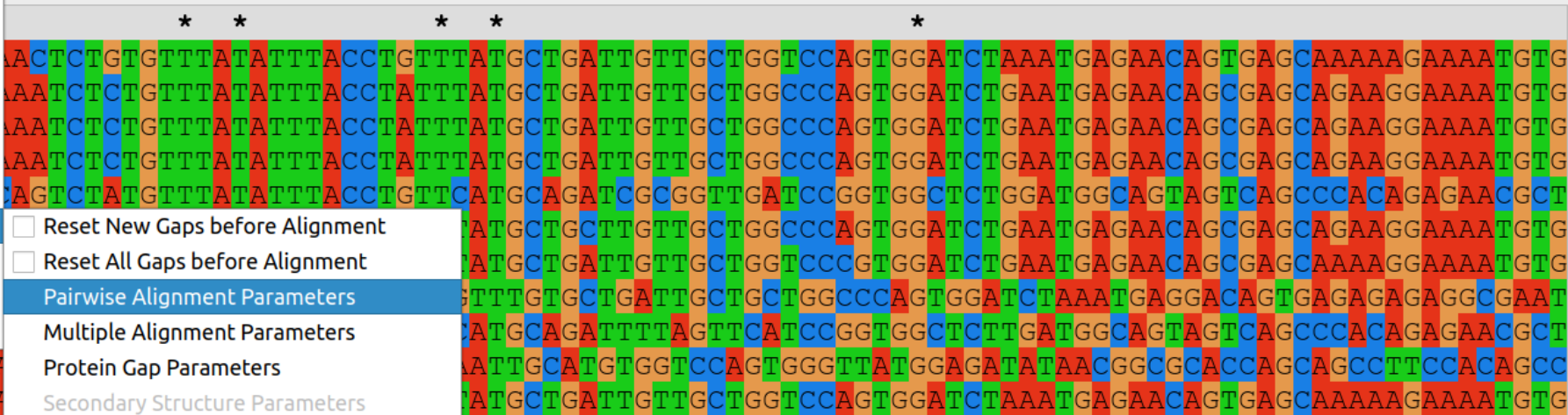
☐ Reset All Gaps before Alignment

Pairwise Alignment Parameters

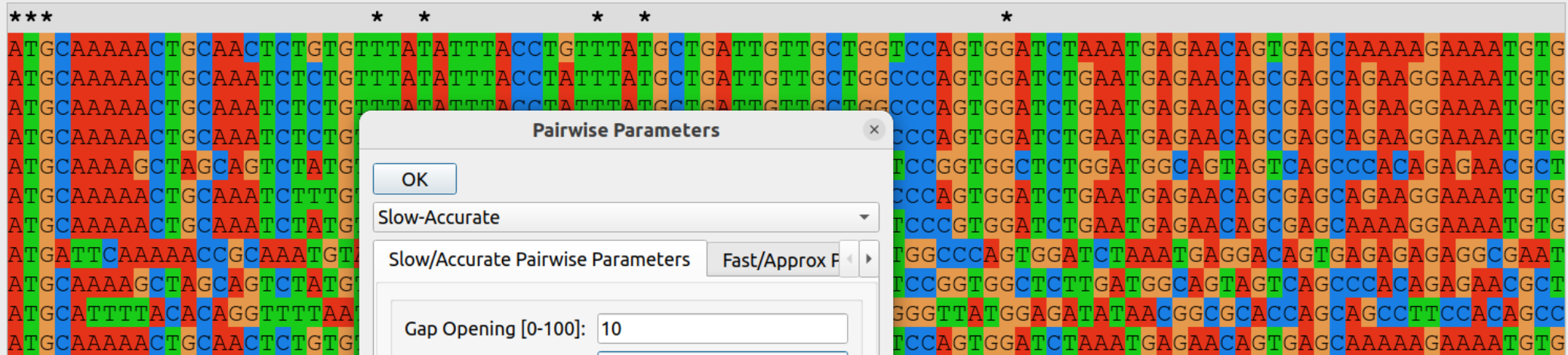
Multiple Alignment Parameters

Protein Gap Parameters

Secondary Structure Parameters



Papio_hamadryas
Bos_taurus
bovine_belgian
bovine_piedmontese
Gallus_gallus
Ovis_aries
Sus_scrofa
Rattus_norvegicus
Meleagris_gallopavo
Danio_rerio
Homo_sapiens



Mode: Multip

Do Complete Alignment Ctrl+L
Do Guide Tree Only Ctrl+G

Do Alignment from Guide Tree

Realign Selected Sequences

Realign Selected Residue Range

Align Profile 2 to Profile 1

Align Profiles from Guide Trees

Align Sequences to Profile 1

Align Sequences to Profile 1 from Tree

Alignment Parameters

Iteration

Output Format Options

Set All Parameters to default

☐ Reset New Gaps before Alignment

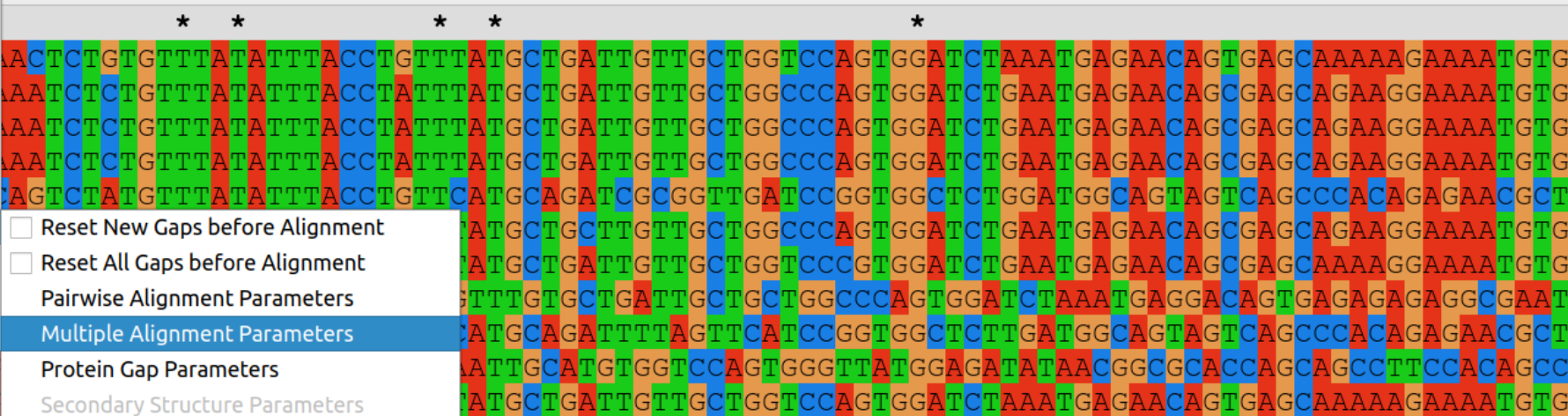
☐ Reset All Gaps before Alignment

Pairwise Alignment Parameters

Multiple Alignment Parameters

Protein Gap Parameters

Secondary Structure Parameters



Mode: Multiple Alignment Mode Font: 24

Papio_hamadryas
Bos_taurus
bovine_belgian
bovine_piedmontese
Gallus_gallus
Ovis_aries
Sus_scrofa
Rattus_norvegicus
Meleagris_gallopavo
Danio_rerio
Homo_sapiens

*** * * * *

ATGCAAAAACCTGCAACTCTGTGTTTATATTTACCTGTTTATGCTGATTGTTGCTGGTCCAGTGGATCTAAATGAGAACAGTGAGCAAAAAGAAAAATGTG
ATGCAAAAACCTGCAAAATCTCTGTTTATATTTACCTATTTATGCTGATTGTTGCTGGCCCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAAATGTG
ATGCAAAAACCTGCAAAATCTCTCTCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAAATGTG
ATGCAAAAACCTGCAAAATCTCTCTCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAAATGTG
ATGCAAAAAGCTAGCAGTCTATCCGGTGGCTCTGGATGGCAGTAGTCAGCCCACAGAGAACGCT
ATGCAAAAACCTGCAAAATCTTTCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAAATGTG
ATGCAAAAACCTGCAAAATCTATCCCGTGGATCTGAATGAGAACAGCGAGCAAAAAGGAAAAATGTG
ATGATTCAAAAACCGCAAAATGCCCCAGTGGATCTAAATGAGGACAGTGAGAGAGAGGCGAAT
ATGCAAAAAGCTAGCAGTCTATCCGGTGGCTCTTGATGGCAGTAGTCAGCCCACAGAGAACGCT
ATGCATTTTACACAGGTTTAAAGGTTATGGAGATATAACGGCGCACCAGCAGCCTTCCACAGCC
ATGCAAAAACCTGCAACTCTGTCCAGTGGATCTAAATGAGAACAGTGAGCAAAAAGAAAAATGTG

Alignment Parameters

OK

Multiple Parameters

Gap Opening [0-100]: 10

Gap Extension [0-100]: 0.1

Delay Divergent Sequences(%): 30

DNA Transition Weight[0-1] 0.5

Use Negative Matrix Off

Protein Weight Matrix

☐ BLOSUM series ☐ PAM series ☐ User defined
☒ Gonnet series ☐ Identity matrix

Load protein matrix:

DNA Weight Matrix

☒ IUB ☐ CLUSTALW(1.6) ☐ User defined

Load DNA matrix:

Ctrl+L

Ctrl+G

Realign Selected Sequences

Realign Selected Residue Range

Align Profile 2 to Profile 1

Align Profiles from Guide Trees

Align Sequences to Profile 1

Align Sequences to Profile 1 from Tree

Alignment Parameters

Iteration

Output Format Options

Set All Parameters to default

Danio rerio

Homo sapiens

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AACTCTGTGTTTATATTTACCTGTTTATGCTGATTGTTGCTGGTCCAGTGGATCTAAATGAGAACAGTGAGCAAAAAGAAAAATGTG
 AATCTCTGTTTATATTTACCTATTTTATGCTGATTGTTGCTGGCCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAAATGTG
 AATCTCTGTTTATATTTACCTATTTTATGCTGATTGTTGCTGGCCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAAATGTG
 AATCTCTGTTTATATTTACCTATTTTATGCTGATTGTTGCTGGCCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAAATGTG
 AGTCTATGTTTATATTTACCTGTTTCATGCAAGATCGCGGTTGATCCGGTGGCTCTGGATGGCAGTAGTTCAGCCCACAGAGAACGCT
 AATCTTTGTTTATATTTACCTATTTTATGCTGCTTGTTGCTGGCCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAAATGTG
 AATCTATGTTTATATTTACCTGTTTATGCTGATTGTTGCTGGTCCCGTGGATCTGAATGAGAACAGCGAGCAAAAGGAAAAATGTG
 GCAAATGTATGTTTATATTTACCTGTTTGTGCTGATTGCTGCTGGCCCAGTGGATCTAAATGAGGACAGTGAGAGAGAGGGCGAAI
 AGTCTATGTTTATATTTACCTGTTTCATGCAAGATTTTAGTTTCATCCGGTGGCTCTTGATGGCAGTAGTTCAGCCCACAGAGAACGCT
 AGGTTTTAATTTCTCTAAGTGTAATTAATTGCATGTGGTCCAGTGGGTTATGGAGATATAACGGCGCACCCAGCAGCCTTCCACAGCC
 AACTCTGTGTTTATATTTACCTGTTTATGCTGATTGTTGCTGGTCCAGTGGATCTAAATGAGAACAGTGAGCAAAAAGAAAAATGTG

Mode: Multiple Alignment Mode Font: 24

Papio_hamadryas
Bos_taurus
bovine_belgian
bovine_piedmontese
Gallus_gallus
Ovis_aries
Sus_scrofa
Rattus_norvegicus
Meleagris_gallopavo
Danio_rerio
Homo_sapiens

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ATGCAAAAACCTGCAACTCTGTGTTTATATTTACCTGTTTATGCTGATTGTTGCTGGTCCAGTGGATCTAAATGAGAACAGTGAGCAAAAAGAAAAATGTG
ATGCAAAAACCTGCAAACTCTCTGTTTATATTTACCTATTTATGCTGATTGTTGCTGGCCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAAATGTG
ATGCAAAAACCTGCAAACTCTCTGTTTATATTTACCTATTTATGCTGATTGTTGCTGGCCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAAATGTG
ATGCAAAAACCTGCAAACTCTCTGTTTATATTTACCTATTTATGCTGATTGTTGCTGGCCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAAATGTG
ATGCAAAAAGCTAGCAGTCTATGTTTCTGGTCCAGTGGCTCTGGATGGCAGTAGTCAGCCCACAGAGAACGCT
CTGGCCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAAATGTG
CTGGTCCCCTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAAATGTG
CTGCTGGCCCAGTGGATCTAAATGAGGACAGTGAGAGAGAGGCGAAT
TTCATCCGGTGGCTCTTGATGGCAGTAGTCAGCCCACAGAGAACGCT
CAGTGGGTTATGGAGATATAACGGCGCACCCAGCAGCCTTCCACAGCC
CTGGTCCAGTGGATCTAAATGAGAACAGTGAGCAAAAAGAAAAATGTG
```

Output Format Options

OK

Output Files

- ☐ CLUSTAL format ☐ NBRF/PIR format
☐ GCG/MSF format ☐ PHYLIP format
☐ GDE format ☐ NEXUS format
☒ FASTA format

GDE output case : Lower ▾
CLUSTALW sequence numbers : Off ▾
Output order : Aligned ▾
Parameters output : Off ▾

Mode: Multip

Ctrl+L

Ctrl+G

Realign Selected Sequences

Align Profile 2 to Profile 1

Align Sequences to Profile 1

Alignment Parameters

Iteration

Output Format Options

Set All Parameters to default

Danio rerio

Homo sapiens

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AACTCTGTGTTTATATTTACCTGTTTATGCTGATTGTTGCTGGTCCAGTGGATCTAAATGAGAACAGTGAGCAAAAAGAAAATGTG
 AAATCTCTGTTTATATTTACCTATTTATGCTGATTGTTGCTGGCCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAATGTG
 AAATCTCTGTTTATATTTACCTATTTATGCTGATTGTTGCTGGCCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAATGTG
 AAATCTCTGTTTATATTTACCTATTTATGCTGATTGTTGCTGGCCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAATGTG
 CAGTCTATGTTTATATTTACCTGTTTCATGCAGATCGCGGTTGATCCGGTGGCTCTGGATGGCAGTAGTCAGCCCACAGAGAACGCT
 AAATCTTTGTTTATATTTACCTATTTATGCTGCTTGTTGCTGGCCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAATGTG
 AAATCTATGTTTATATTTACCTGTTTATGCTGATTGTTGCTGGTCCCGTGGATCTGAATGAGAACAGCGAGCAAAAGGAAAATGTG
 CGCAAATGTATGTTTATATTTACCTGTTTGTGCTGATTGCTGCTGGCCCAGTGGATCTAAATGAGGACAGTGAGAGAGAGGGCGAAI
 CAGTCTATGTTTATATTTACCTGTTTCATGCAGATTTTAGTTTCATCCGGTGGCTCTTGATGGCAGTAGTCAGCCCACAGAGAACGCT
 AGGTTTTTAATTTCTCTAAGTGTATTAATTGCATGTGGTCCAGTGGGTTATGGAGATATAACGGCGCACCAGCAGCCTTCCACAGCC
 AACTCTGTGTTTATATTTACCTGTTTATGCTGATTGTTGCTGGTCCAGTGGATCTAAATGAGAACAGTGAGCAAAAAGAAAATGTG

Mode: Multiple Alignment Mode Font: 24

Papio_hamadryas
Bos_taurus
bovine_belgian
bovine_piedmontese
Gallus_gallus
Ovis_aries
Sus_scrofa
Rattus_norvegicus
Meleagris_gallopavo
Danio_rerio
Homo_sapiens

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ATGCAAAA	ACTGCAAA	CTCTGT	TTTTAT	ATTTAC	CTGTTT	ATGCTG	ATTGTT	GCTGGT	TCCAGT	TGGATC	TAAATG	GAGAAC	CAGTGAG	CAAAA	AGAAAA	TGTG
ATGCAAAA	ACTGCAAA	CTCTGT	TTTTAT	ATTTAC	CTATTT	ATGCTG	ATTGTT	GCTGGC	CCCAGT	TGGATC	TGAATG	GAGAAC	CAGCGAG	CAGA	AGAAAA	TGTG
ATGCAAAA	ACTGCAAA	CTCTGT	TTTTAT	ATTTAC	CTATTT	ATGCTG	ATTGTT	GCTGGC	CCCAGT	TGGATC	TGAATG	GAGAAC	CAGCGAG	CAGA	AGAAAA	TGTG
ATGCAAAA	ACTGCAAA	CTCTGT	TTTTAT	ATTTAC	CTATTT	ATGCTG	ATTGTT	GCTGGC	CCCAGT	TGGATC	TGAATG	GAGAAC	CAGCGAG	CAGA	AGAAAA	TGTG
ATGCAAAA	AGCTAGC	AGTCTA	TGTTT	ATATTT	ACCTGT	TCATGC	AGATCG	CGGTG	TGATCC	GGTGG	CTCTG	GATGG	CAGTAG	TCAGC	CCACAG	AGAACG
ATGCAAAA	ACTGCAAA	CTTTGT	TTTTAT	ATTTAC	CTATTT	ATGCTG	CTTGTG	TGCTGG	CCCAGT	TGGATC	TGAATG	GAGAAC	CAGCGAG	CAGA	AGAAAA	TGTG
ATGCAAAA	ACTGCAAA	CTATGT	TTTTAT	ATTTAC	CTGTTT	ATGCTG	ATTGTT	GCTGGT	TCCCGT	TGGATC	TGAATG	GAGAAC	CAGCGAG	CAAAA	AGAAAA	TGTG
ATGATT	CAAAA	CCGCAAA														
ATGCAAAA	AGCTAGC	AGTCTA														
ATGCAT	TTTTA	CACAGG	TTT													
ATGCAAAA	ACTGCAAA	CTCT														

Complete Alignment

Output Guide Tree:

Browse

Output Alignment Files

Fasta:

Browse

OK

Cancel

Prima

	***	*	*	*	*	*
Papio_hamadryas	ATGCAAAAAC	TGCAACTCT	GTGTTTAT	ATTTACCTG	TTTATGCT	GATTGTTGCTGGTCCAGTGGATCTAAATGAGAACAGTGAGCAAAAAGAAAAATGTG
Bos_taurus	ATGCAAAAAC	TGCAAACTCT	GTGTTTAT	ATTTACCTA	TTTATGCT	GATTGTTGCTGGCCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAAATGTG
bovine_belgian	ATGCAAAAAC	TGCAAACTCT	GTGTTTAT	ATTTACCTA	TTTATGCT	GATTGTTGCTGGCCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAAATGTG
bovine_piedmontese	ATGCAAAAAC	TGCAAACTCT	GTGTTTAT	ATTTACCTA	TTTATGCT	GATTGTTGCTGGCCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAAATGTG
Gallus_gallus	ATGCAAAAAG	CTAGCAGTCT	ATGTTTAT	ATTTACCTG	TTTATGCT	GATTGCTGCGGTTGATCCGGTGGCTCTGGATGGCAGTAGTCAGCCCACAGAGAACGCT
Ovis_aries	ATGCAAAAAC	TGCAAACTCT	TGTTTAT	ATTTACCTA	TTTATGCT	GCTTGTGCTGGCCCAGTGGATCTGAATGAGAACAGCGAGCAGAAGGAAAAATGTG
Sus_scrofa	ATGCAAAAAC	TGCAAACTCT	ATGTTTAT	ATTTACCTG	TTTATGCT	GATTGTTGCTGGTCCCCTGGATCTGAATGAGAACAGCGAGCAAAAGGAAAAATGTG
Rattus_norvegicus	ATGATTCAAAAAC	CGCAAACTGT	ATGTTTAT	ATTTACCTG	TTTGTGCT	GATTGCTGCTGGCCCAGTGGATCTAAATGAGGACAGTGAGAGAGAGGGCGAA
Meleagris_gallopavo	ATGCAAAAAG	CTAGCAGTCT	ATGTTTAT	ATTTACCTG	TTTATGCT	GATTGCTGCTGGCCCAGTGGATCTAAATGAGGACAGTGAGAGAGAGGGCGAA
Danio_rerio	ATGCATTTTAC	ACAGGTTTTAA	TTTCTCTAAG	TGTATTAA	TGTGCATGTGGT	TCCAGTGGGTTATGGAGATATAACGGCGCACCAGCAGCCTTCCACAGCC
Homo_sapiens	ATGCAAAAAC	TGCAACTCT	GTGTTTAT	ATTTACCTG	TTTATGCT	GATTGTTGCTGGTCCAGTGGATCTAAATGAGAACAGTGAGCAAAAAGAAAAATGTG

dopo

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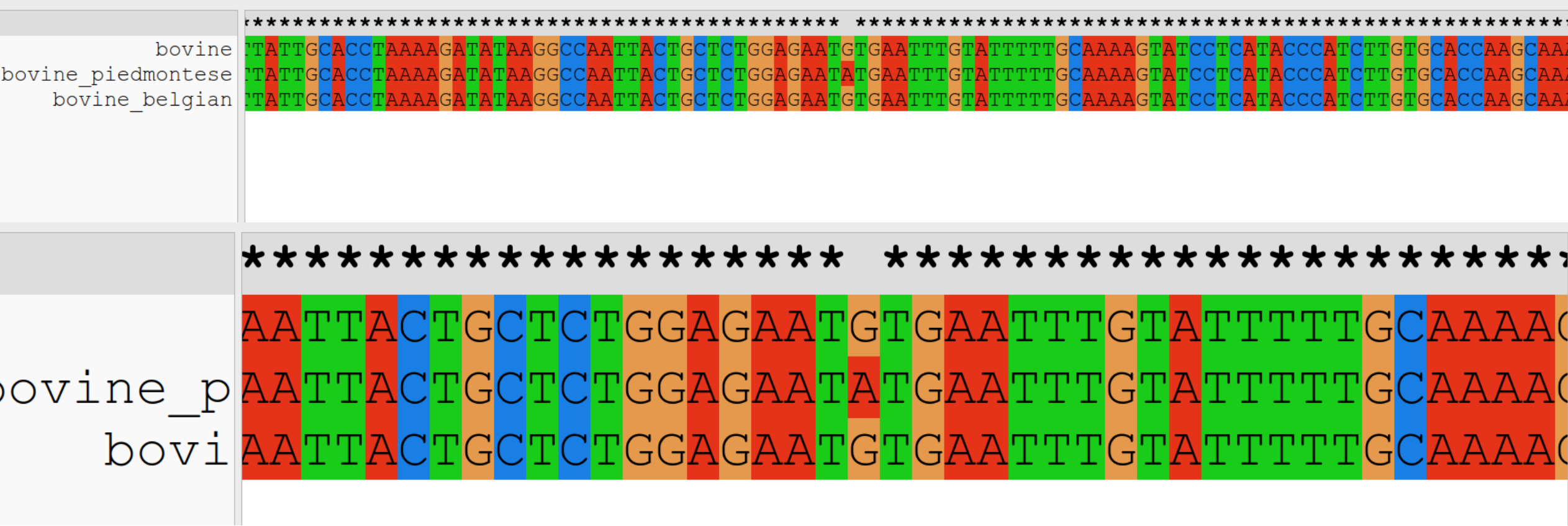
TGAACACTCCA

[illegible]

La razza Belga e la miostatina

La razza Piemontese e la miostatina

Transizione G → A

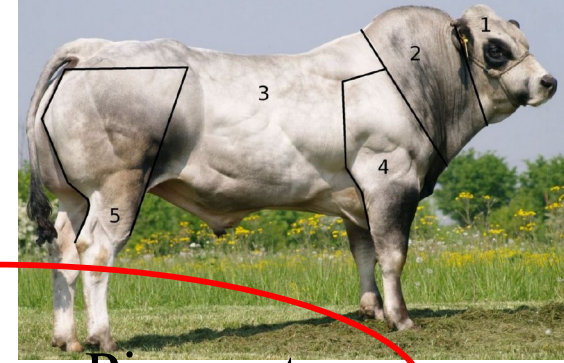


Le razze bovine Belga e Piemontese, se venissero classificate nella stessa categoria in base alla somiglianza fenotipica dovuta all'iperplasia muscolare congenita, formerebbero un **gruppo polifiletico**, ossia artificiale.

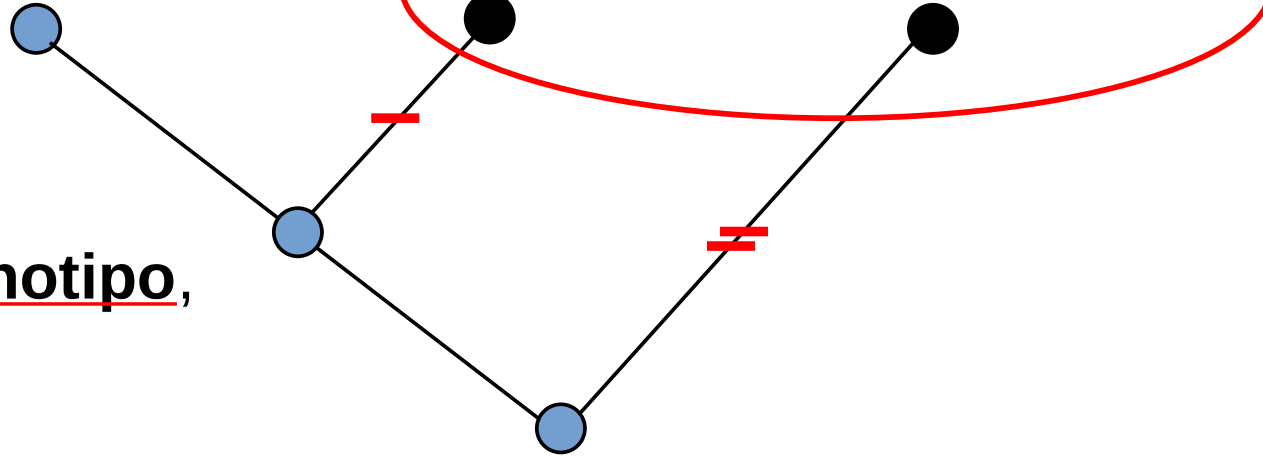
Ciò evidenzia i limiti della classificazione basata sul fenotipo, ossia sui caratteri morfologici.



Belga



Piemontese



github.com/pierorivoira/ESERCITAZIONI/blob/main/myostatin_species_names.fas

Accedi

pierorivoira / ESERCITAZIONI

<> Code

Issues

Pull requests

Actions

Projects 1

Wiki

← Files

main

ESERCITAZIONI / myostatin_species_names.fas

pierorivoira

Add files via upload

cdecaa6 · 45 minutes ago

2906 lines (2906 loc) · 218 KB

Download raw file

Code

Blame

Raw

1 >baboon

2 atgcaaaaactgcaactctgtgttttatatttacctgtttatgctgattgTTgctggTccagtggatctaataGagaacag

3 tgagcaaaaagaaaaatgtggaaaaagaggggctgtgtaatgcAtgtacttgGagacaaaacactaaatCttcaagaatag

4 aagccattaaaatacaaatcctcagtaaacttcgtctggaaacagctcctaacaTcacgcaaagatgctataagacaactt

Miostatina: allineamento multi-sequenza

Mode: Multiple Alignment Mode Font: 10

Cancel

Load Sequence File

Q Open

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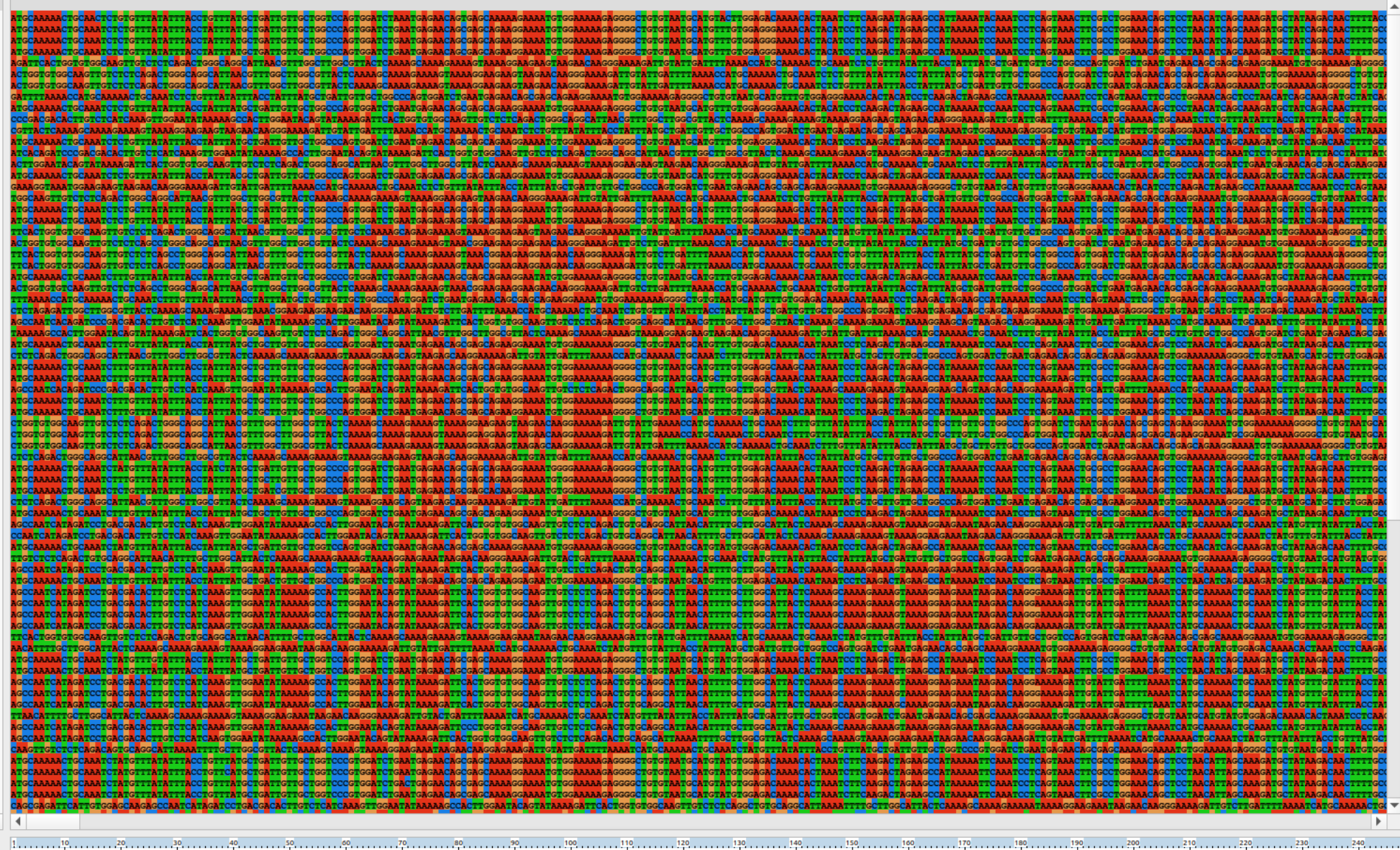
DNA_SEQUENCES

+ Altre posizioni

< piero DNA_SEQUENCES >

Nome	Dimensione	Tipo	Data di modifica
.RData	10,2 MB	R Data File	leri
.Rhistory	15,5 kB	Testo	leri
allineamento_miostatina_blast.txt	480,1 kB	Testo	11 gen 2024
Bos_taurus_myostatin.fas	2,9 kB	Testo	15 nov 2024
Bovidae_myostatin.dnd	75 byte	Testo	15 nov 2024
Bovidae_myostatin.fas	8,6 kB	Testo	15 nov 2024
Bovidae_myostatin.fasta	8,6 kB	Testo	15 nov 2024
database_completo_miostatina.txt	480,1 kB	Testo	13 gen 2024
dexter.fasta	88,6 kB	Testo	3 set 2024
R fcodons.R	210 byte	Testo	11 nov 2024
link_Cervidae.txt	344 byte	Testo	10 gen 2024
mcpherron-lee-1997-double-muscling-in-cattle-due-to-mu...	537,5 kB	Documento	27 set 2024
MT_Cervidae	333,7 kB	Testo	10 gen 2024
MT_Cervidae.fasta	448,4 kB	Testo	11 gen 2024
my_data.txt	98,9 kB	Testo	27 dic 2023
myostatin.dnd	3,7 kB	Testo	19 nov 2024
myostatin.fas	220,0 kB	Testo	leri
myostatin.fasta	714,6 kB	Testo	19 nov 2024
myostatin.nexus	1,2 MB	Testo	1 ott 2024
myostatin.RData	8,8 MB	R Data File	28 set 2024
MYOSTATIN_ACCESSION_NUMBERS.txt	1,7 kB	Testo	29 set 2024
MYOSTATIN_DATA.txt	17,3 kB	Testo	29 set 2024
myostatin_mod.aln	0 byte	Testo	leri
myostatin_mod.dnd	3,8 kB	Testo	leri
myostatin_mod.fas	222,3 kB	Testo	leri
myostatin_mod.fasta	727,8 kB	Testo	leri
myostatin_species_names.fas	222,9 kB	Testo	21:40
myostatin_x.fas	10,4 kB	Testo	17:03
myostatin_x_mod.dnd	265 byte	Testo	17:39
myostatin_x_mod.fas	11,5 kB	Testo	17:38

All Files

[illegible]

Miostatina: allineamento multi-sequenza

Mode: Multiple Alignment Mode Font: 10

[illegible]

Complete Alignment

Output Guide Tree:

hier/DNA_SEQUENCES/myostatin_species_names.dnd

Browse

Output Alignment Files

Fasta: myostatin_species_names.fasta

Browse

OK

Cancel

This figure displays a comprehensive set of genomic alignments for 50 different species. The species names are listed on the left, grouped by taxonomic level. The alignments are shown as horizontal tracks, each representing a species. Within each track, individual sequencing reads are aligned to a reference sequence. Mismatches between the reads and the reference are indicated by colored markers (red, green, blue, orange, etc.). The tracks are organized into columns, with each column representing a specific genomic region. A scale bar at the bottom of the figure provides a reference for the genomic positions, ranging from 330 to 570. The overall layout allows for a comparative analysis of genomic variations across a wide range of species.

La razza Belga e la miostatina

```
> setwd('~ /DNA_SEQUENCES/')
> load('myostatin.RData')
# [sostituire <myostatin.RData> con il nome del file che avete creato (es.: <piero.Rdata>)]

> library(ape)
> library(seqinr)

> myostatin.fas = read.dna("myostatin_species_names.fas", format="fasta")
> myostatin.fasta = read.dna("myostatin_species_names.fasta", format="fasta")

> myostatin.fas

> myostatin.fasta
```

La razza Belga e la miostatina

```
> myostatin.fasta
```

```
111 DNA sequences in binary format stored in a matrix.
```

```
all sequences of same length: 6434
```

```
Labels:
```

```
chicken
```

```
turkey
```

```
baboon
```

```
human
```

```
Manis_pentadactyla
```

```
Capra_hircus_breed_qianbei_ma_goat
```

```
...
```

```
Base composition:
```

```
      a      c      g      t  
a.337 a.181 a.189 a.293  
(Total: 714.17 kb)
```

La razza Belga e la miostatina

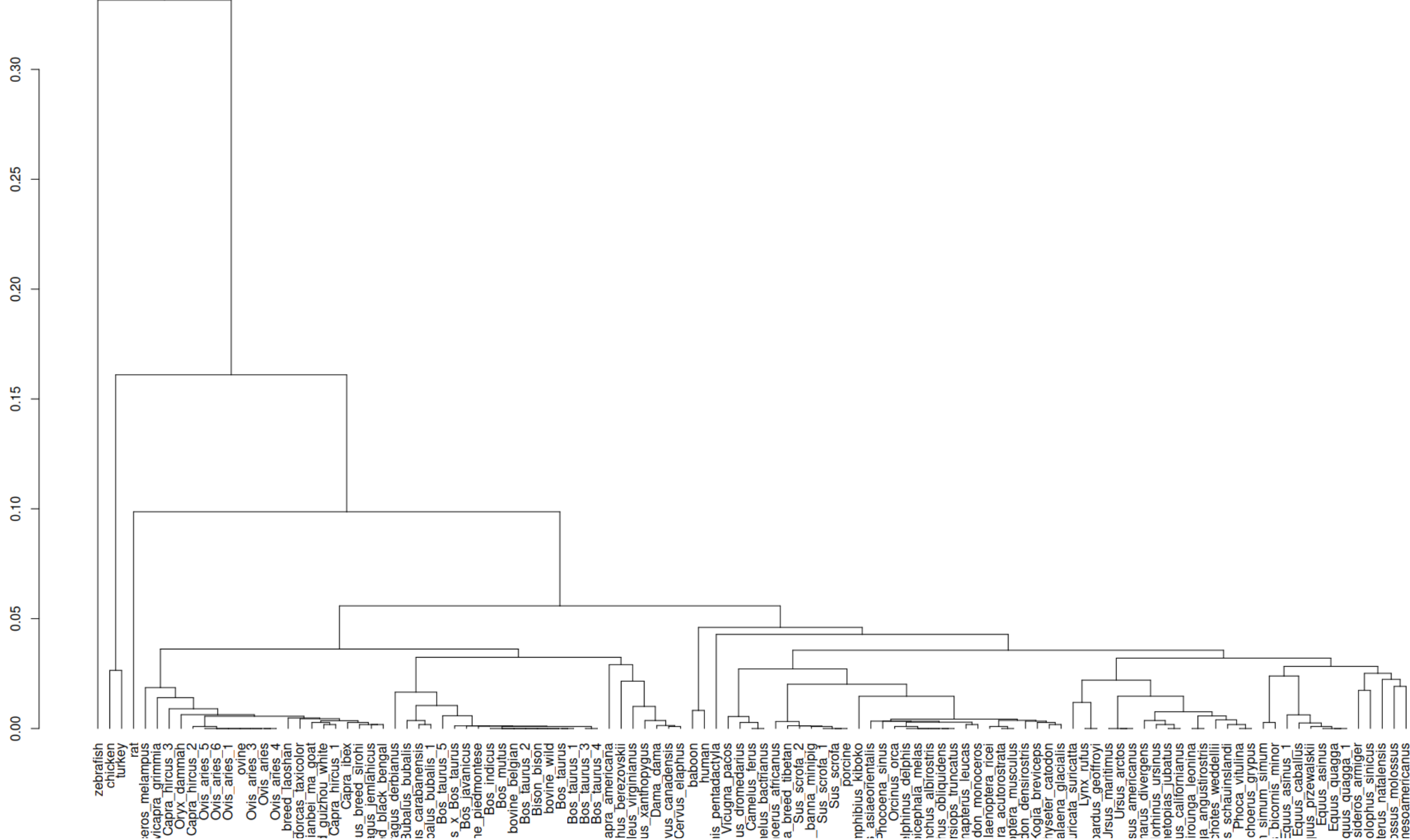
```
> str(myostatin.fasta)
'DNabin' raw [1:111, 1:6434] - - - - ...
- attr(*, "dimnames")=List of 2
..$ : chr [1:111] "chicken" "turkey" "baboon" "human" ...
..$ : NULL
```

```
> myostatin.dist=dist.dna(myostatin.fasta,model="raw")
> myostatin.dist
```

	chicken	turkey	baboon
turkey	a.a26484a18		
baboon	a.15a684932	a.143378995	
human	a.149771689	a.142465753	a.aa8219178
Manis_pentadactyla	a.1652968a4	a.153424658	a.a54794521
Capra_hircus_breed_qianbei_ma_goat	a.175342466	a.168a3653a	a.a76712329
Capra_hircus_breed_laoshan	a.1762557a8	a.169863a14	a.a78538813
Capra_hircus_breed_guizhou_white	a.1762557a8	a.168949772	a.a77625571
Capra_hircus_1	a.174429224	a.167123288	a.a75799a87
Capra_hircus_3	a.173515982	a.168a3653a	a.a76712329
Budorcas_taxicolor	a.1762557a8	a.167123288	a.a739726a3

La razza Belga e la miostatina

```
> myostatin.tree = hclust(myostatin.dist,"average")  
> plot(as.dendrogram(myostatin.tree))
```



La razza Belga e la miostatina

- Scegliere 20 specie, max 2 della stessa famiglia
- Uno dei taxa dev'essere **Bovine Belgian**, un altro **Zebra fish**
- Costruire l'albero UPGMA
- Come si distribuiscono i vari taxa sui rami dell'albero? Perché?
(rispondere sul quaderno)

```
> library(ape)
> setwd('~ /MATERIALE_DIDATTICO_A.S._2024-2025/ESERCITAZIONI/MIOSTATINA/')
> myostatin.fasta = read.dna("myostatin_5_species_100_bp.fasta",
format="fasta")
> X = myostatin.fasta
> myostatin.dist = dist.dna(X, model="raw")
```

raw: This is simply the proportion or the number of sites that differ between each pair of sequences.

```
> library(ape)
> setwd('~\\MATERIALE_DIDATTICO_A.S._2024-2025\\ESERCITAZIONI\\MIOSTATINA\\')
> X = read.dna("myostatin_5_species_100_bp.fasta", format="fasta",
as.character = FALSE)
> str(X)
'DNABin' raw [1:5, 1:100] a a a a ...
- attr(*, "dimnames")=List of 2
 ..$ : chr [1:5] "baboon" "bovine" "chicken" "ovine" ...
 ..$ : NULL
> X
5 DNA sequences in binary format stored in a matrix.
```

All sequences of same length: 100

Labels:
baboon
bovine
chicken
ovine
zebra_fish

Base composition:

a	c	g	t
0.286	0.178	0.240	0.296

(Total: 500 bases)

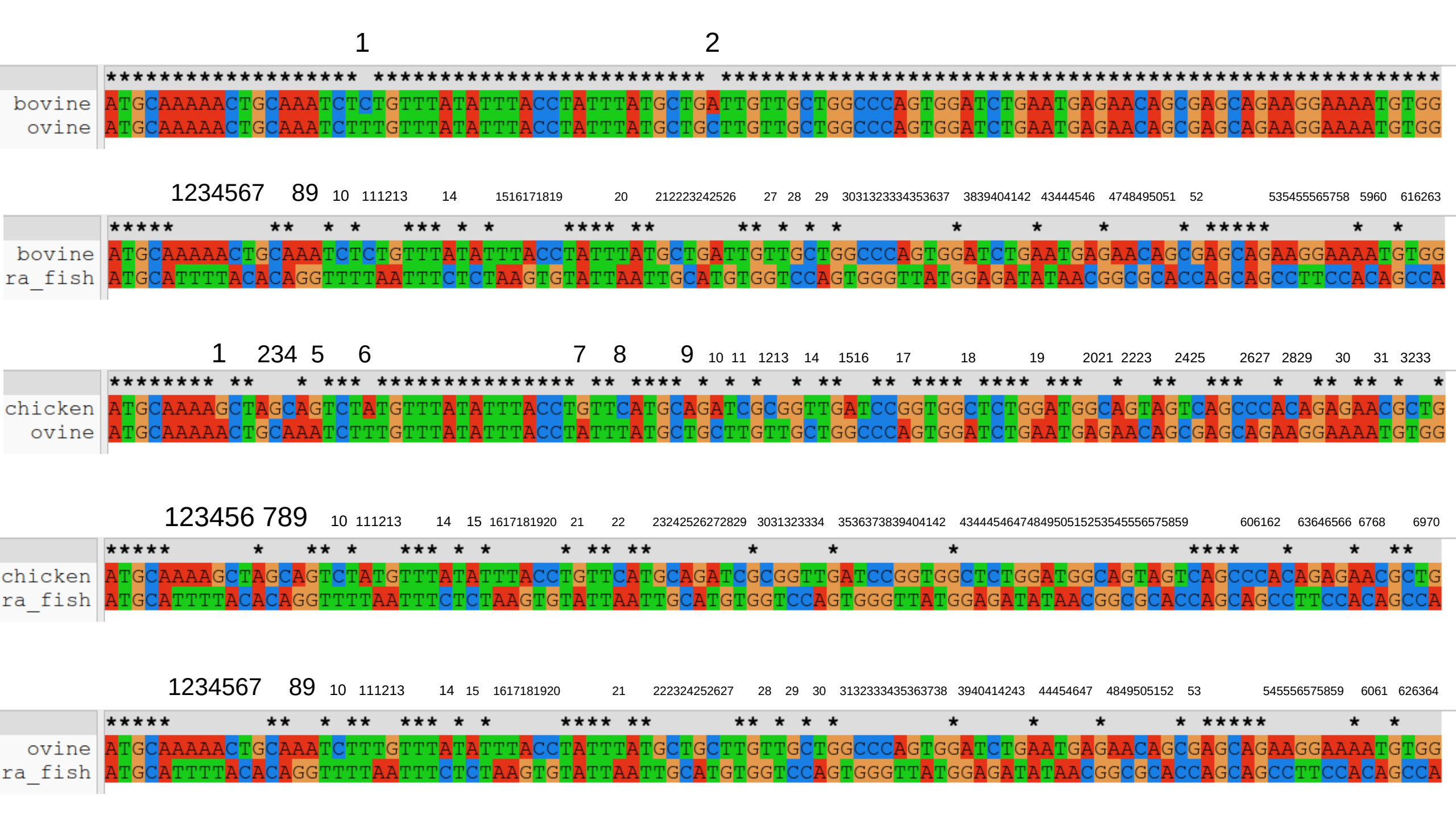
```

> myostatin.dist = dist.dna(X, model="raw")
> myostatin.dist

```

	baboon	bovine	chicken	ovine
bovine	0.08			
chicken	0.29	0.32		
ovine	0.09	0.02	0.33	
zebra_fish	0.67	0.64	0.70	0.63





```
> myostatin.UPGMA = hclust(myostatin.dist, "average")
```

```
# simple UPGMA
```

```
> myostatin.UPGMA
```

```
Call:
```

```
hclust(d = myostatin.dist, method = "average")
```

```
Cluster method      : average
```

```
Distance             : raw
```

```
Number of objects: 5
```

La funzione hclust effettua una *cluster analysis* gerarchica utilizzando, in questo caso, l'algoritmo **UPGMA** (*Unweighted Pair Group Method with Arithmetic Mean*).

L'idea alla base della *cluster analysis* gerarchica è di rappresentare graficamente quali elementi (singoli esemplari di una stessa o di specie diverse, sequenze di DNA ecc.) di un certo campione sono più simili tra di loro, raggruppandoli a due a due sullo stesso ramo dell'albero.

Gli oggetti che sono più diversi fra loro vengono collocati in rami differenti.

```
> myostatin.dist
      baboon bovine chicken ovine
bovine      0.08
chicken      0.29    0.32
ovine        0.09    0.02    0.33
zebra_fish   0.67    0.64    0.70    0.63
```

Matrice **D1** di distanze genetiche
(n° di nucleotidi differenti su 100)
fra coppie di specie

Il bovino e la pecora sono le specie
tra le quali la D1 è <, quindi le
mettiamo insieme

Quindi calcoliamo la distanza media
tra bov-ovi e ciascuna delle altre
specie

$$D2((\text{bov}, \text{ovi}), \text{bab}) = [D1(\text{bov}, \text{bab}) + D1(\text{ovi}, \text{bab})] / 2 = (8 + 9) / 2 = 8,5$$
$$D2((\text{bov}, \text{ovi}), \text{gal}) = [D1(\text{bov}, \text{gal}) + D1(\text{ovi}, \text{gal})] / 2 = (32 + 33) / 2 = 32,5$$
$$D2((\text{bov}, \text{ovi}), \text{zeb}) = [D1(\text{bov}, \text{zeb}) + D1(\text{ovi}, \text{zeb})] / 2 = (64 + 63) / 2 = 63,5$$

	bab	bov	gal	ovi	zeb
bab	0	8	29	9	67
bov	8	0	32	2	64
gal	29	32	0	33	70
ovi	9	2	33	0	63
zeb	67	64	70	63	0

	bab	bov- ovi	gal	zeb
bab	0	8,5	29	67
bov- ovi	8,5	0	32,5	63,5
gal	29	32,5	0	70
zeb	67	63,5	70	0

	(bov-ovi)bab	gal	zeb
(bov-ovi)bab	0	30,75	65,25
gal	30,75	0	70
zeb	65,25	70	0

$D3(((\text{bov}, \text{ovi}), \text{bab}), \text{gal}) = [D2((\text{bov}, \text{ovi}), \text{gal}) + (D2(\text{bab}, \text{gal}))] / 2 = (32,5 + 29) / 2 = 30,75$
 $D3(((\text{bov}, \text{ovi}), \text{bab}), \text{zeb}) = [D2((\text{bov}, \text{ovi}), \text{zeb}) + (D2(\text{bab}, \text{zeb}))] / 2 = (63,5 + 67) / 2 = 65,25$

	(bov-ovi)bab	gal	zeb
(bov-ovi)bab	0	30,75	65,25
gal	30,75	0	70
zeb	65,25	70	0

$$D4((((bov,ovi),bab),gal),zeb) = [D3((((bov,ovi),bab),zeb)+D3(gal,zeb)]/2 = (65,25+70)/2 = 67,625$$

	((bov-ovi)bab)ga	zeb
((bov-ovi)bab)gal	0	67,625
zeb	67,625	0

```
> plot(myostatin.UPGMA, main = 'UPGMA distances')
```

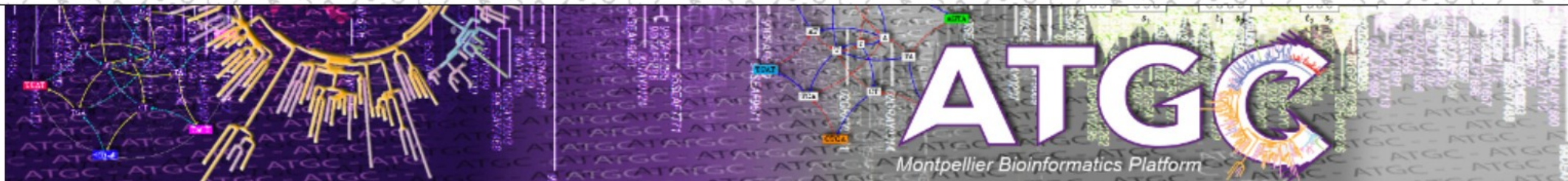


Scelta del modello di sostituzione nucleotidica

L'analisi filogenetica più semplice si basa sul calcolo delle distanze fra coppie di sequenze nucleotidiche, le quali diventano gli ingressi di una matrice che, tramite algoritmi di clusterizzazione, viene convertita in un albero filogenetico.

La distanza fra due sequenze è definita come numero atteso di sostituzioni nucleotidiche per sito (generalmente < 1) e può essere espressa come semplice proporzione dei siti variabili o distanza **p**.

```
> myostatin.dist
      baboon bovine chicken ovine
bovine      0.08
chicken      0.29    0.32
ovine        0.09    0.02    0.33
zebra_fish   0.67    0.64    0.70    0.63
```



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PhyML 3.0: new algorithms, methods and utilities

Please cite:

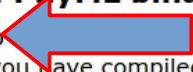
"New Algorithms and Methods to Estimate Maximum-Likelihood Phylogenies: Assessing the Performance of PhyML 3.0."

Guindon S., Dufayard J.F., Lefort V., Anisimova M., Hordijk W., Gascuel O.

Systematic Biology, 59(3):307-21, 2010.

Get the latest PhyML releases from [GitHub](#).

Download PhyML binaries

Click [here](#) to  binaries

The binaries you have compiled on various Mac, Windows or LINUX systems are much welcome ([Stéphane Guindon](#)) and will be displayed on this web page.

Il formato PHYLIP interleaved

```
5 100
baboon
atgcaaaaactgcaactctgtgtttatatttacctgtttatgctgattgttgctgggtccagtggtatctaaatgagaacagtgagcaaaaagaaaatgtgg
bovine
atgcaaaaactgcaaactctgtgtttatatttacctatttatgctgattgttgctggcccagtggtatctgaatgagaacagcgagcagaaggaaaatgtgg
chicken
atgcaaaagctagcagtcctatgttttatatttacctgttcatgcagatcgcggttgatccggtggctctggatggcagtagtcagccacagagaaacgctg
ovine
atgcaaaaactgcaaactcttggttttatatttacctatttatgctgcttggttgctggcccagtggtatctgaatgagaacagcgagcagaaggaaaatgtgg
zebrafish
atgcattttacacagggttttaatttctctaagtgtattaattgcatgtggtccagtggggttatggagatatataacggcgcaccagcagccttccacagcca
```

```
> substitution_model_test = phymltest('myostatin_5_species_100_bp.phy', format
= "interleaved", itree = NULL, exclude = NULL, execname = '/home/piero/PhyML-
3.1/PhyML-3.1_linux64', append = TRUE)
```

```
. Sequence filename:          myostatin_5_species_100_bp.phy
. Data set:                #1
. Tree topology search :    NNIs
. Initial tree:            BioNJ
. Model of nucleotides substitution:    GTR
. Number of taxa:          5
. Log-likelihood:          -402.43191
. Unconstrained likelihood:    -343.80642
. Parsimony:                100
. Tree size:                4.55423
. Nucleotides frequencies:
  - f(A)= 0.28600
  - f(C)= 0.17800
  - f(G)= 0.24000
  - f(T)= 0.29600
. GTR relative rate parameters :
A <-> C    11.85779
A <-> G    14.05957
A <-> T     0.00993
C <-> G     8.42143
C <-> T    13.46606
G <-> T     1.00000
```

```
. Instantaneous rate matrix :
[A-----C-----G-----T-----]
-1.01259    0.38945    0.62260    0.00054
 0.62574   -1.73412    0.37293    0.73546
 0.74193    0.27659   -1.07313    0.05462
 0.00052    0.44227    0.04428   -0.48708
```

L'output del test

```
> substitution_model_test
      nb.free.para      loglik      AIC
JC69      1 -414.3124  830.6248
JC69+I     2 -414.1910  832.3821
JC69+G     2 -413.8407  831.6814
JC69+I+G   3 -413.8313  833.6625
K80        2 -411.0496  826.0993
K80+I      3 -408.3180  822.6359
K80+G      3 -408.8849  823.7699
K80+I+G    4 -407.9957  823.9915
F81        4 -414.1411  836.2823
F81+I      5 -414.0996  838.1992
F81+G      5 -413.7568  837.5136
F81+I+G    6 -413.7486  839.4972
F84        5 -409.8965  829.7931
F84+I      6 -407.9673  827.9346
F84+G      6 -408.0924  828.1848
F84+I+G    7 -407.5650  829.1301
HKY85      5 -409.8806  829.7612
HKY85+I    6 -407.9661  827.9323
HKY85+G    6 -408.0732  828.1464
HKY85+I+G  7 -407.5730  829.1460
TN93       6 -409.8804  831.7607
TN93+I     7 -407.9637  829.9274
TN93+G     7 -408.0700  830.1400
TN93+I+G   8 -407.5648  831.1297
GTR        9 -402.4319  822.8638
GTR+I     10 -401.2392  822.4784
GTR+G     10 -400.2603  820.5205
GTR+I+G    11 -400.2379  822.4758
>
```

Il file <myostatin_5_species_100_bp.phy_phyml_tree.txt>

```
(bovine:0.00000202,(ovine:0.00002963,zebrafish:9.99805710)0.000000:0.02049504,
(baboon:0.00008628,chicken:0.50886609)0.842000:0.09067509);
(baboon:0.01048862,(bovine:0.00000001,(zebrafish:1.37441488,ovine:0.00000508):0.02026739):0.07447686,chicken:0.35829670);
(baboon:0.00972686,chicken:0.36497752,(bovine:0.00000118,(ovine:0.00000029,zebrafish:1.54101437):0.02030498):0.07560156);
(baboon:0.00007584,chicken:0.44219573,(ovine:0.02056245,(bovine:0.00000001,zebrafish:3.54999086):0.00000001):0.08826723);
(baboon:0.00000910,chicken:0.45531080,(bovine:0.00000001,(zebrafish:4.16221975,ovine:0.00001874):0.02059383):0.08887950);
(bovine:0.00000099,(ovine:0.02028732,zebrafish:3.09237767):0.00003742,(baboon:0.01122146,chicken:0.36941408):0.07398107);
(zebrafish:8.82019763,(bovine:0.00000175,ovine:0.02039573):0.02277632,(chicken:0.39408351,baboon:0.01090682):0.05227845);
(bovine:0.00000099,(ovine:0.02044652,zebrafish:10.00000000):0.00002146,(baboon:0.00002629,chicken:0.50356626):0.08929782);
(chicken:0.43223177,baboon:0.00713230,(bovine:0.00000126,(zebrafish:8.89223736,ovine:0.01973263):0.00074093):0.08047846);
(baboon:0.01115834,chicken:0.36143405,(bovine:0.00000181,(zebrafish:1.42016394,ovine:0.00000268):0.02033915):0.07448850);
(bovine:0.00000091,(chicken:0.36579093,baboon:0.01074355):0.07520598,(zebrafish:1.52463974,ovine:0.00000001):0.02037077);
(bovine:0.00000117,(zebrafish:3.58786252,(chicken:0.44228493,baboon:0.00171289):0.08712725):0.00000872,ovine:0.02063859);
(baboon:0.00000824,chicken:0.46731497,(bovine:0.00000001,(zebrafish:4.82711651,ovine:0.00000756):0.02070614):0.08999625);
(bovine:0.00000101,(ovine:0.02042293,zebrafish:3.77229889):0.00003150,(baboon:0.01338516,chicken:0.37881952):0.07317730);
(zebrafish:8.98214792,(bovine:0.00000182,ovine:0.02045418):0.02675688,(chicken:0.39647730,baboon:0.01296603):0.04725382);
(bovine:0.00000204,(ovine:0.00002615,zebrafish:9.99812034):0.02054687,(baboon:0.00002661,chicken:0.50940308):0.09100487);
(chicken:0.44286719,zebrafish:9.08352240,(baboon:0.00942811,(bovine:0.00000132,ovine:0.02057252):0.07969971):0.00014779);
(bovine:0.00000099,(ovine:0.02039898,zebrafish:3.70535167):0.00003159,(baboon:0.01300483,chicken:0.37844433):0.07338886);
(zebrafish:8.98429384,(bovine:0.00000180,ovine:0.02042441):0.02673440,(chicken:0.39638088,baboon:0.01260347):0.04745109);
(bovine:0.00000202,(ovine:0.00002963,zebrafish:9.99805710):0.02049504,(baboon:0.00008628,chicken:0.50886609):0.09067509);
(chicken:0.44246753,zebrafish:9.08528623,(baboon:0.00898618,(bovine:0.00000130,ovine:0.02053515):0.07990025):0.00016885);
(bovine:0.00000099,(ovine:0.02039408,zebrafish:3.68224905):0.00003166,(baboon:0.01293915,chicken:0.37829051):0.07342438);
(zebrafish:8.98269671,(bovine:0.00000180,ovine:0.02044070):0.03061041,(chicken:0.39640554,baboon:0.01278578):0.04346390);
(bovine:0.00000205,(ovine:0.00003700,zebrafish:9.99683005):0.02045292,(baboon:0.00008704,chicken:0.50879952):0.09053445);
(chicken:0.44280285,zebrafish:9.08381557,(baboon:0.00936629,(bovine:0.00000130,ovine:0.02056699):0.07972586):0.00014726);
(bovine:0.00000001,(zebrafish:4.05071318,(baboon:0.00593667,chicken:0.39542302):0.08161689):0.00000004,ovine:0.02053827);
(zebrafish:4.34749500,(bovine:0.00000057,(chicken:0.40630281,baboon:0.00316606):0.08445039):0.02051856,ovine:0.00000251);
(zebrafish:10.00000000,(bovine:0.00000056,(baboon:0.00001018,chicken:0.54408882):0.09498755):0.02116598,ovine:0.00001086);
(baboon:0.00000993,chicken:0.53213267,(bovine:0.00000001,(zebrafish:9.76620951,ovine:0.00000001):0.02114857):0.09443561);
```

Scelta del modello di sostituzione nucleotidica

```
> r = '(baboon:0.00000993,chicken:0.53213267,(bovine:0.00000001,  
(zebrafish:9.76620951,ovine:0.00000001):0.02114857):0.09443561);'  
> albero = read.tree(text = r)  
> albero
```

Phylogenetic tree with 5 tips and 3 internal nodes.

Tip labels:

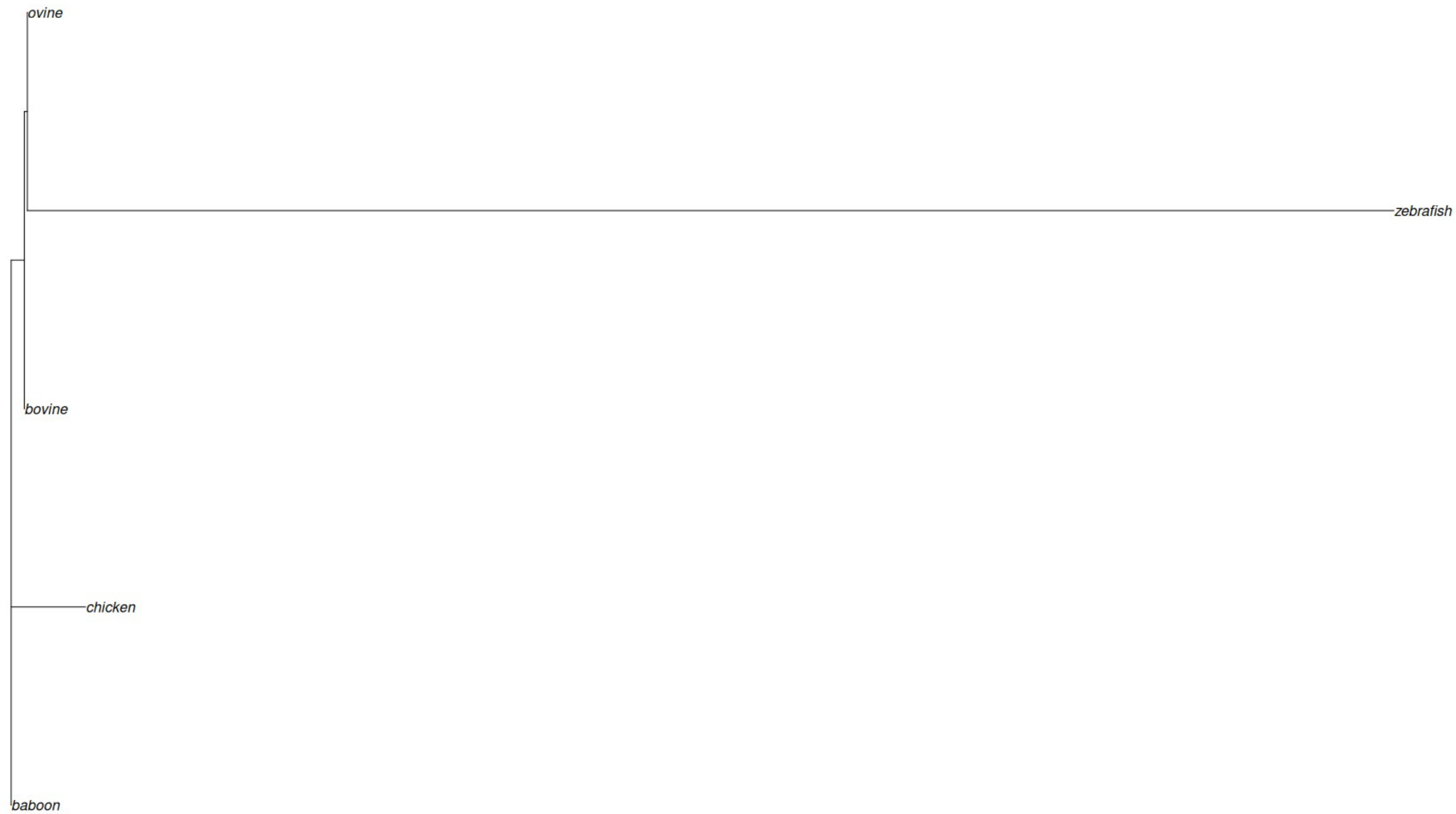
baboon, chicken, bovine, zebrafish, ovine

Unrooted; includes branch lengths.

```
> plot(albero, main = 'GTR model')
```

L'albero più verosimile

GTR model



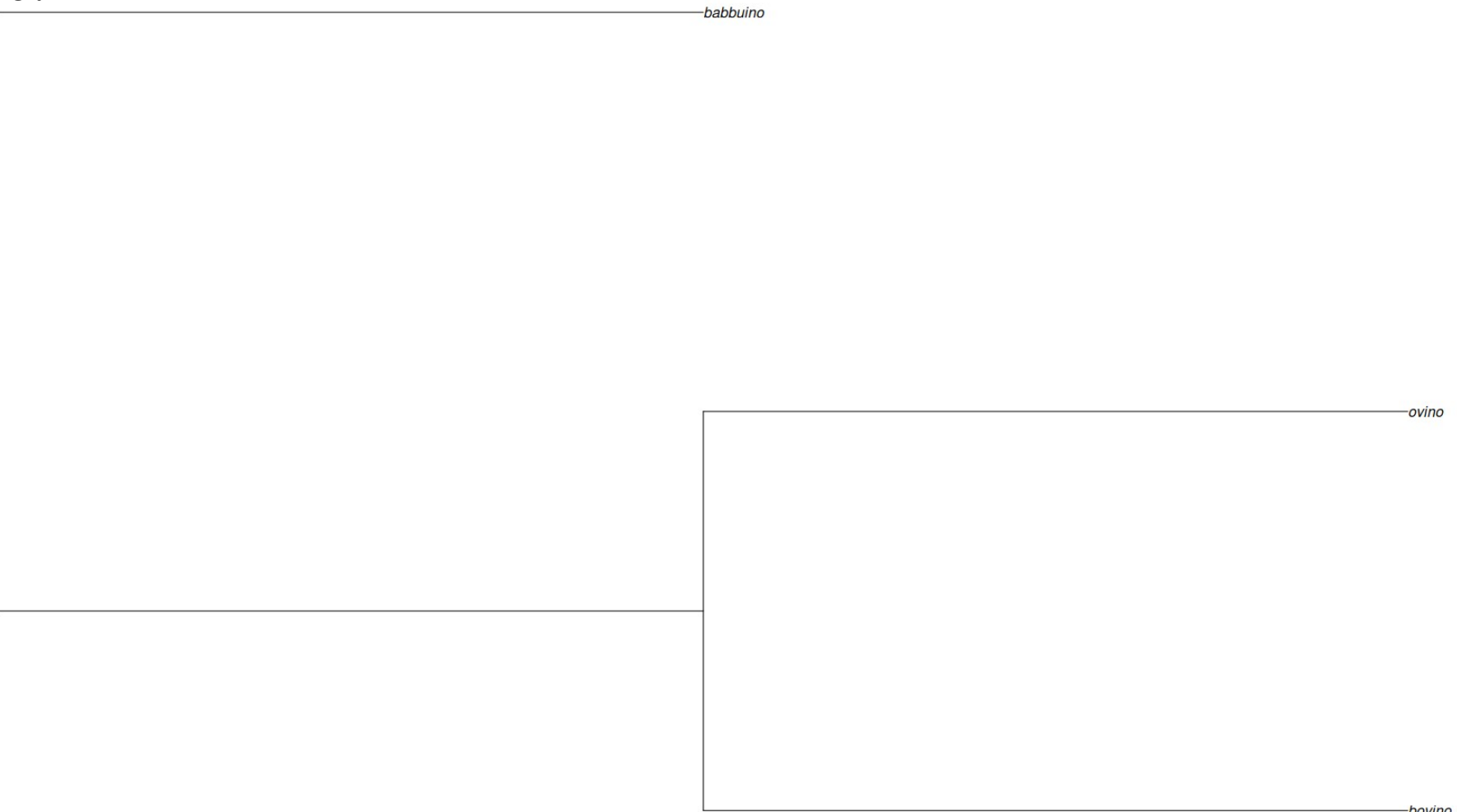
```
> alb = read.tree(text = "((bovino:1, ovino:1):1, babbuino:1);")
> alb
```

Phylogenetic tree with 3 tips and 2 internal nodes.

Tip labels:
bovino, ovino, babbuino

Rooted; includes branch lengths.

```
> plot(alb)
```



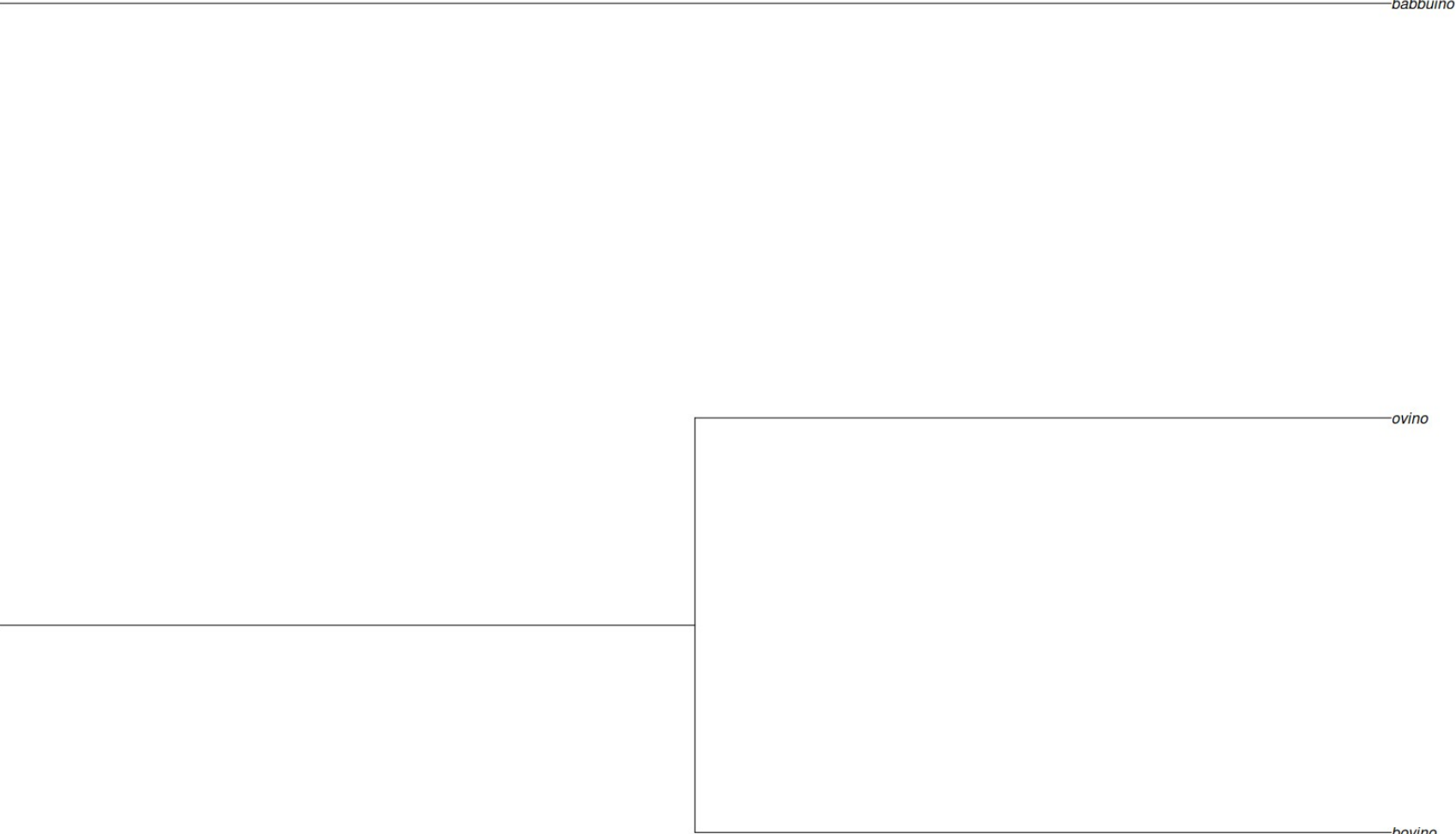
```
> alb = read.tree(text = "((bovino:1, ovino:1):1, babbuino:2);")
> alb
```

Phylogenetic tree with 3 tips and 2 internal nodes.

Tip labels:
bovino, ovino, babbuino

Rooted; includes branch lengths.

```
> plot(alb)
```



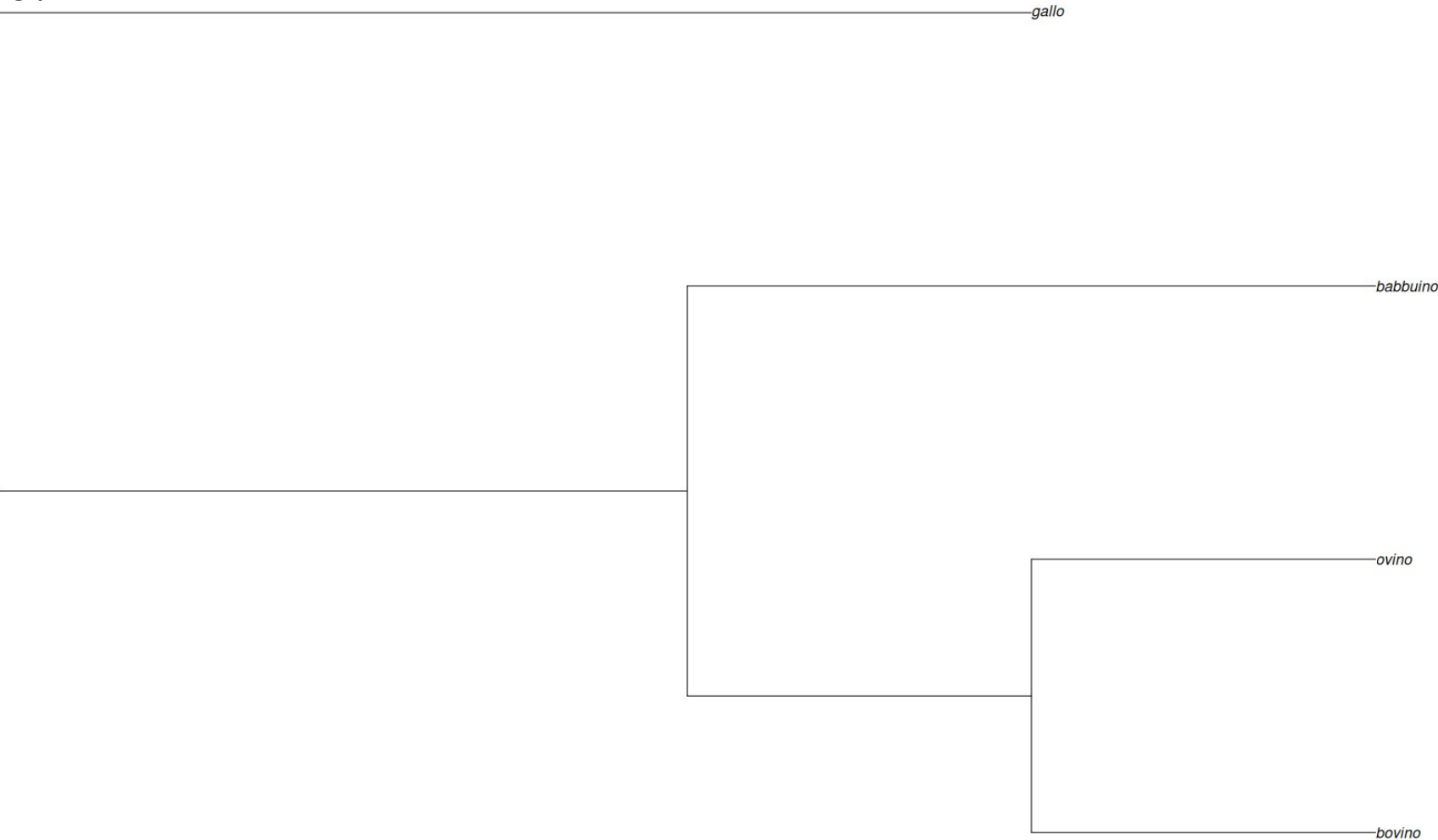
```
> alb = read.tree(text = "((bovino:1, ovino:1):1, babbuino:2):2, gallo:3);")
> alb
```

Phylogenetic tree with 4 tips and 3 internal nodes.

Tip labels:
bovino, ovino, babbuino, gallo

Rooted; includes branch lengths.

```
> plot(alb)
```



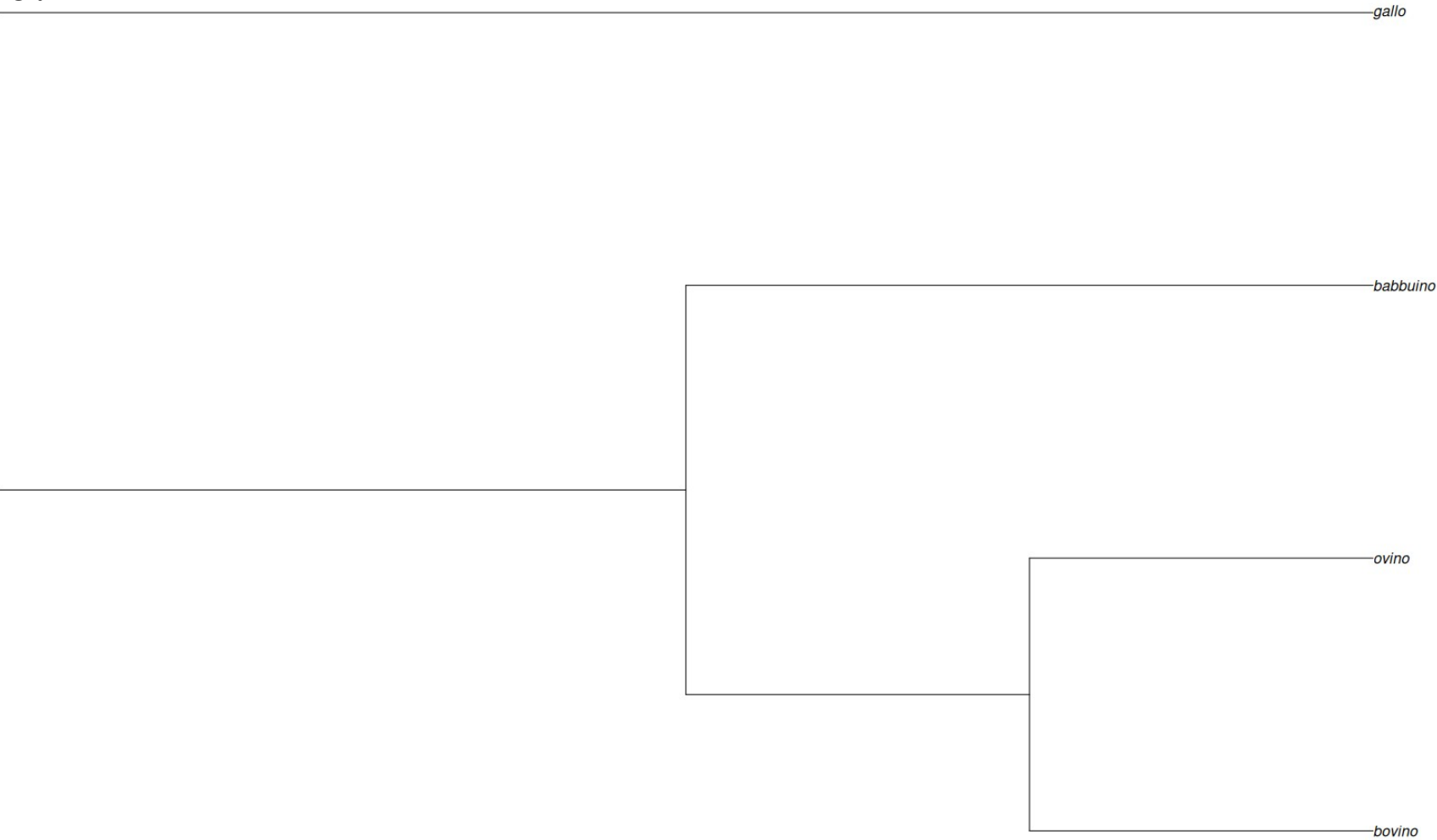
```
> alb = read.tree(text = "((bovino:1, ovino:1):1, babbuino:2):2, gallo:4);")
> alb
```

Phylogenetic tree with 4 tips and 3 internal nodes.

Tip labels:
bovino, ovino, babbuino, gallo

Rooted; includes branch lengths.

```
> plot(alb)
```



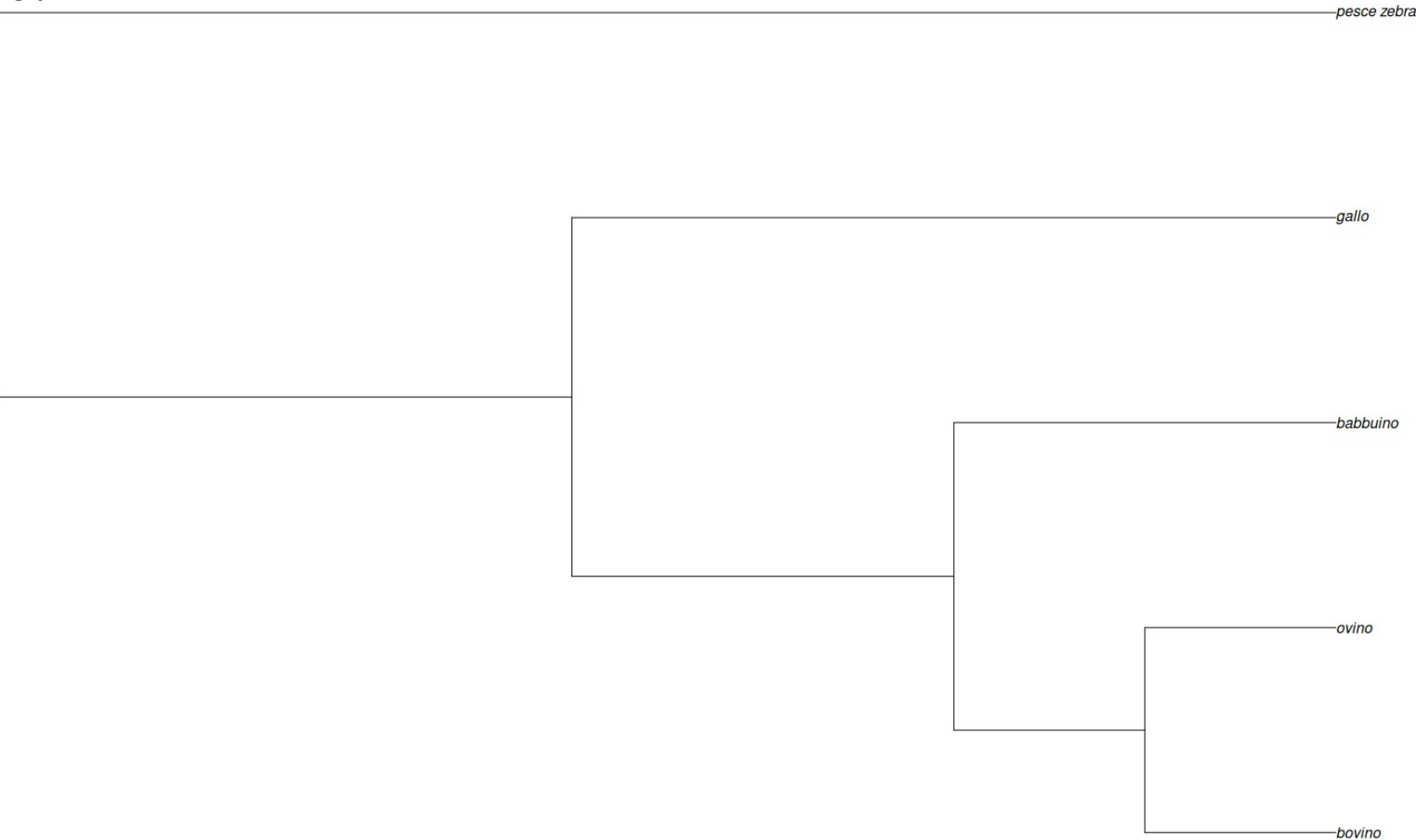
```
> alb = read.tree(text = "(((bovino:1, ovino:1):1, babbuino:2):2, gallo:4):3, pesce_zebra:7);")
> alb
```

Phylogenetic tree with 5 tips and 4 internal nodes.

Tip labels:
 bovino, ovino, babbuino, gallo, pesce_zebra

Rooted; includes branch lengths.

```
> plot(alb)
```



STATISTICA

L'effetto additivo dei fattori genetici ed ambientali

gene normale A -> effetto fenotipico = 8 $p = 0,5$

allele mutante A' -> effetto fenotipico = 4 $q = 0,5$

omozigoti AA -> valore fenotipico = 16 $F = 0,25$

omozigoti A'A' -> valore fenotipico = 8 $F = 0,25$

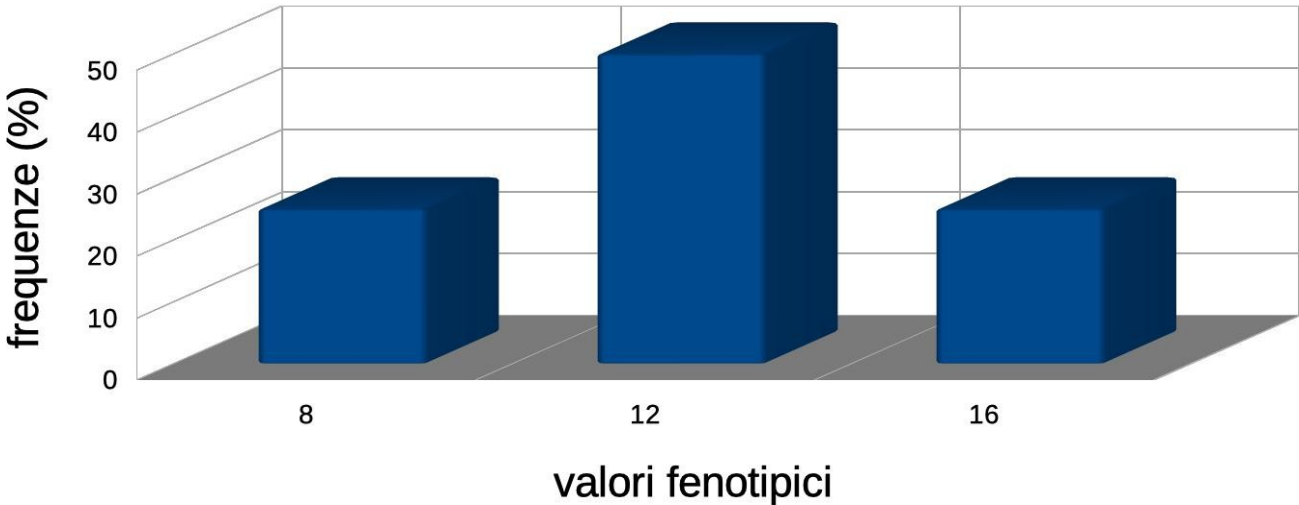
eterozigoti AA' -> valore fenotipico = 12 $F = 0,25$

eterozigoti A'A -> valore fenotipico = 12 $F = 0,25$

L'effetto additivo dei fattori genetici ed ambientali

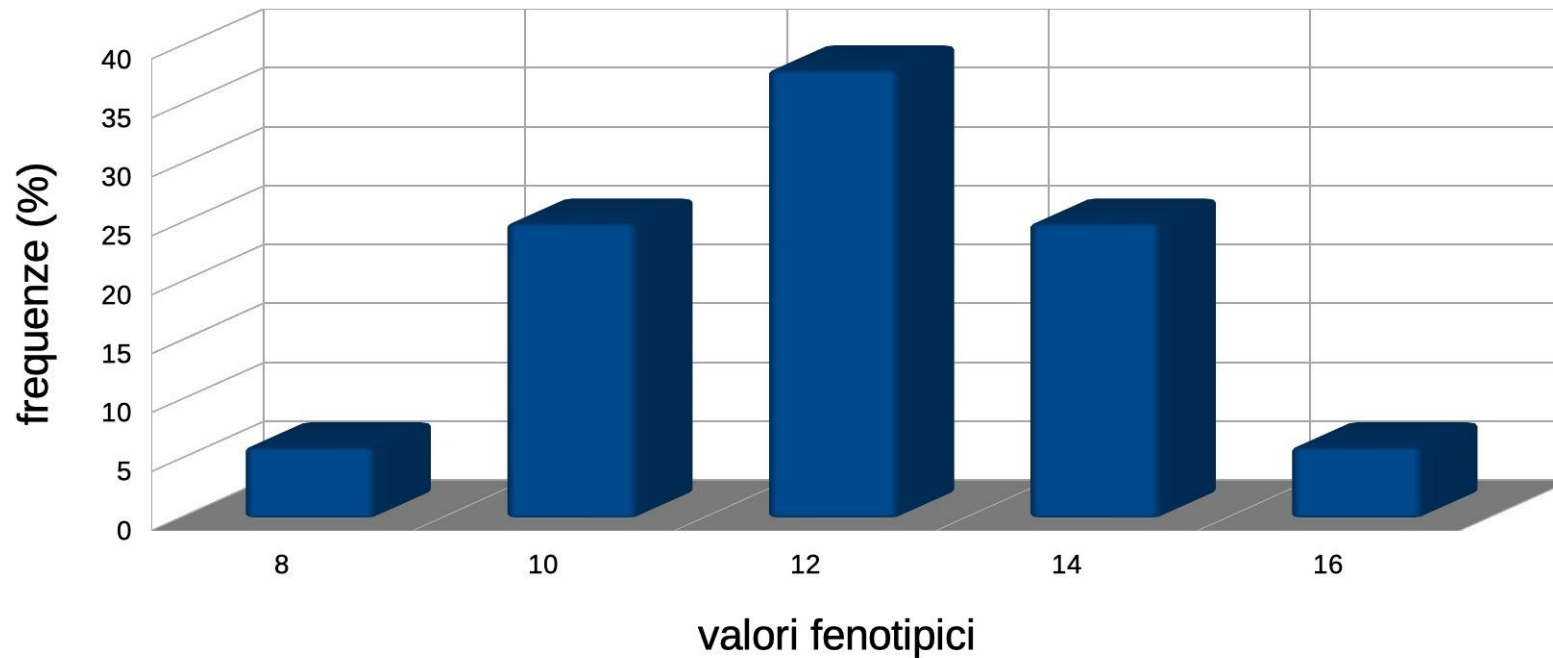
CLASSE GENOTIPICA	VALORE FENOTIPICO	FREQUENZA
AA	16	1/4 = 25%
AA' - A'A	12	2/4 = 50%
A'A'	8	1/4 = 25%

**DISTRIBUZIONE DEI VALORI FENOTIPICI
CON 1 COPPIA ALLELICA**



Due loci

DISTRIBUZIONE DEI VALORI FENOTIPICI CON 2 COPPIE ALLELICHE



Valori e frequenze dei possibili genotipi nel caso in cui un carattere quantitativo è controllato da due *loci* («a» e «B»)

Tre loci

aumentando ulteriormente il numero di coppie alleliche interessate, aumenta in numero delle classi di valori ma diminuiscono le differenze fra i valori delle varie classi

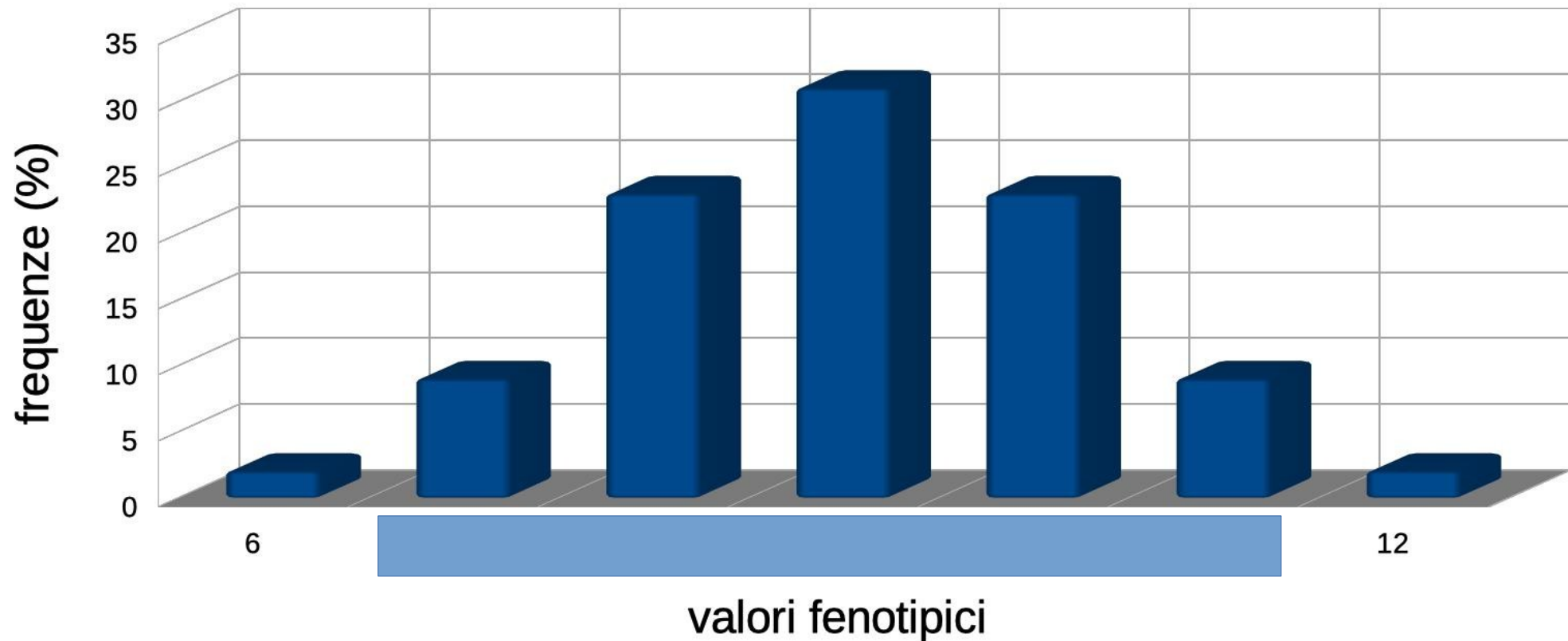
Se si considerano tre coppie di geni, $a-a'$, $B-B'$ e $C-C'$, dove gli alleli «normali» a , B e C hanno effetto 2 e gli alleli «mutanti» a' , B' e C' effetto 1, le classi genotipiche e i relativi valori fenotipici saranno:

Tre loci

Quali sono i
valori fenotipici
intermedi?

CLaSSE GENOTIPICA	VaLORE FENOTIPICO	FREQUENZA
aaBBCC	12	1/64 = 2%
aaBBCC' – aaBBC'C – aaBB'CC aaB'BCC – aa'BBCC – a'aBBCC		6/64 = 9%
aaBBC'C' – aaBB'CC' – aaB'BCC' aa'BBCC' – a'aBBCC' – aaBB'C'C aaB'BC'C – aa'BBC'C – a'aBBC'C aaB'B'CC – aa'BB'CC – a'aBB'CC aa'B'BCC – a'aB'BCC – a'a'BBCC		15/64 = 23%
aaBB'C'C' – aaB'BC'C' – aaB'B'CC' aaB'B'C'C – aa'BBC'C' – aa'BB'CC' aa'B'BCC' – aa'BB'C'C – aa'B'BC'C aa'B'B'CC – a'aBBC'C' – a'aBB'CC' a'aB'BCC' – a'aBB'C'C – a'aB'BC'C a'aB'B'CC – a'a'B'BCC – a'a'BB'CC, a'a'BBC'C – a'a'BBCC'		2a/64 = 31%
aaB'B'C'C' – aa'BB'C'C' – a'aBB'C'C' a'aBBC'C' – aa'B'BC'C' – a'aB'BC'C' a'a'B'BCC' – aa'B'B'CC' – a'aB'B'CC' a'a'B'BC'C – aa'B'B'C'C – a'aB'B'C'C a'a'B'B'CC – a'a'BB'CC' – a'a'BB'C'C		15/64 = 23%
aa'B'B'C'C' – a'aB'B'C'C' – a'a'B'B'C'C a'a'B'B'CC' – a'a'BB'C'C' – a'a'B'BC'C'		6/64 = 9%
a'a'B'B'C'C'	6	1/64 = 2%

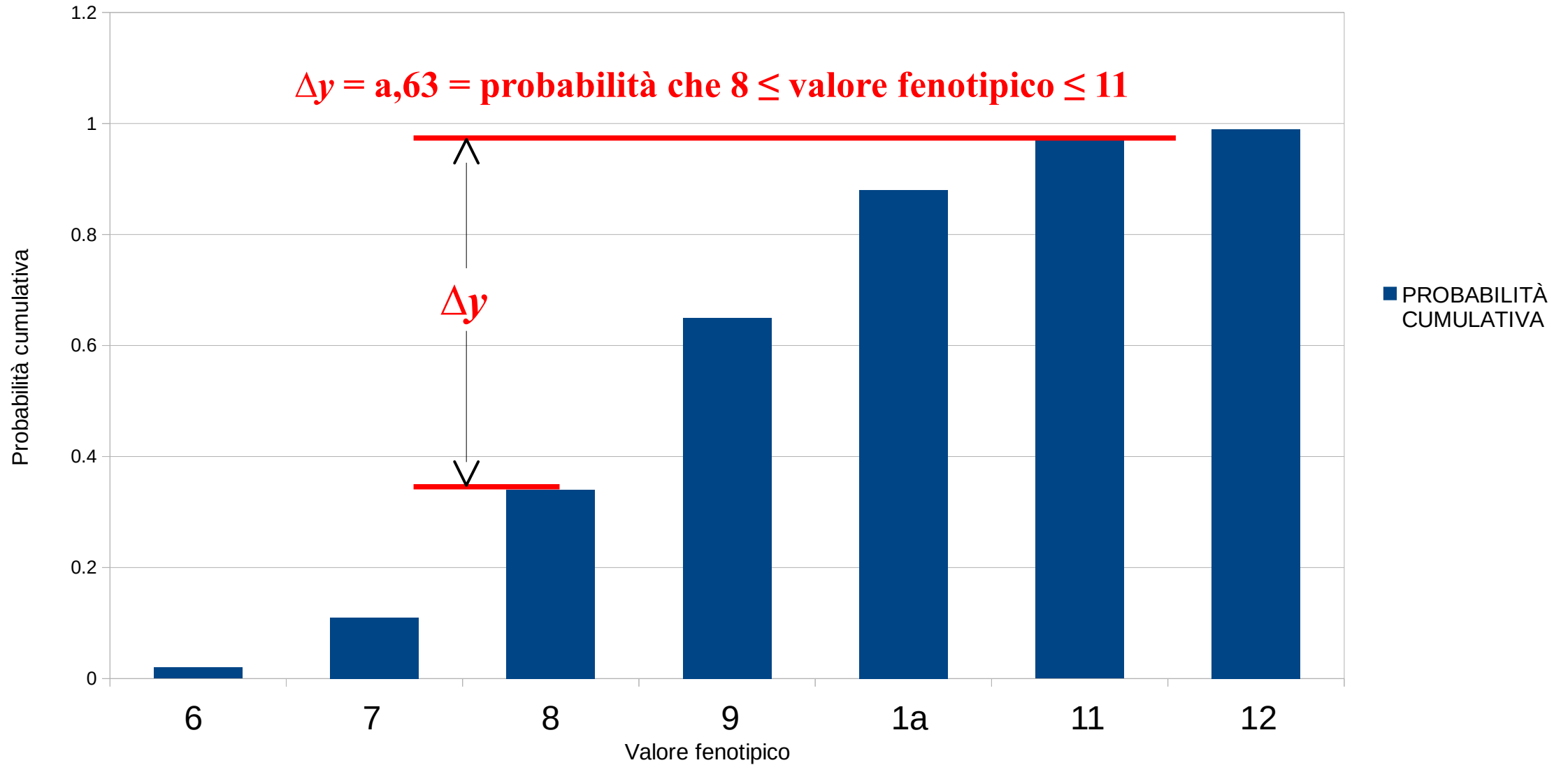
DISTRIBUZIONE DEI VALORI FENOTIPICI CON 3 COPPIE ALLELICHE



Tre loci

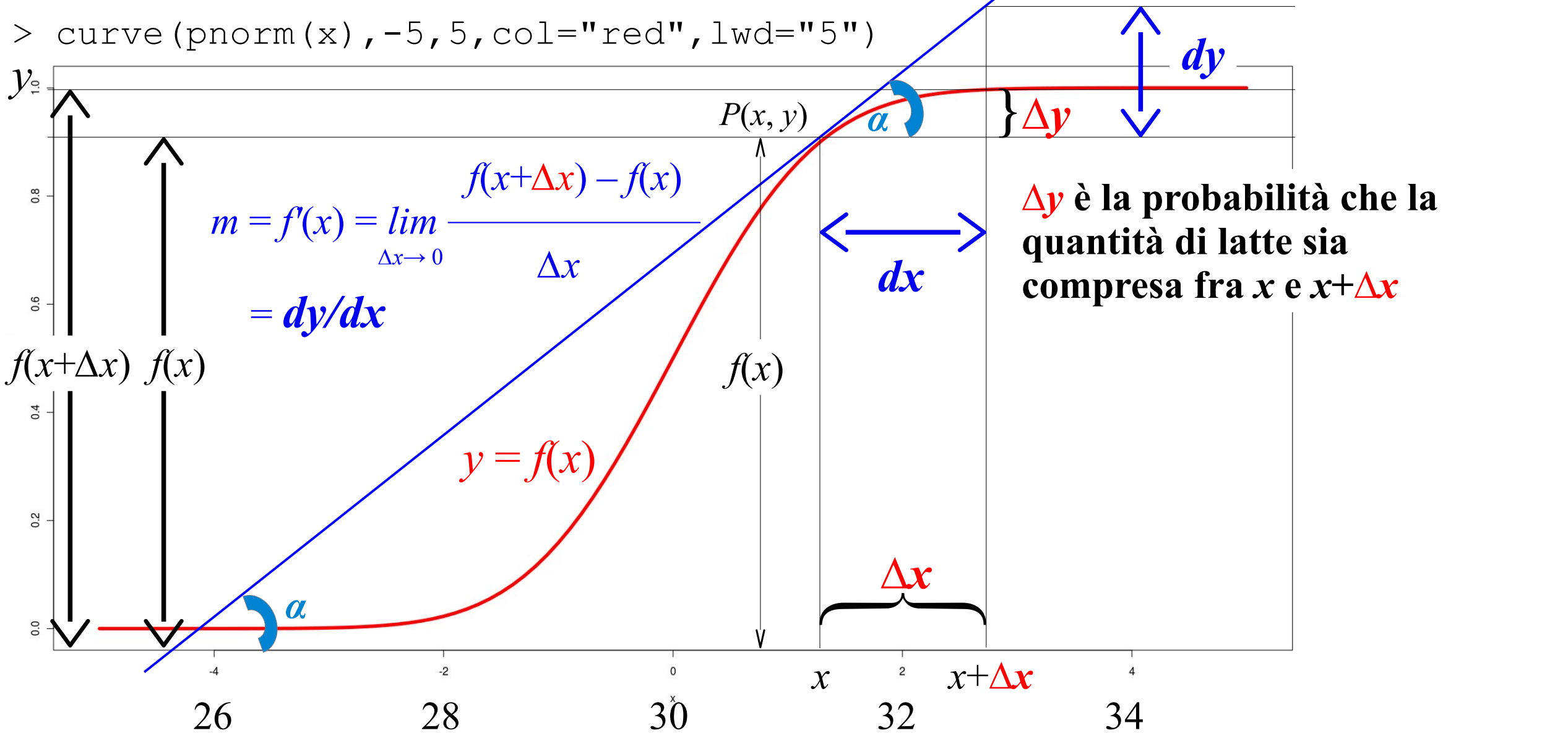
Distribuzione di probabilità cumulativa

Carattere quantitativo controllato da 3 loci



Un numero (quasi) infinito di fattori ambientali e genetici

```
> curve(pnorm(x), -5, 5, col="red", lwd="5")
```

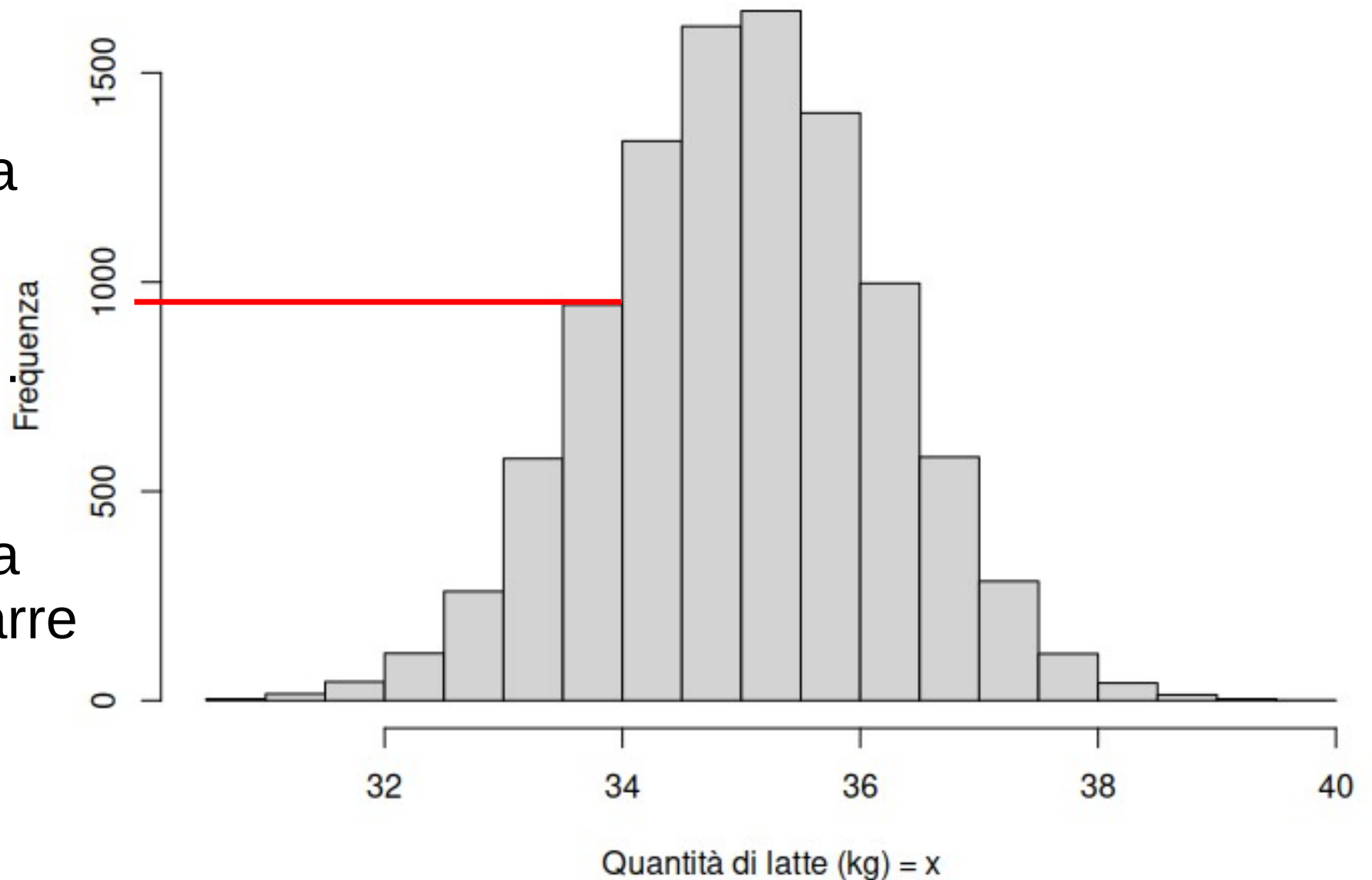


Un numero (quasi) infinito di fattori ambientali e genetici

Il N° di campioni compresi fra 33,5 e 34 è uguale a ≈ 950

Quindi l'h di ogni barra rappresenta...

Se ci sono 1aaaa campioni, la somma delle h di tutte le barre sarà uguale a...

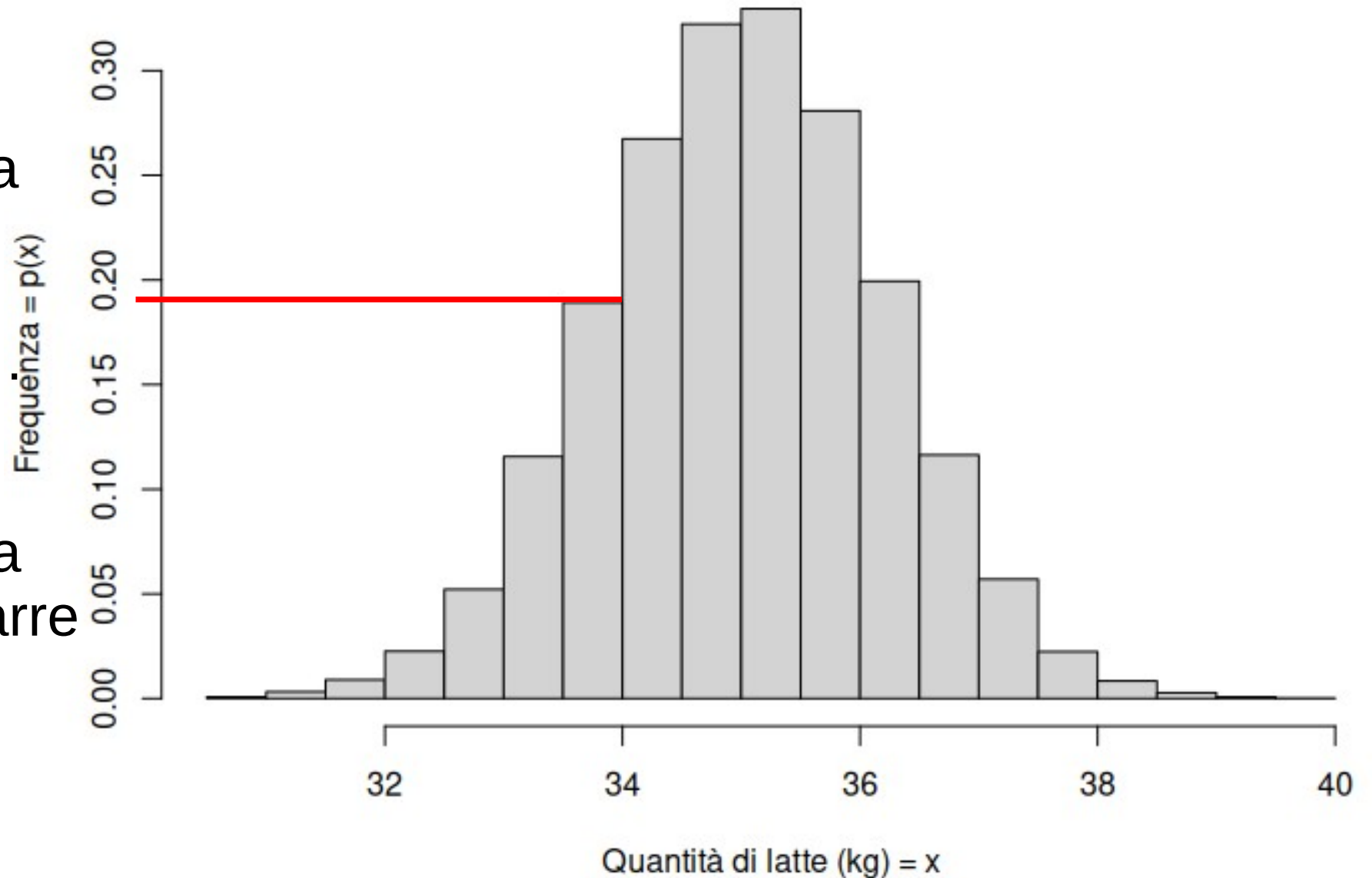


Un numero (quasi) infinito di fattori ambientali e genetici

Il N° di campioni compresi fra 33,5 e 34 è uguale a $\approx 95a$

Quindi l'h di ogni barra rappresenta...

Se ci sono 1aaaa campioni, la somma delle h di tutte le barre sarà uguale a...

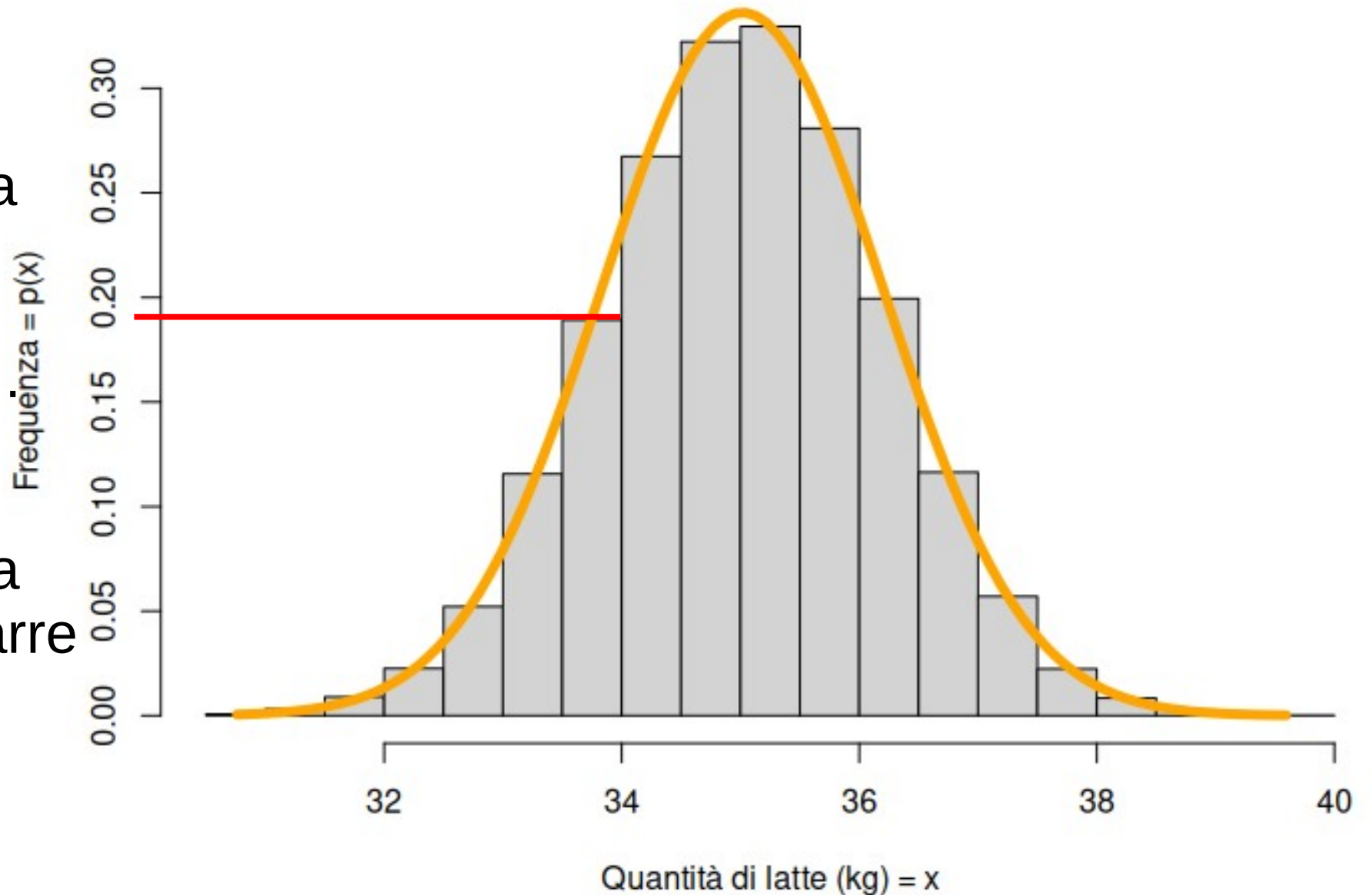


Un numero (quasi) infinito di fattori ambientali e genetici

Il N° di campioni compresi fra 33,5 e 34 è uguale a $\approx 95a$

Quindi l'h di ogni barra rappresenta...

Se ci sono 1aaaaa campioni, la somma delle h di tutte le barre sarà uguale a...



Un numero (quasi) infinito di fattori ambientali e genetici

per disegnare l'istogramma

```
> x = rnorm(1aaaa, mean = 35, sd = 1.2)
> hist(x, xlab = 'Quantità di latte (kg) = x', ylab = 'Frequenza')
> hist(x, prob = TRUE, xlab = 'Quantità di latte (kg) = x', ylab = 'Frequenza
= p(x)')
> x_values = seq(min(x), max(x), length = 1aa)
> fun = dnorm(x_values, mean = mean(x), sd = sd(x))
> lines(x_values, fun, col = 'orange', lwd = 5)
```

per disegnare la densità

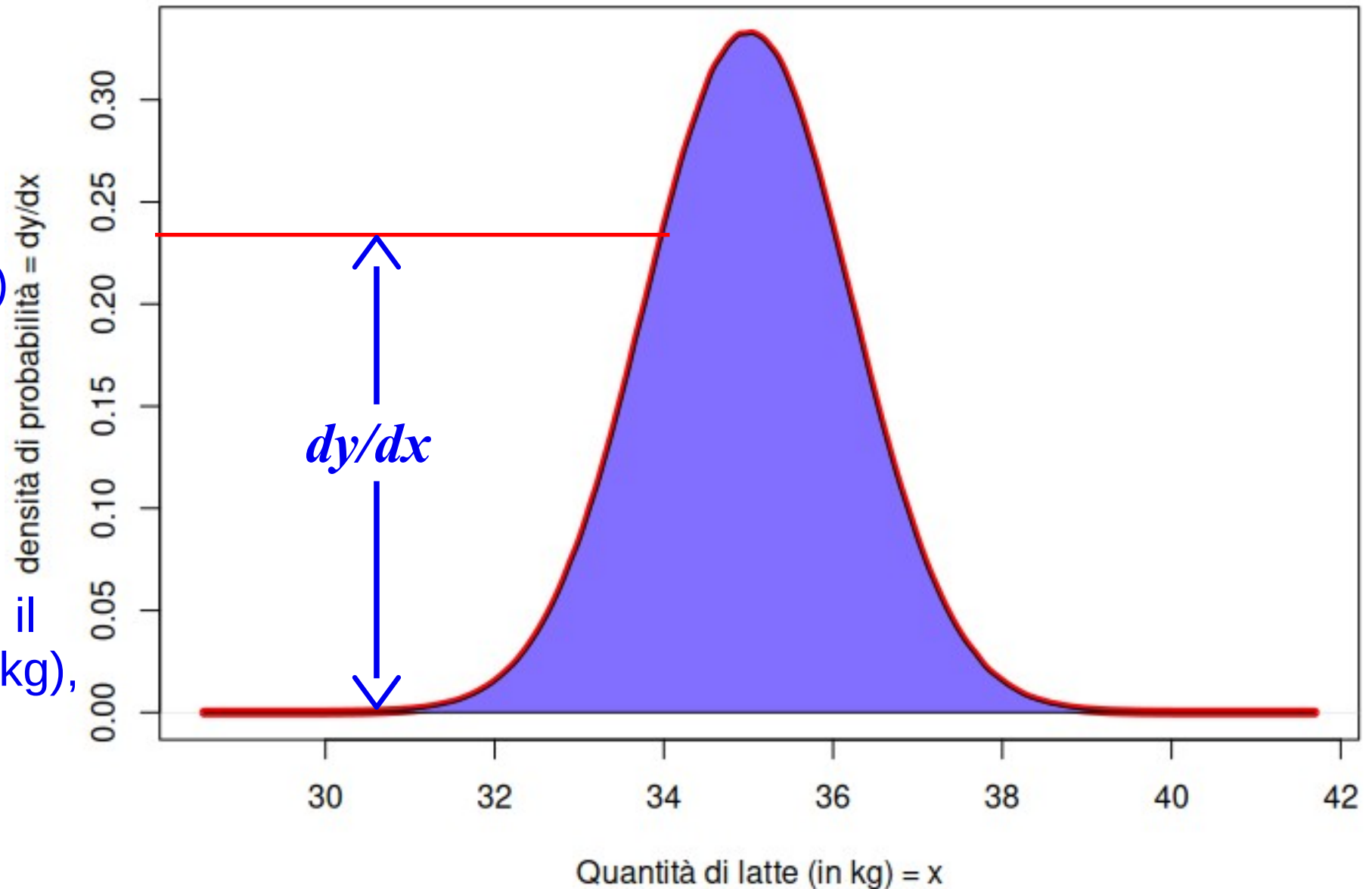
attenzione: x ha un significato diverso

```
> y = rnorm(1aaaaaaaa, mean = 35, sd = 1.2)
> den = density(y)
> plot(den, col = 'red', lwd = 5.a, xlab = 'Quantità di latte (in kg) = x',
ylab = 'densità di probabilità = dy/dx')
> polygon(den, col = 'slateblue1')
> value1 = 33
> value2 = 34
> l = min(which(den$x >= value1))
> h = max(which(den$x < value2))
> polygon(c(den$x[c(l, 1:h, h)]), c(a, den$y[1:h], a), col = 'slateblue1')
```

Un numero (quasi) infinito di fattori ambientali e genetici

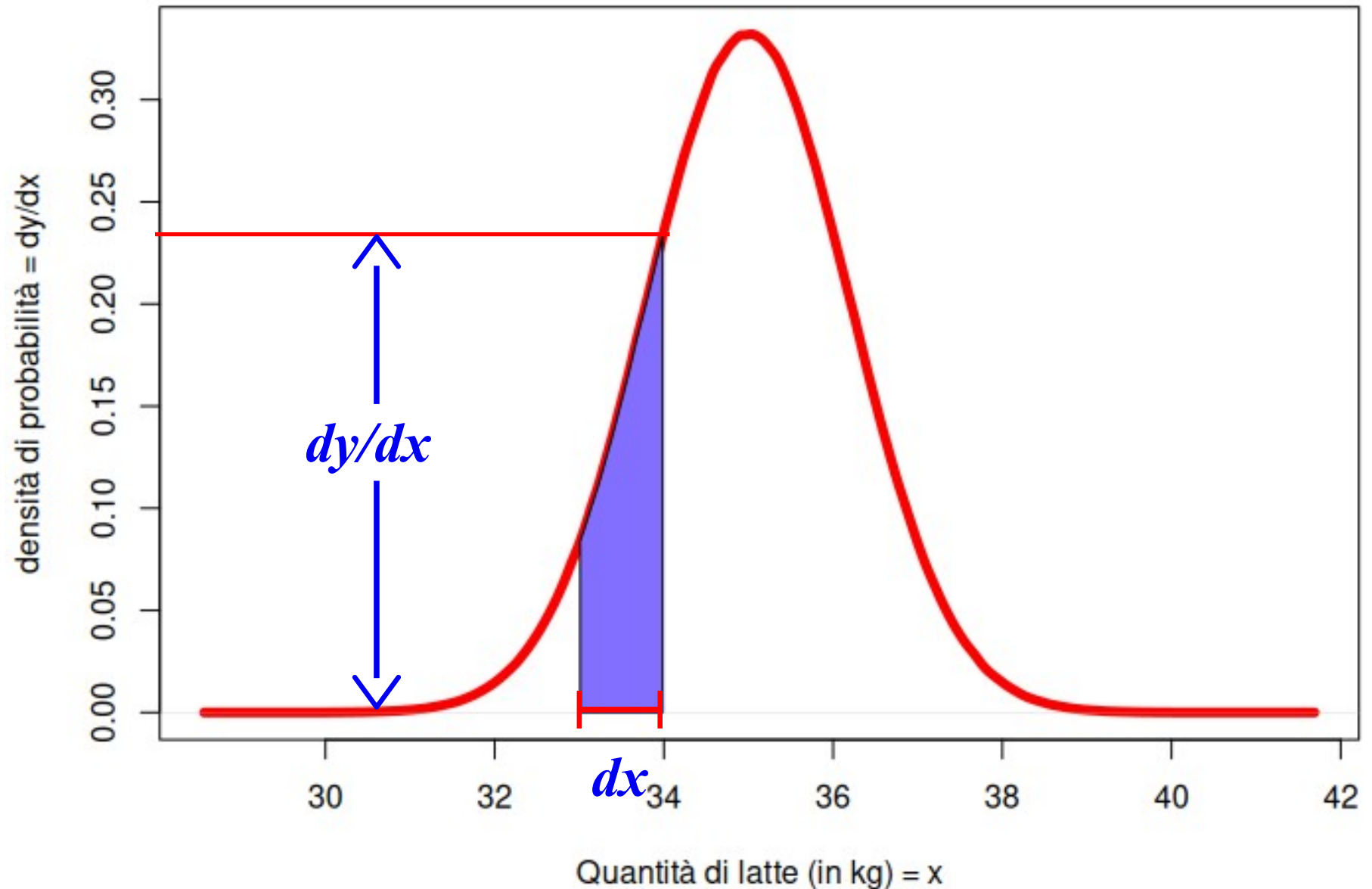
dy = probabilità che la quantità di latte sia compresa fra due valori, x e $x+dx$, in cui $dx \rightarrow a$
(densità di probabilità)

L'area sotto la curva rappresenta la probabilità che x sia compreso tra il suo valore min ($\approx 3a$ kg) e il suo valore max ($\approx 4a$ kg), che equivale al 100% (evento certo).



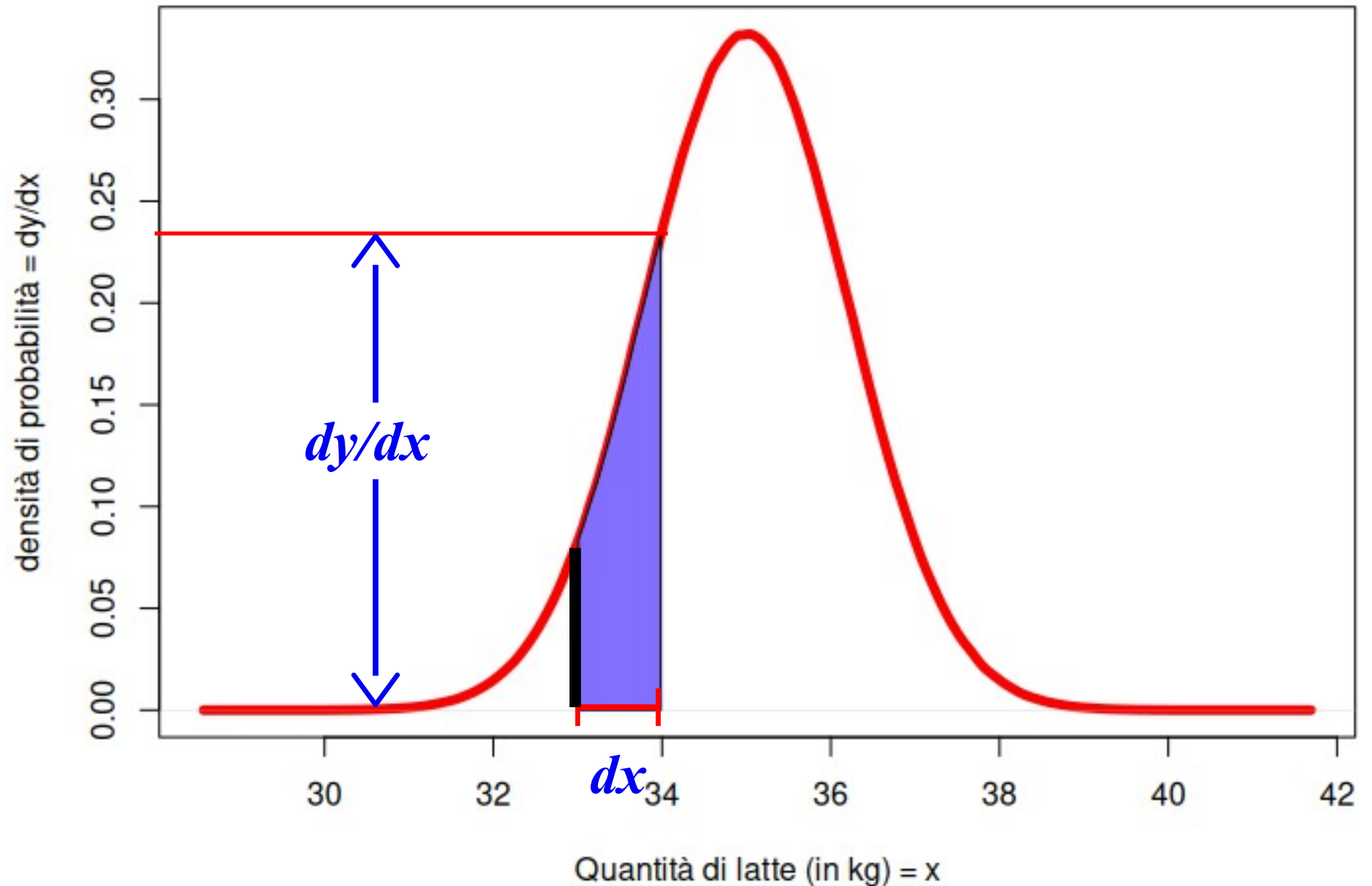
Un numero (quasi) infinito di fattori ambientali e genetici

La probabilità che la quantità di latte sia compresa fra 33 e 34 Kg è data dall'area blu, che può essere calcolata utilizzando un'operazione che prende il nome di «**integrazione**».



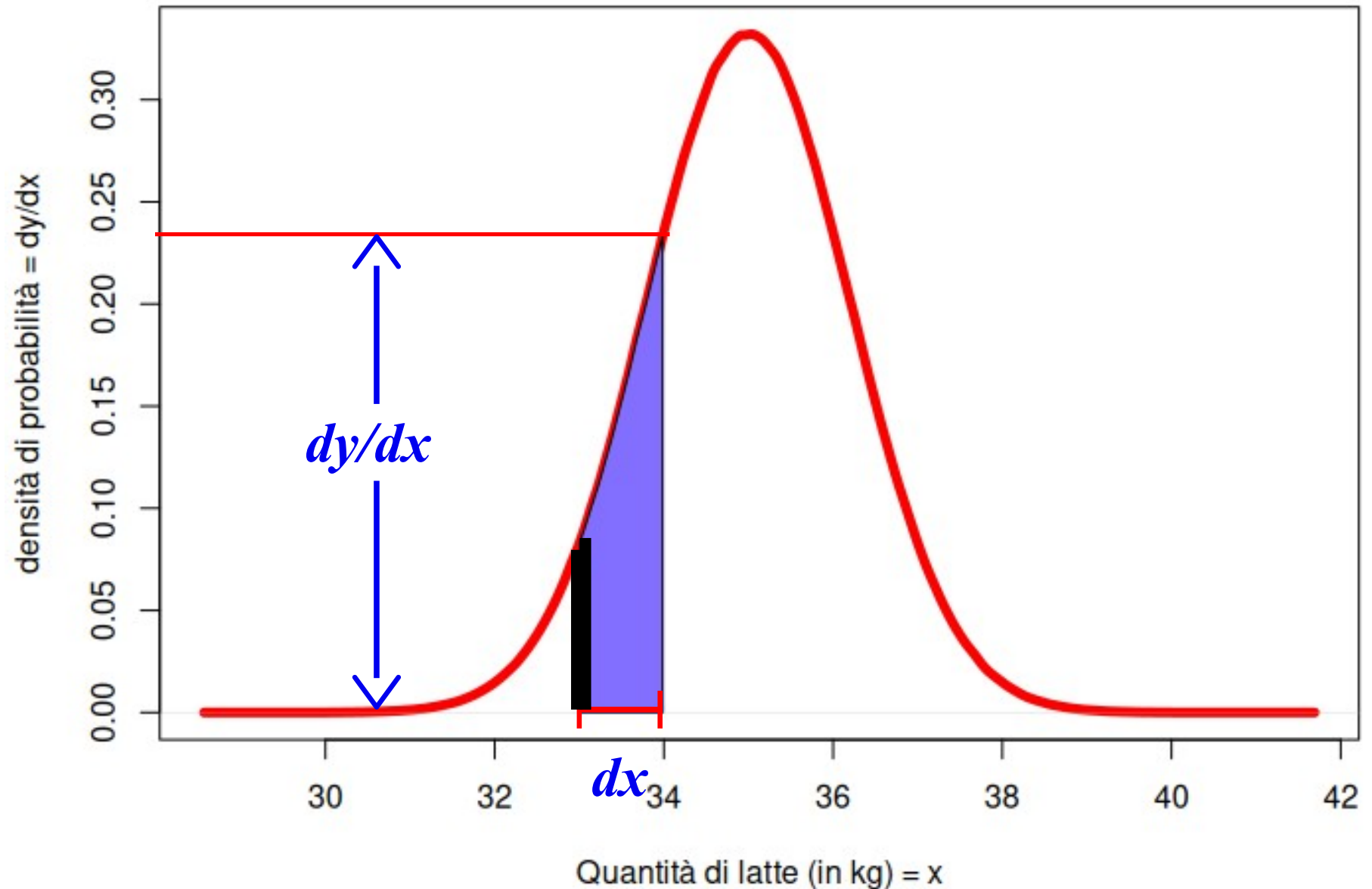
Un numero (quasi) infinito di fattori ambientali e genetici

La probabilità che la quantità di latte sia compresa fra 33 e 34 Kg è data dall'area blu, che può essere calcolata utilizzando un'operazione che prende il nome di «**integrazione**».



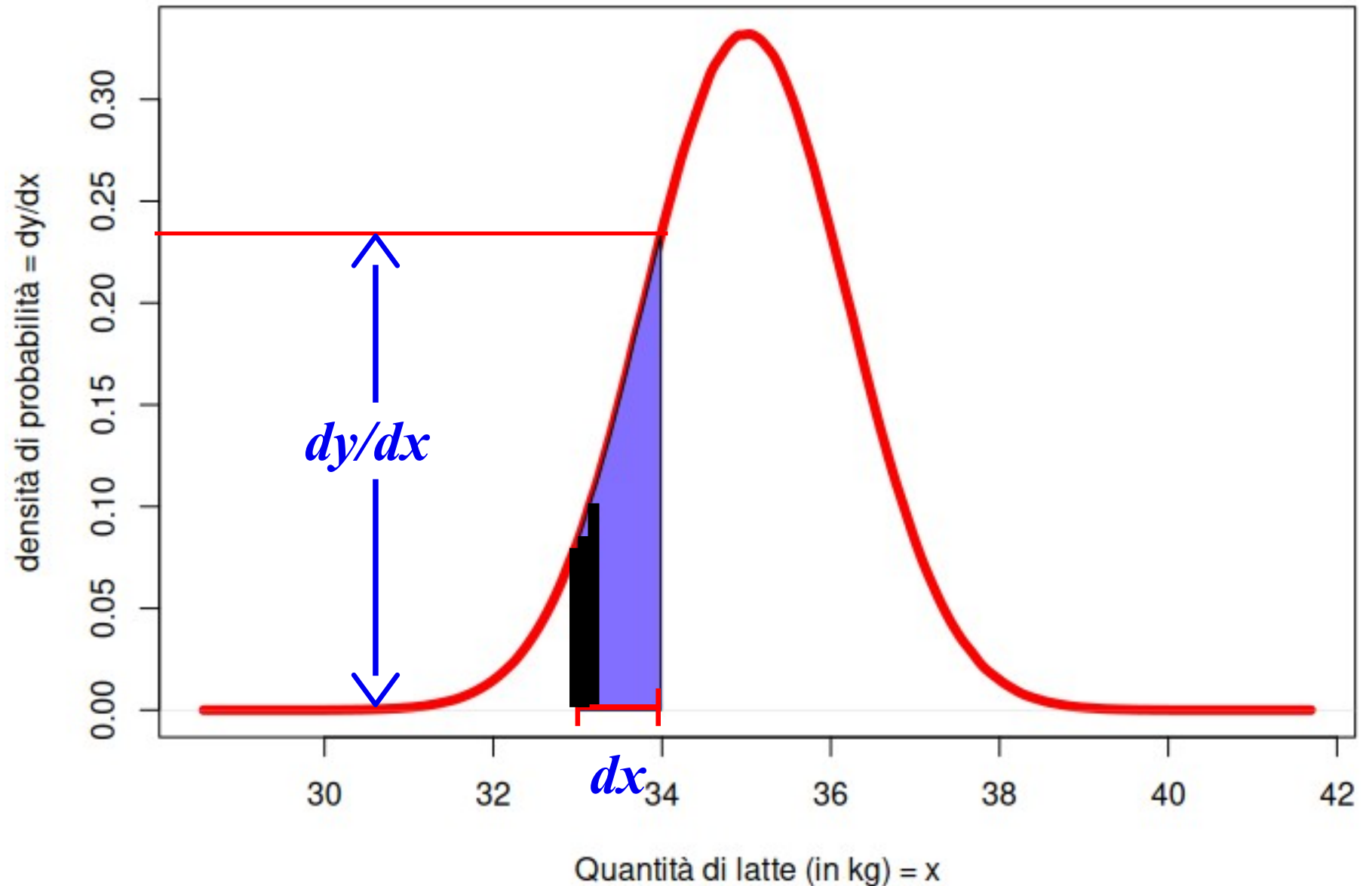
Un numero (quasi) infinito di fattori ambientali e genetici

La probabilità che la quantità di latte sia compresa fra 33 e 34 Kg è data dall'area blu, che può essere calcolata utilizzando un'operazione che prende il nome di «**integrazione**».



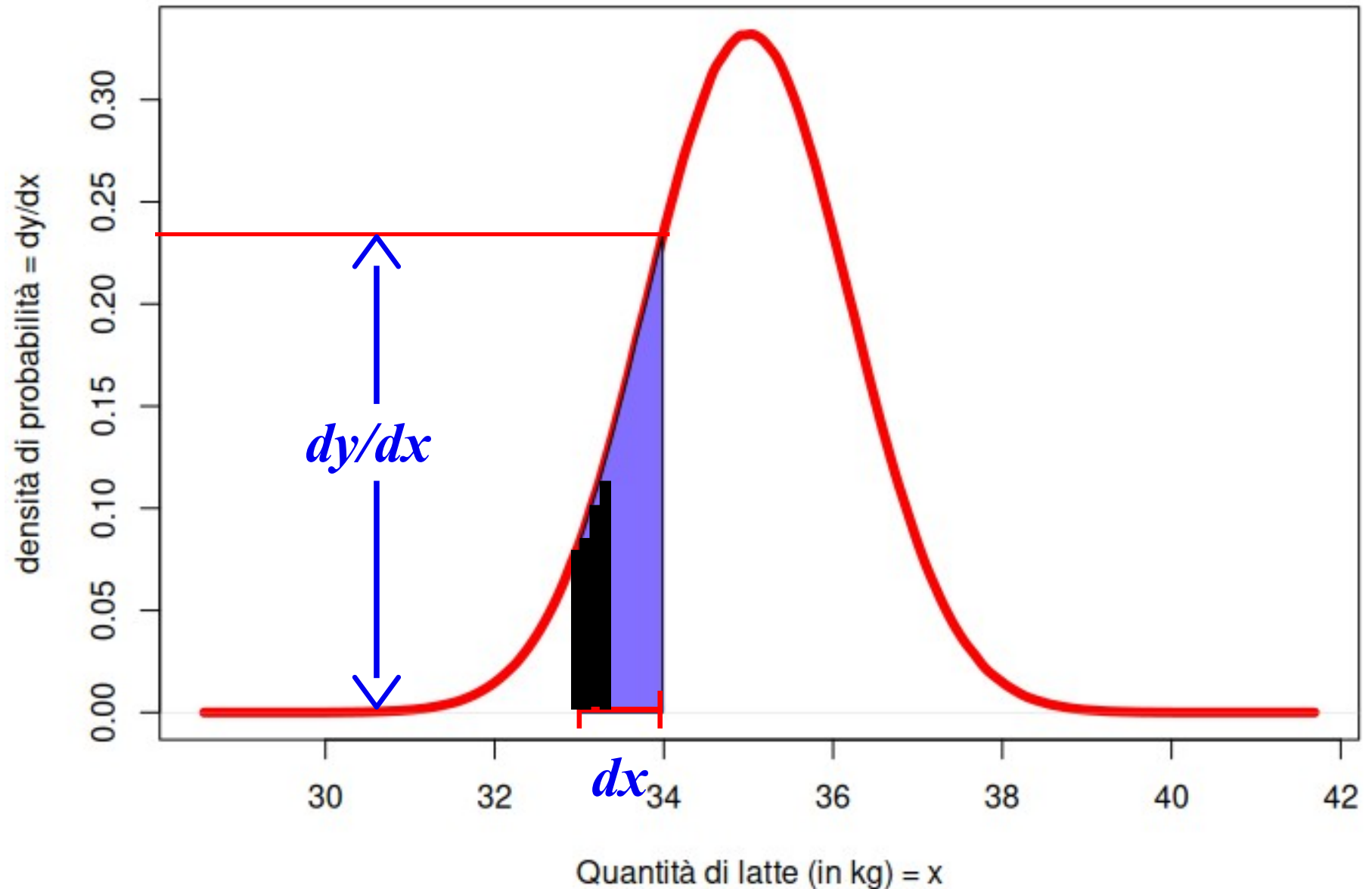
Un numero (quasi) infinito di fattori ambientali e genetici

La probabilità che la quantità di latte sia compresa fra 33 e 34 Kg è data dall'area blu, che può essere calcolata utilizzando un'operazione che prende il nome di «**integrazione**».



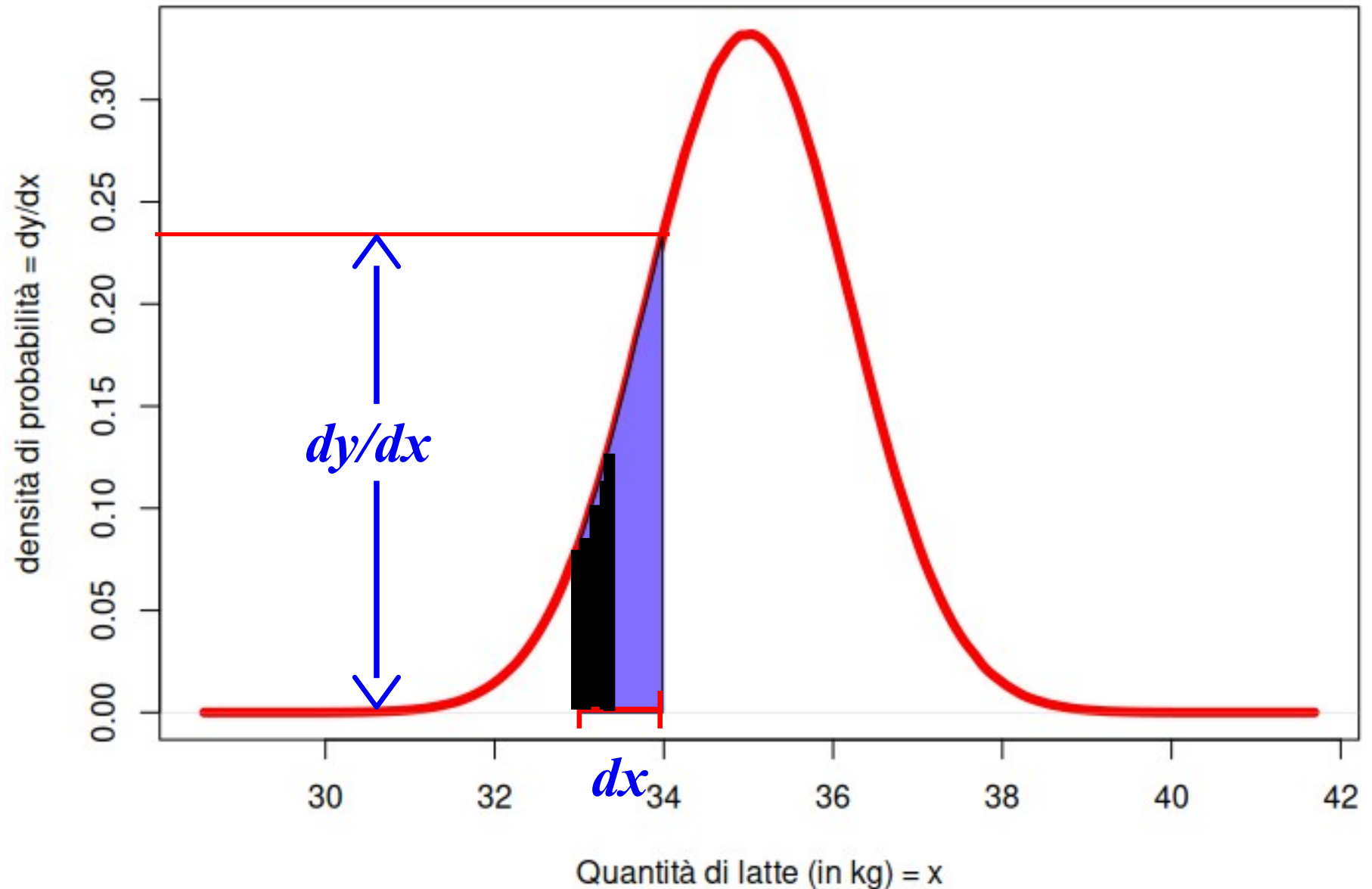
Un numero (quasi) infinito di fattori ambientali e genetici

La probabilità che la quantità di latte sia compresa fra 33 e 34 Kg è data dall'area blu, che può essere calcolata utilizzando un'operazione che prende il nome di «**integrazione**».



Un numero (quasi) infinito di fattori ambientali e genetici

La probabilità che la quantità di latte sia compresa fra 33 e 34 Kg è data dall'area blu, che può essere calcolata utilizzando un'operazione che prende il nome di «**integrazione**».




```
. Instantaneous rate matrix :
[A-----C-----G-----T-----]
-0.99555    0.39721    0.59777    0.00057
 0.63822   -1.84503    0.45935    0.74746
 0.71234    0.34068   -1.05494    0.00191
 0.00055    0.44949    0.00155   -0.45159
```

```
> x = seq(0, 10, by = 0.1)
> y = dgamma(x, shape = 0.997)
> plot(y, type = 'l', lwd = 5, col = 'red', xlab = 'Tasso di
sostituzione nucleotidica', ylab = 'Densità di probabilità')
```

Densità di probabilità

0.8

0.6

0.4

0.2

0.0

0

20

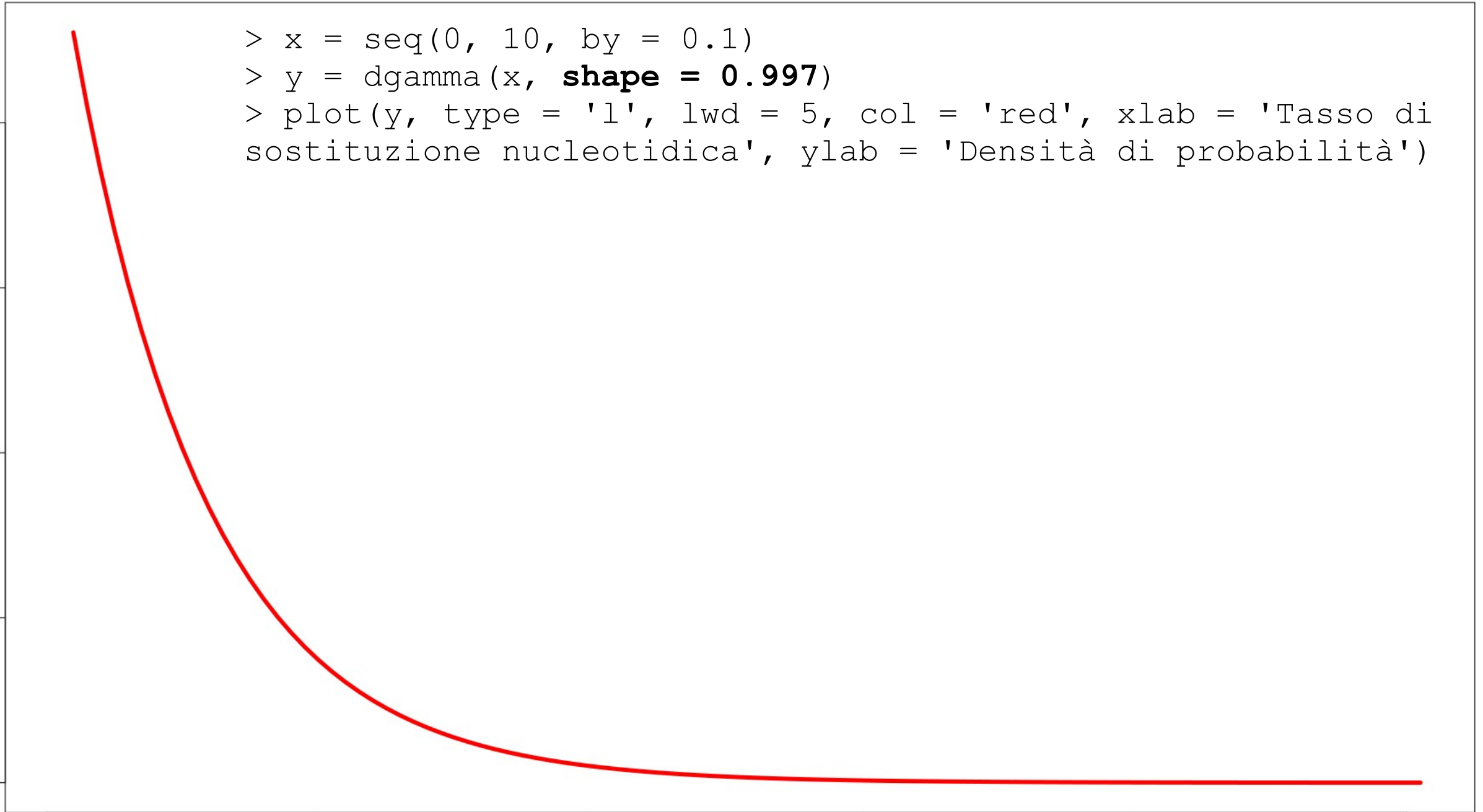
40

60

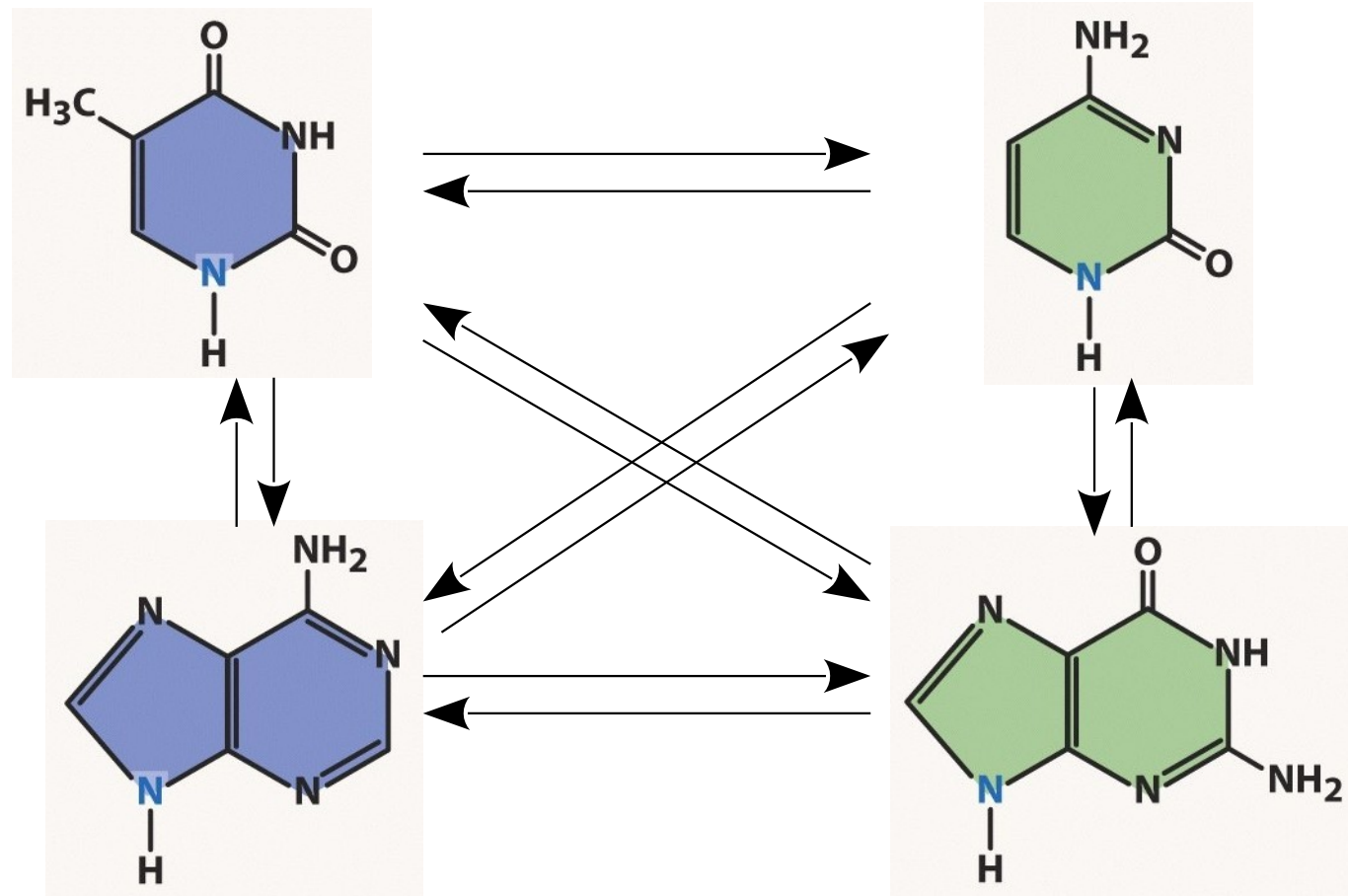
80

100

Tasso di sostituzione nucleotidica



Il modello JC69

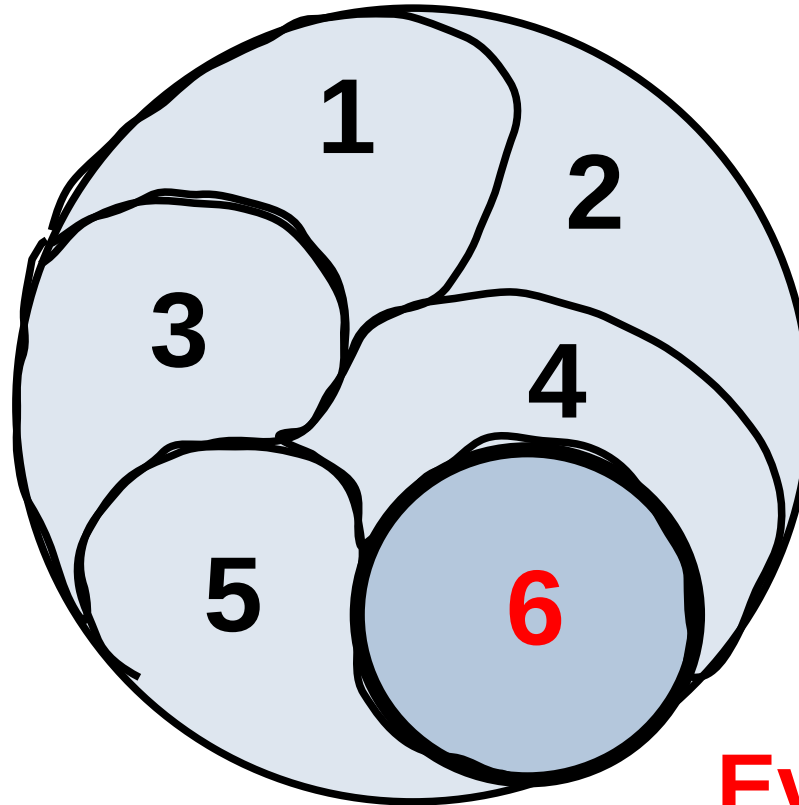


STATISTICA: concetti fondamentali

- **Esperimento:** processo che genera un'osservazione (per es., il lancio di un dado o di una moneta)
- **Evento:** ognuno dei possibili risultati di un esperimento (contrassegnati da lettere maiuscole)
- **Evento elementare:** evento che non può essere decomposto e che corrisponde ad un «punto campione» indivisibile

STATISTICA: spazio campione Ω

- Insieme di tutti i possibili risultati di un esperimento
Es. lancio di un dado
- 6 risultati equiprobabili (se il dado non è truccato)



$$\Omega = 1$$

Evento certo

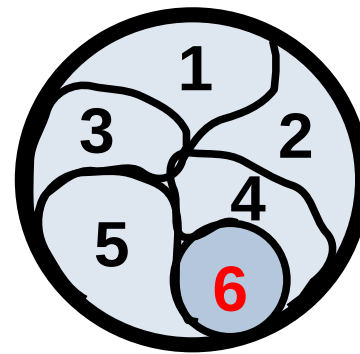
Evento elementare a

Cosa succede se lancio il dado una seconda volta?

- Vinco se esce di nuovo il numero 6: che probabilità ho di vincere?
- Lo spazio campione Ω diventa uguale ad $1/6$
- Poiché si tratta di EVENTI INDIPENDENTI, la $P(6, 6) = 1/6 * 1/6 = 1/36$
- Probabilità congiunta di eventi indipendenti = = prodotto delle probabilità dei singoli eventi

$$\Omega = 1/6$$

Evento certo



Evento elementare B

STATISTICA: probabilità di due eventi indipendenti

- **a** = evento elementare esce 6 al primo lancio
- **B** = evento elementare esce 6 al secondo lancio

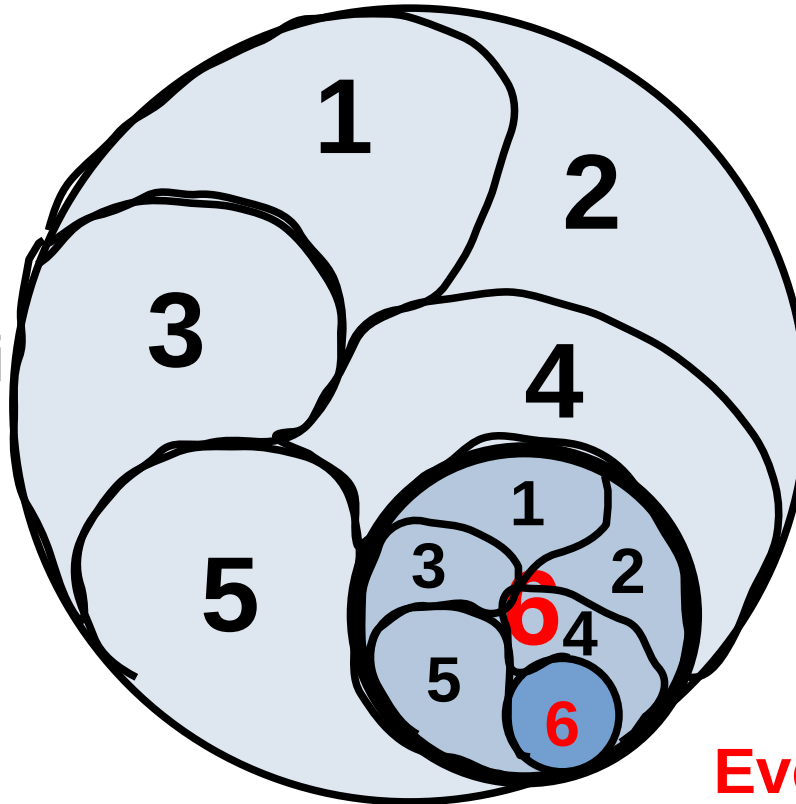
Quindi, noi vogliamo calcolare la Prob dell'intersezione fra il cerchio medio e il cerchio piccolo:

$$P(a \cap B) = P(a) * P(B|a)$$

Ma, poiché i due eventi sono indipendenti, $P(B|a) = P(B)$ e si applica la regola del prodotto

$$P(a \cap B) = P(a) * P(B)$$

$$P(a \cap B) = 1/6 * 1/6 = 1/36$$



$$\Omega = 1/6$$

Evento certo

Evento
elementare **a**

Evento elementare **B**

STATISTICA: la probabilità condizionata

Da

$$P(a \cap B) = P(a) * P(B|a)$$

si ricava:

$$P(B|a) = P(a \cap B) / P(a)$$

La sostituzione nucleotidica è un processo di Markov

Nel corso del tempo sono stati elaborati vari modelli di sostituzione nucleotidica, più complessi, i quali prevedono che i quattro differenti nucleotidi, che possono essere presenti ad ogni sito, siano gli stati di una catena di Markov.

Ogni nucleotide può essere sostituito da ogni altro ed anche da un nucleotide identico. Le probabilità di tali cambiamenti si chiamano **probabilità di transizione** e sono gli ingressi di una matrice quadrata $\mathbf{P}(t)$ di dimensioni 4 X 4:

$$\mathbf{P} = \begin{matrix} & \begin{matrix} p_{11} & p_{12} & p_{13} & p_{14} \end{matrix} \\ \begin{matrix} p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \\ p_{41} & p_{42} & p_{43} & p_{44} \end{matrix} & \end{matrix}$$

La sostituzione nucleotidica è un processo di Markov

I **processi di Markov**, chiamati così in onore del grande matematico russo Andrey Andreyevich Markov, sono catene di eventi che si presentano uno dopo l'altro in istanti successivi di tempo.

Nel nostro caso gli eventi sono le sostituzioni nucleotidiche casuali, in seguito alle quali lo stato X del carattere che si osserva ad un determinato sito nucleotidico, ossia la base azotata, può cambiare al trascorrere del tempo.

Esso dipende, però, soltanto dall'ultimo stato in cui il processo si trovava e non da quelli precedenti: la Prob che una certa base azotata venga sostituita da ciascuna delle altre tre dipende dalla base “attuale” e non da quelle che si trovavano in quella posizione della sequenza in passato.

$$P\{X_{t+1}=j \mid X_t=i, X_{t-1}=i-1, \dots, X_0=0\} = P\{X_{t+1}=j \mid X_t=i\} = P_{ij}$$

per tutti i valori di j , i , X_{t-1} , ..., X_0 e t . P_{ij} è costante nel tempo e non dipende da alcuno stato precedente; in altri termini, il processo non ha memoria degli stati precedenti a quello corrente, ad eccezione dell'ultimo (*proprietà di Markov*).

La sostituzione nucleotidica è un processo di Markov

$$P\{X_{t+1}=j \mid X_t=i, X_{t-1}=i-1, \dots, X_0=0\} = P\{X_{t+1}=j \mid X_t=i\} = P_{ij}$$

per tutti i valori di j , i , X_{t-1} , ..., X_0 e t .

P_{ij} = **probabilità di transizione** dallo stato i allo stato j (probabilità che il processo si sposti da uno stato ad un altro in un singolo passaggio) è costante nel tempo e non dipende da alcuno stato precedente; in altri termini, il processo non ha memoria degli stati precedenti a quello corrente, ad eccezione dell'ultimo (*proprietà di Markov*).

X_{t+1} = stato del processo nell'istante $t+1$

Ogni processo di Markov ha uno *spazio campione* Ω , definito come l'insieme, finito, dei possibili stati del processo (spazio degli stati):

$$\Omega = \{1, 2, \dots, n\}$$

con n stati possibili.

Stati transitori e stati definitivi

Gli **stati transitori**, dopo essere stati visitati, possono essere abbandonati mentre da quelli **definitivi** la catena, una volta entrata, non può più uscire. Gli stati definitivi, per cui $P_{ii} = 1$, si presentano nella matrice di transizione $\mathbf{P}$ come altrettanti 1 (uno) collocati nella diagonale principale, mentre in tutti gli altri ingressi ci sono solo zeri.

Per gli stati transitori $P_{ii} < 1$ cosicché almeno qualche altro ingresso della i -esima riga è > 0 .

Per definizione, la catena deve trovarsi in uno degli stati possibili ad ogni *step*; per qualsiasi valore di i e j , $0 \leq P_{ij} \leq 1$ e

$$\sum_{j=1}^n P_{ij} = 1$$

per tutti gli i , ossia la somma delle probabilità che il processo si sposti dallo stato attuale i ad uno stato j qualsiasi (che può anche coincidere con i stesso) è uguale a 1.

La sostituzione nucleotidica è un processo di Markov

```
> mdat = matrix(c(2,1,3,4,1,3,1,5,7,1,1,1,4,2,1,3), nrow = 4, ncol = 4, byrow = TRUE)
```

```
> mdat
```

	[,1]	[,2]	[,3]	[,4]
[1,]	2	1	3	4
[2,]	1	3	1	5
[3,]	7	1	1	1
[4,]	4	2	1	3

```
>
```

Come passare da i a j in due passaggi

Le probabilità di transizione per $t = 1$ sono date dagli ingressi della matrice $\mathbf{P}$.

$$\mathbf{P} = \begin{matrix} & \begin{matrix} p_{11} & p_{12} & p_{13} & p_{14} \end{matrix} \\ \begin{matrix} p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \\ p_{41} & p_{42} & p_{43} & p_{44} \end{matrix} & \end{matrix}$$

Per $t = 2$, invece, occorre tener conto del fatto che il processo può passare dallo stato i allo stato j attraversando un qualsivoglia stato intermedio k :

$$P_{ij}(2) = \sum_{k=1}^n P_{ik} P_{kj}$$

per tutti gli stati i, j .

Per ogni possibile stato intermedio k si calcola la probabilità congiunta che il processo passi da i a k e, successivamente, da k a j , quindi si sommano le singole probabilità congiunte, tante quanti sono i possibili stati k , ossia n

Come passare da i a j in due passaggi

Per calcolare la probabilità che il sistema passi dallo stato iniziale $i = \textcircled{1}$ allo stato finale $j = \textcircled{3}$ passando per un qualsivoglia stato intermedio k occorre applicare la regola della **moltiplicazione di due matrici**: riga X colonna

$$\mathbf{P} = \begin{matrix} & \begin{matrix} p_{11} & p_{12} & p_{13} & p_{14} \end{matrix} \\ \begin{matrix} p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \\ p_{41} & p_{42} & p_{43} & p_{44} \end{matrix} & \end{matrix}$$

k					j				
i					k				
$\textcircled{1}$	P_{11}	P_{12}	P_{13}	P_{14}	1	P_{11}	P_{12}	P_{13}	P_{14}
2	P_{21}	P_{22}	P_{23}	P_{24}	2	P_{21}	P_{22}	P_{23}	P_{24}
3	P_{31}	P_{32}	P_{33}	P_{34}	3	P_{31}	P_{32}	P_{33}	P_{34}
4	P_{41}	P_{42}	P_{43}	P_{44}	4	P_{41}	P_{42}	P_{43}	P_{44}

Come passare da A a C in due passaggi

	A	T	C	G		A	T	C	G		
	[,1]	[,2]	[,3]	[,4]		[,1]	[,2]	[,3]	[,4]		
A[1,]	2	1	3	4	*	A[1,]	2	1	3	= Q _{AC}	
T[2,]	1	3	1	5		T[2,]	1	3	1		5
C[3,]	7	1	1	1		C[3,]	7	1	1		1
G[4,]	4	2	1	3		G[4,]	4	2	1		3

$2*3+1*1+3*1+4*1 = \mathbf{14}$

```
> mdat%*%mdat
      [,1] [,2] [,3] [,4]
[1,]   42   16  14   28
[2,]   32   21   12   35
[3,]   26   13   24   37
[4,]   29   17   18   36
>
```

Come passare da A a C in due passaggi

```
piero@piero-XPS-9320:~$ octave
QSocketNotifier: Can only be used with threads started with QThread
GNU Octave, version 8.3.0
Copyright (C) 1993-2023 The Octave Project Developers.
This is free software; see the source code for copying conditions.
There is ABSOLUTELY NO WARRANTY; not even for MERCHANTABILITY or
FITNESS FOR A PARTICULAR PURPOSE.  For details, type 'warranty'.

Octave was configured for "x86_64-pc-linux-gnu".

Additional information about Octave is available at https://www.octave.org.

Please contribute if you find this software useful.
For more information, visit https://www.octave.org/get-involved.html

Read https://www.octave.org/bugs.html to learn how to submit bug reports.
For information about changes from previous versions, type 'news'.

octave:1>
```

Come passare da A a C in due passaggi

```
octave:1> a = [2, 1, 3, 4; 1, 3, 1, 5; 7, 1, 1, 1; 4, 2, 1, 3]
a =
```

2	1	3	4
1	3	1	5
7	1	1	1
4	2	1	3

```
octave:2> a^2
ans =
```

42	16	14	28
32	21	12	35
26	13	24	37
29	17	18	36

```
octave:3>
```

Stati transitori e stati definitivi

Gli **stati transitori**, dopo essere stati visitati, possono essere abbandonati mentre da quelli **definitivi** la catena, una volta entrata, non può più uscire. Gli stati definitivi, per cui $P_{ii} = 1$, si presentano nella matrice di transizione $\mathbf{P}$ come altrettanti 1 (uno) collocati nella diagonale principale, mentre in tutti gli altri ingressi ci sono solo zeri.

Per gli stati transitori $P_{ii} < 1$ cosicché almeno qualche altro ingresso della i -esima riga è > 0 .

Per definizione, la catena deve trovarsi in uno degli stati possibili ad ogni *step*; per qualsiasi valore di i e j , $0 \leq P_{ij} \leq 1$ e

$$\sum_{j=1}^n P_{ij} = 1$$

per tutti gli i , ossia la somma delle probabilità che il processo si sposti dallo stato attuale i ad uno stato j qualsiasi (che può anche coincidere con i stesso) è uguale a 1.

Il magico numero <e>

```
> 10*9*8*7*6*5*4*3*2*1
```

```
[1] 3628800
```

```
> factorial(10)
```

```
[1] 3628800
```

```
> Euler = function(n,z)
```

```
+ {
```

```
+ result = NULL
```

```
+ numbers = seq(1:n)
```

```
+ for (i in numbers)
```

```
+ {
```

```
+ result[i] = (z^i)/factorial(i)
```

```
+ }
```

```
+ return(1 + sum(result))
```

```
+ }
```

```
> Euler(100,1)
```

```
[1] 2.718282
```

```
> Euler(100,10)
```

```
[1] 22026.47
```

```
> exp(10)
```

```
[1] 22026.47
```

$$e = 1 + 1/1! + 1/2! + 1/3! + 1/4! + \dots \cong 2.718282$$

$$e^z = 1 + z/1! + z^2/2! + z^3/3! + z^4/4! + \dots \cong 2.718282$$

Per $z < 1$:

$$e^z \cong 1 + z$$

Es.

$$\text{se } z = 0.1$$

$$e^{0.1} \cong 1 + 0.1 \cong 1.1$$

```
> exp(0.1)
```

```
[1] 1.105171
```

```
> Euler(10,0.1)
```

```
[1] 1.105171
```

Da
 $e^z = 1 + z/1! + z^2/2! + z^3/3! + z^4/4! + \dots \cong 1 + z$
 per $z < 1$ (al limite per z che tende a a)

Si ricava che:
 $e^{\mathbf{A}} = \mathbf{I} + \mathbf{A}/1! + \mathbf{A}^2/2! + \mathbf{A}^3/3! + \mathbf{A}^4/4! + \dots \cong (\mathbf{I} + \mathbf{A}) = \mathbf{P}$

$$\mathbf{P} = \begin{pmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \\ p_{41} & p_{42} & p_{43} & p_{44} \end{pmatrix}$$

$$\mathbf{I} + \mathbf{A} = \mathbf{P}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix} = \begin{pmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \\ p_{41} & p_{42} & p_{43} & p_{44} \end{pmatrix}$$